

2 Gravity, the figure of the Earth and geodynamics

2.1 THE EARTH'S SIZE AND SHAPE

2.1.1 Earth's size

The philosophers and savants in ancient civilizations could only speculate about the nature and shape of the world they lived in. The range of possible travel was limited and only simple instruments existed. Unrelated observations might have suggested that the Earth's surface was upwardly convex. For example, the Sun's rays continue to illuminate the sky and mountain peaks after its disk has already set, departing ships appear to sink slowly over the horizon, and the Earth's shadow can be seen to be curved during partial eclipse of the Moon. However, early ideas about the heavens and the Earth were intimately bound up with concepts of philosophy, religion and astrology. In Greek mythology the Earth was a disk-shaped region embracing the lands of the Mediterranean and surrounded by a circular stream, *Oceanus*, the origin of all the rivers. In the sixth century BC the Greek philosopher Anaximander visualized the heavens as a celestial sphere that surrounded a flat Earth at its center. Pythagoras (582–507 BC) and his followers were apparently the first to speculate that the Earth was a sphere. This idea was further propounded by the influential philosopher Aristotle (384–322 BC). Although he taught the scientific principle that theory must follow fact, Aristotle is responsible for the logical device called syllogism, which can explain correct observations by apparently logical accounts that are based on false premises. His influence on scientific methodology was finally banished by the scientific revolution in the seventeenth century.

The first scientifically sound estimate of the size of the terrestrial sphere was made by Eratosthenes (275–195 BC), who was the head librarian at Alexandria, a Greek colony in Egypt during the third century BC. Eratosthenes had been told that in the city of Syene (modern Aswan) the Sun's noon rays on midsummer day shone vertically and were able to illuminate the bottoms of wells, whereas on the same day in Alexandria shadows were cast. Using a sun-dial Eratosthenes observed that at the summer solstice the Sun's rays made an angle of one-fiftieth of a circle (7.2°) with the vertical in Alexandria (Fig. 2.1). Eratosthenes believed that Syene and Alexandria were on the same meridian. In fact they are slightly displaced; their geographic coordinates are $24^\circ 5'N$ $32^\circ 56'E$ and

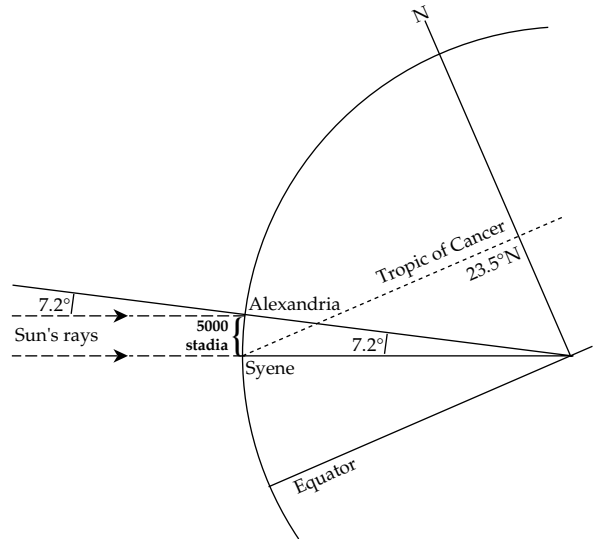


Fig. 2.1 The method used by Eratosthenes (275–195 BC) to estimate the Earth's circumference used the 7.2° difference in altitude of the Sun's rays at Alexandria and Syene, which are 5000 *stadia* apart (after Strahler, 1963).

$31^\circ 13'N$ $29^\circ 55'E$, respectively. Syene is actually about half a degree north of the tropic of Cancer. Eratosthenes knew that the approximate distance from Alexandria to Syene was 5000 *stadia*, possibly estimated by travellers from the number of days ("10 camel days") taken to travel between the two cities. From these observations Eratosthenes estimated that the circumference of the global sphere was 250,000 *stadia*. The Greek *stadium* was the length (about 185 m) of the U-shaped racecourse on which footraces and other athletic events were carried out. Eratosthenes' estimate of the Earth's circumference is equivalent to 46,250 km, about 15% higher than the modern value of 40,030 km.

Estimates of the length of one meridian degree were made in the eighth century AD during the Tang dynasty in China, and in the ninth century AD by Arab astronomers in Mesopotamia. Little progress was made in Europe until the early seventeenth century. In 1662 the Royal Society was founded in London and in 1666 the Académie Royale des Sciences was founded in Paris. Both organizations provided support and impetus to the scientific revolution. The invention of the telescope enabled more precise geodetic surveying. In 1671 a French astronomer, Jean Picard

(1620–1682), completed an accurate survey by triangulation of the length of a degree of meridian arc. From his results the Earth’s radius was calculated to be 6372 km, remarkably close to the modern value of 6371 km.

2.1.2 Earth’s shape

In 1672 another French astronomer, Jean Richer, was sent by Louis XIV to make astronomical observations on the equatorial island of Cayenne. He found that an accurate pendulum clock, which had been adjusted in Paris precisely to beat seconds, was losing about two and a half minutes per day, i.e., its period was now too long. The error was much too large to be explained by inaccuracy of the precise instrument. The observation aroused much interest and speculation, but was only explained some 15 years later by Sir Isaac Newton in terms of his laws of universal gravitation and motion.

Newton argued that the shape of the rotating Earth should be that of an *oblate ellipsoid*; compared to a sphere, it should be somewhat flattened at the poles and should bulge outward around the equator. This inference was made on logical grounds. Assume that the Earth does not rotate and that holes could be drilled to its center along the rotation axis and along an equatorial radius (Fig. 2.2). If these holes are filled with water, the hydrostatic pressure at the center of the Earth sustains equal water columns along each radius. However, the rotation of the Earth causes a centrifugal force at the equator but has no effect on the axis of rotation. At the equator the outward centrifugal force of the rotation opposes the inward gravitational attraction and pulls the water column upward. At the same time it reduces the hydrostatic pressure produced by the water column at the Earth’s center. The reduced central pressure is unable to support the height of the water column along the polar radius, which subsides. If the Earth were a hydrostatic sphere, the form of the rotating Earth should be an oblate ellipsoid of revolution. Newton assumed the Earth’s density to be constant and calculated that the flattening should be about 1:230 (roughly 0.5%). This is somewhat larger than the actual flattening of the Earth, which is about 1:298 (roughly 0.3%).

The increase in period of Richer’s pendulum could now be explained. Cayenne was close to the equator, where the larger radius placed the observer further from the center of gravitational attraction, and the increased distance from the rotational axis resulted in a stronger opposing centrifugal force. These two effects resulted in a lower value of gravity in Cayenne than in Paris, where the clock had been calibrated.

There was no direct proof of Newton’s interpretation. A corollary of his interpretation was that the degree of meridian arc should subtend a longer distance in polar regions than near the equator (Fig. 2.3). Early in the eighteenth century French geodesists extended the standard meridian from border to border of the country and found a puzzling result. In contrast to the prediction of Newton, the degree

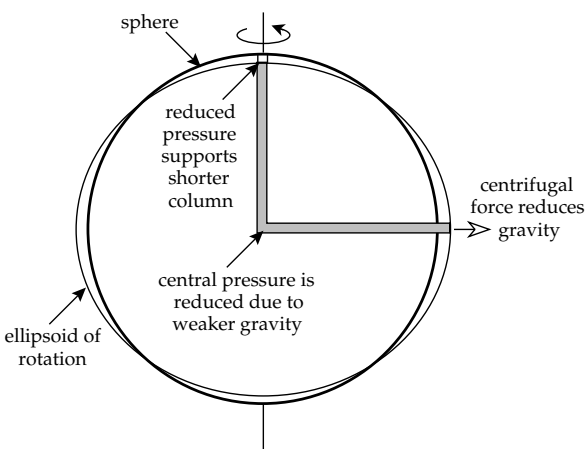


Fig. 2.2 Newton’s argument that the shape of the rotating Earth should be flattened at the poles and bulge at the equator was based on hydrostatic equilibrium between polar and equatorial pressure columns (after Strahler, 1963).

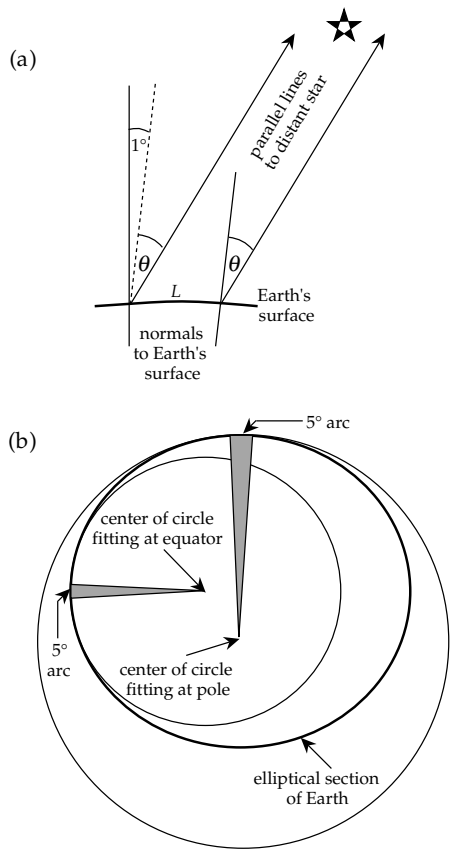


Fig. 2.3 (a) The length of a degree of meridian arc is found by measuring the distance between two points that lie one degree apart on the same meridian. (b) The larger radius of curvature at the flattened poles gives a longer arc distance than is found at the equator where the radius of curvature is smaller (after Strahler, 1963).

of meridian arc *decreased* northward. The French interpretation was that the Earth’s shape was a *prolate* ellipsoid, elongated at the poles and narrowed at the equator, like the shape of a rugby football. A major scientific controversy arose between the “flatteners” and the “elongators.”

To determine whether the Earth's shape was oblate or prolate, the Académie Royale des Sciences sponsored two scientific expeditions. In 1736–1737 a team of scientists measured the length of a degree of meridian arc in Lapland, near the Arctic Circle. They found a length appreciably longer than the meridian degree measured by Picard near Paris. From 1735 to 1743 a second party of scientists measured the length of more than 3 degrees of meridian arc in Peru, near the equator. Their results showed that the equatorial degree of latitude was shorter than the meridian degree in Paris. Both parties confirmed convincingly the prediction of Newton that the Earth's shape is that of an oblate ellipsoid.

The ellipsoidal shape of the Earth resulting from its rotation has important consequences, not only for the variation with latitude of gravity on the Earth's surface, but also for the Earth's rate of rotation and the orientation of its rotational axis. These are modified by torques that arise from the gravitational attractions of the Sun, Moon and planets on the ellipsoidal shape.

2.2 GRAVITATION

2.2.1 The law of universal gravitation

Sir Isaac Newton (1642–1727) was born in the same year in which Galileo died. Unlike Galileo, who relished debate, Newton was a retiring person and avoided confrontation. His modesty is apparent in a letter written in 1675 to his colleague Robert Hooke, famous for his experiments on elasticity. In this letter Newton made the famous disclaimer “if I have seen further (than you and Descartes) it is by standing upon the shoulders of Giants.” In modern terms Newton would be regarded as a theoretical physicist. He had an outstanding ability to synthesize experimental results and incorporate them into his own theories. Faced with the need for a more powerful technique of mathematical analysis than existed at the time, he invented differential and integral calculus, for which he is credited equally with Gottfried Wilhelm von Leibnitz (1646–1716) who discovered the same method independently. Newton was able to resolve many issues by formulating logical thought experiments; an example is his prediction that the shape of the Earth is an oblate ellipsoid. He was one of the most outstanding synthesizers of observations in scientific history, which is implicit in his letter to Hooke. His three-volume book *Philosophiae Naturalis Principia Mathematica*, published in 1687, ranks as the greatest of all scientific texts. The first volume of the *Principia* contains Newton's famous *Laws of Motion*, the third volume handles the *Law of Universal Gravitation*.

The first two laws of motion are generalizations from Galileo's results. As a corollary Newton applied his laws of motion to demonstrate that forces must be added as vectors and showed how to do this geometrically with a parallelogram. The second law of motion states that the

rate of change of momentum of a mass is proportional to the force acting upon it and takes place in the direction of the force. For the case of constant mass, this law serves as the definition of force (**F**) in terms of the acceleration (**a**) given to a mass (*m*):

$$\mathbf{F} = m\mathbf{a} \quad (2.1)$$

The unit of force in the SI system of units is the newton (N). It is defined as the force that gives a mass of one kilogram (1 kg) an acceleration of 1 m s^{-2} .

His celebrated observation of a falling apple may be a legend, but Newton's genius lay in recognizing that the type of gravitational field that caused the apple to fall was the same type that served to hold the Moon in its orbit around the Earth, the planets in their orbits around the Sun, and that acted between minute particles characterized only by their masses. Newton used Kepler's empirical third law (see Section 1.1.2 and Eq. (1.2)) to deduce that the force of attraction between a planet and the Sun varied with the “quantities of solid matter that they contain” (i.e., their masses) and with the inverse square of the distance between them. Applying this law to two particles or point masses *m* and *M* separated by a distance *r* (Fig. 2.4a), we get for the gravitational attraction **F** exerted by *M* on *m*

$$\mathbf{F} = -G \frac{mM}{r^2} \hat{\mathbf{r}} \quad (2.2)$$

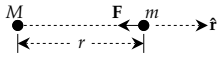
In this equation **f** is a unit vector in the direction of increase in coordinate *r*, which is directed away from the center of reference at the mass *M*. The negative sign in the equation indicates that the force **F** acts in the opposite direction, toward the attracting mass *M*. The constant *G*, which converts the physical law to an equation, is the constant of universal gravitation.

There was no way to determine the gravitational constant experimentally during Newton's lifetime. The method to be followed was evident, namely to determine the force between two masses in a laboratory experiment. However, seventeenth century technology was not yet up to this task. Experimental determination of *G* was extremely difficult, and was first achieved more than a century after the publication of *Principia* by Lord Charles Cavendish (1731–1810). From a set of painstaking measurements of the force of attraction between two spheres of lead, Cavendish in 1798 determined the value of *G* to be $6.754 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$. A modern value (Mohr and Taylor, 2005) is $6.674 210 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$. It has not yet been possible to determine *G* more precisely, due to experimental difficulty. Although other physical constants are now known with a relative standard uncertainty of much less than 1×10^{-6} , the gravitational constant is known to only 150×10^{-6} .

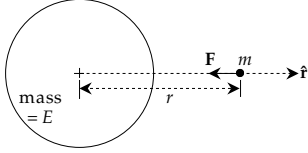
2.2.1.1 Potential energy and work

The law of conservation of energy means that the total energy of a closed system is constant. Two forms of

(a) point masses



(b) point mass and sphere



(c) point mass on Earth's surface

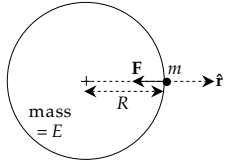


Fig. 2.4 Geometries for the gravitational attraction on (a) two point masses, (b) a point mass outside a sphere, and (c) a point mass on the surface of a sphere.

energy need be considered here. The first is the potential energy, which an object has by virtue of its position relative to the origin of a force. The second is the work done against the action of the force during a change in position.

For example, when Newton's apple is on the tree it has a higher potential energy than when it lies on the ground. It falls because of the downward force of gravity and loses potential energy in doing so. To compute the change in potential energy we need to raise the apple to its original position. This requires that we apply a force equal and opposite to the gravitational attraction on the apple and, because this force must be moved through the distance the apple fell, we have to expend energy in the form of work. If the original height of the apple above ground level was h and the value of the force exerted by gravity on the apple is F , the force we must apply to put it back is $(-F)$. Assuming that F is constant through the short distance of its fall, the work expended is $(-F)h$. This is the increase in potential energy of the apple, when it is on the tree.

More generally, if the constant force F moves through a small distance dr in the same direction as the force, the work done is $dW = Fdr$ and the change in potential energy dE_p is given by

$$dE_p = -dW = -Fdr \quad (2.3)$$

In the more general case we have to consider motions and forces that have components along three orthogonal axes. The displacement dr and the force F no longer need to be parallel to each other. We have to treat F and dr as vectors. In Cartesian coordinates the displacement vector dr has components (dx, dy, dz) and the force has components (F_x, F_y, F_z) along each of the respective axes. The

work done by the x -component of the force when it is displaced along the x -axis is $F_x dx$, and there are similar expressions for the displacements along the other axes. The change in potential energy dE_p is now given by

$$dE_p = -dW = -(F_x dx + F_y dy + F_z dz) \quad (2.4)$$

The expression in brackets is called the *scalar product* of the vectors F and dr . It is equal to $F dr \cos \theta$, where θ is the angle between the vectors.

2.2.2 Gravitational acceleration

In physics the *field* of a force is often more important than the absolute magnitude of the force. The field is defined as the force exerted on a material unit. For example, the electrical field of a charged body at a certain position is the force it exerts on a unit of electrical charge at that location. The *gravitational field* in the vicinity of an attracting mass is the force it exerts on a unit mass. Equation (2.1) shows that this is equivalent to the acceleration vector.

In geophysical applications we are concerned with accelerations rather than forces. By comparing Eq. (2.1) and Eq. (2.2) we get the gravitational acceleration $\mathbf{a}_G$ of the mass m due to the attraction of the mass M :

$$\mathbf{a}_G = -G \frac{M}{r^2} \hat{\mathbf{r}} \quad (2.5)$$

The SI unit of acceleration is the m s^{-2} ; this unit is unpractical for use in geophysics. In the now superseded c.g.s. system the unit of acceleration was the cm s^{-2} , which is called a *gal* in recognition of the contributions of Galileo. The small changes in the acceleration of gravity caused by geological structures are measured in thousandths of this unit, i.e., in milligal (*mgal*). Until recently, gravity anomalies due to geological structures were surveyed with field instruments accurate to about one-tenth of a milligal, which was called a *gravity unit*. Modern instruments are capable of measuring gravity differences to a millionth of a gal, or microgal (*μgal*), which is becoming the practical unit of gravity investigations. The value of gravity at the Earth's surface is about 9.8 m s^{-2} , and so the sensitivity of modern measurements of gravity is about 1 part in 10^9 .

2.2.2.1 Gravitational potential

The gravitational potential is the potential energy of a unit mass in a field of gravitational attraction. Let the potential be denoted by the symbol U_G . The potential energy E_p of a mass m in a gravitational field is thus equal to (mU_G) . Thus, a change in potential energy (dE_p) is equal to $(m dU_G)$. Equation (2.3) becomes, using Eq. (2.1),

$$m dU_G = Fdr = -ma_G dr \quad (2.6)$$

Rearranging this equation we get the gravitational acceleration

$$\mathbf{a}_G = -\frac{dU_G}{dr} \hat{\mathbf{r}} \quad (2.7)$$

In general, the acceleration is a three-dimensional vector. If we are using Cartesian coordinates (x, y, z) , the acceleration will have components (a_x, a_y, a_z) . These may be computed by calculating separately the derivatives of the potential with respect to x, y and z :

$$a_x = -\frac{\partial U_G}{\partial x} \quad a_y = -\frac{\partial U_G}{\partial y} \quad a_z = -\frac{\partial U_G}{\partial z} \quad (2.8)$$

Equating Eqs. (2.3) and (2.7) gives the gravitational potential of a point mass M :

$$\frac{dU_G}{dr} = G \frac{M}{r^2} \quad (2.9)$$

the solution of which is

$$U_G = -G \frac{M}{r} \quad (2.10)$$

2.2.2.2 Acceleration and potential of a distribution of mass

Until now, we have considered only the gravitational acceleration and potential of point masses. A solid body may be considered to be composed of numerous small particles, each of which exerts a gravitational attraction at an external point P (Fig. 2.5a). To calculate the gravitational acceleration of the object at the point P we must form a vector sum of the contributions of the individual discrete particles. Each contribution has a different direction. Assuming m_i to be the mass of the particle at distance r_i from P , this gives an expression like

$$\mathbf{a}_G = -G \frac{m_1}{r_1^2} \hat{\mathbf{r}}_1 - G \frac{m_2}{r_2^2} \hat{\mathbf{r}}_2 - G \frac{m_3}{r_3^2} \hat{\mathbf{r}}_3 - \dots \quad (2.11)$$

Depending on the shape of the solid, this vector sum can be quite complicated.

An alternative solution to the problem is found by first calculating the gravitational potential, and then differentiating it as in Eq. (2.5) to get the acceleration. The expression for the potential at P is

$$U_G = -G \frac{m_1}{r_1} - G \frac{m_2}{r_2} - G \frac{m_3}{r_3} - \dots \quad (2.12)$$

This is a scalar sum, which is usually more simple to calculate than a vector sum.

More commonly, the object is not represented as an assemblage of discrete particles but by a continuous mass distribution. However, we can subdivide the volume into discrete elements; if the density of the matter in each volume is known, the mass of the small element can be calculated and its contribution to the potential at the external point P can be determined. By integrating over the volume of the body its gravitational potential at P can be calculated. At a point in the body with coordinates (x, y, z) let the density be $\rho(x, y, z)$ and let its distance from P

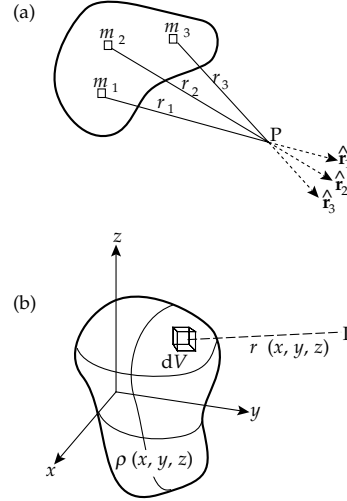


Fig. 2.5 (a) Each small particle of a solid body exerts a gravitational attraction in a different direction at an external point P . (b) Computation of the gravitational potential of a continuous mass distribution.

be $r(x, y, z)$ as in Fig. 2.5b. The gravitational potential of the body at P is

$$U_G = -G \iiint_V \frac{\rho(x, y, z)}{r(x, y, z)} dx dy dz \quad (2.13)$$

The integration readily gives the gravitational potential and acceleration at points inside and outside a hollow or homogeneous solid sphere. The values outside a sphere at distance r from its center are the same as if the entire mass E of the sphere were concentrated at its center (Fig. 2.4b):

$$U_G = -G \frac{E}{r} \quad (2.14)$$

$$\mathbf{a}_G = -G \frac{E}{r^2} \hat{\mathbf{r}} \quad (2.15)$$

2.2.2.3 Mass and mean density of the Earth

Equations (2.14) and (2.15) are valid everywhere outside a sphere, including on its surface where the distance from the center of mass is equal to the mean radius R (Fig. 2.4c). If we regard the Earth to a first approximation as a sphere with mass E and radius R , we can estimate the Earth's mass by rewriting Eq. (2.15) as a scalar equation in the form

$$E = \frac{R^2 a_G}{G} \quad (2.16)$$

The gravitational acceleration at the surface of the Earth is only slightly different from mean gravity, about 9.81 m s^{-2} , the Earth's radius is 6371 km, and the gravitational constant is $6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$. The mass of the Earth is found to be $5.974 \times 10^{24} \text{ kg}$. This large number is not so meaningful as the mean density of the Earth, which may be calculated by dividing the Earth's mass by its volume ($\frac{4}{3}\pi R^3$). A mean density of 5515 kg m^{-3} is obtained,

which is about double the density of crustal rocks. This indicates that the Earth's interior is not homogeneous, and implies that density must increase with depth in the Earth.

2.2.3 The equipotential surface

An equipotential surface is one on which the potential is constant. For a sphere of given mass the gravitational potential (Eq. (2.15)) varies only with the distance r from its center. A certain value of the potential, say U_1 , is realized at a constant radial distance r_1 . Thus, the equipotential surface on which the potential has the value U_1 is a sphere with radius r_1 ; a different equipotential surface U_2 is the sphere with radius r_2 . The equipotential surfaces of the original spherical mass form a set of concentric spheres (Fig. 2.6a), one of which (e.g., U_0) coincides with the surface of the spherical mass. This particular equipotential surface describes the *figure* of the spherical mass.

By definition, no change in potential takes place (and no work is done) in moving from one point to another on an equipotential surface. The work done by a force F in a displacement dr is $Fdr\cos\theta$ which is zero when $\cos\theta$ is zero, that is, when the angle θ between the displacement and the force is 90° . If no work is done in a motion along a gravitational equipotential surface, the force and acceleration of the gravitational field must act perpendicular to the surface. This normal to the equipotential surface defines the *vertical*, or plumb-line, direction (Fig. 2.6b). The plane tangential to the equipotential surface at a point defines the *horizontal* at that point.

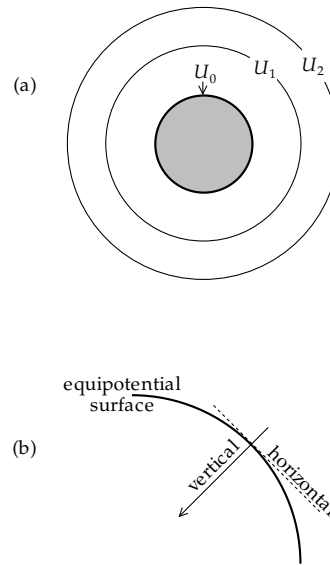


Fig. 2.6 (a) Equipotential surfaces of a spherical mass form a set of concentric spheres. (b) The normal to the equipotential surface defines the vertical direction; the tangential plane defines the horizontal.

straight line unless compelled to change that state by forces acting on it. The continuation of a state of motion is by virtue of the *inertia* of the body. A framework in which this law is valid is called an *inertial system*. For example, when we are travelling in a car at constant speed, we feel no disturbing forces; reference axes fixed to the moving vehicle form an inertial frame. If traffic conditions compel the driver to apply the brakes, we experience decelerating forces; if the car goes around a corner, even at constant speed, we sense sideways forces toward the outside of the corner. In these situations the moving car is being forced to change its state of uniform rectilinear motion and reference axes fixed to the car form a *non-inertial system*.

Motion in a circle implies that a force is active that continually changes the state of rectilinear motion. Newton recognized that the force was directed inwards, towards the center of the circle, and named it the *centripetal* (meaning “center-seeking”) force. He cited the example of a stone being whirled about in a sling. The inward centripetal force exerted on the stone by the sling holds it in a circular path. If the sling is released, the restraint of the centripetal force is removed and the inertia of the stone causes it to continue its motion at the point of release. No longer under the influence of the restraining force, the stone flies off in a straight line. Arguing that the curved path of a projectile near the surface of the Earth was due to the effect of gravity, which caused it constantly to fall toward the Earth, Newton postulated that, if the speed of the projectile were exactly right, it might never quite reach the Earth's surface. If the projectile fell toward the center of the Earth at the same rate as the curved surface of the Earth fell away from it, the projectile would go into orbit around the Earth. Newton suggested that the Moon was held in

2.3 THE EARTH'S ROTATION

2.3.1 Introduction

The rotation of the Earth is a vector, i.e., a quantity characterized by both magnitude and direction. The Earth behaves as an elastic body and deforms in response to the forces generated by its rotation, becoming slightly flattened at the poles with a compensating bulge at the equator. The gravitational attractions of the Sun, Moon and planets on the spinning, flattened Earth cause changes in its rate of rotation, in the orientation of the rotation axis, and in the shape of the Earth's orbit around the Sun. Even without extra-terrestrial influences the Earth reacts to tiny displacements of the rotation axis from its average position by acquiring a small, unsteady wobble. These perturbations reflect a balance between gravitation and the forces that originate in the Earth's rotational dynamics.

2.3.2 Centripetal and centrifugal acceleration

Newton's first law of motion states that every object continues in its state of rest or of uniform motion in a

orbit around the Earth by just such a centripetal force, which originated in the gravitational attraction of the Earth. Likewise, he visualized that a centripetal force due to gravitational attraction restrained the planets in their circular orbits about the Sun.

The passenger in a car going round a corner experiences a tendency to be flung outwards. He is restrained in position by the frame of the vehicle, which supplies the necessary centripetal acceleration to enable the passenger to go round the curve in the car. The inertia of the passenger's body causes it to continue in a straight line and pushes him outwards against the side of the vehicle. This outward force is called the *centrifugal* force. It arises because the car does not represent an inertial reference frame. An observer outside the car in a fixed (inertial) coordinate system would note that the car and passenger are constantly changing direction as they round the corner. The centrifugal force feels real enough to the passenger in the car, but it is called a pseudo-force, or *inertial force*. In contrast to the centripetal force, which arises from the gravitational attraction, the centrifugal force does not have a physical origin, but exists only because it is being observed in a non-inertial reference frame.

2.3.2.1 Centripetal acceleration

The mathematical form of the centripetal acceleration for circular motion with constant angular velocity ω about a point can be derived as follows. Define orthogonal Cartesian axes x and y relative to the center of the circle as in Fig. 2.7a. The linear velocity v at any point where the radius vector makes an angle $\theta = (\omega t)$ with the x -axis has components

$$\begin{aligned} v_x &= -v \sin(\omega t) = -r\omega \sin(\omega t) \\ v_y &= v \cos(\omega t) = r\omega \cos(\omega t) \end{aligned} \quad (2.17)$$

The x - and y -components of the acceleration are obtained by differentiating the velocity components with respect to time. This gives

$$\begin{aligned} a_x &= -\omega v \cos(\omega t) = -r\omega^2 \cos(\omega t) \\ a_y &= -\omega v \sin(\omega t) = -r\omega^2 \sin(\omega t) \end{aligned} \quad (2.18)$$

These are the components of the centripetal acceleration, which is directed radially inwards and has the magnitude $\omega^2 r$ (Fig. 2.7b).

2.3.2.2 Centrifugal acceleration and potential

In handling the variation of gravity on the Earth's surface we must operate in a non-inertial reference frame attached to the rotating Earth. Viewed from a fixed, external inertial frame, a stationary mass moves in a circle about the Earth's rotation axis with the same rotational speed as the Earth.

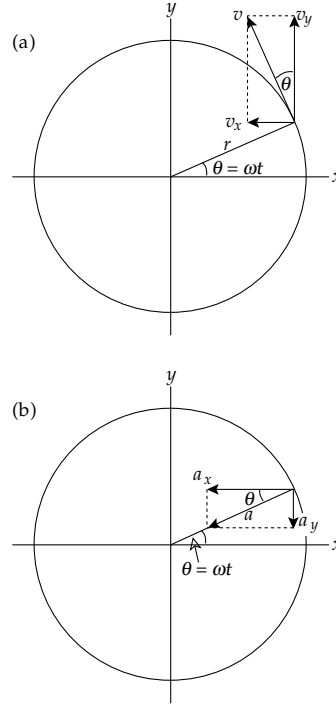


Fig. 2.7 (a) Components v_x and v_y of the linear velocity v where the radius makes an angle $\theta = (\omega t)$ with the x -axis, and (b) the components a_x and a_y of the centripetal acceleration, which is directed radially inward.

However, within a rotating reference frame attached to the Earth, the mass is stationary. It experiences a centrifugal acceleration (a_c) that is exactly equal and opposite to the centripetal acceleration, and which can be written in the alternative forms

$$\begin{aligned} a_c &= \omega^2 r \\ a_c &= \frac{v^2}{r} \end{aligned} \quad (2.19)$$

The centrifugal acceleration is not a centrally oriented acceleration like gravitation, but instead is defined relative to an axis of rotation. Nevertheless, potential energy is associated with the rotation and it is possible to define a *centrifugal potential*. Consider a point rotating with the Earth at a distance r from its center (Fig. 2.8). The angle θ between the radius to the point and the axis of rotation is called the colatitude; it is the angular complement of the latitude λ . The distance of the point from the rotational axis is $x (= r \sin \theta)$, and the centrifugal acceleration is $\omega^2 x$ outwards in the direction of increasing x . The centrifugal potential U_c is defined such that

$$\mathbf{a}_c = -\frac{\partial U_c}{\partial \mathbf{x}} \hat{\mathbf{x}} = (\omega^2 x) \hat{\mathbf{x}} \quad (2.20)$$

where $\hat{\mathbf{x}}$ is the outward unit vector. On integrating, we obtain

$$U_c = -\frac{1}{2} \omega^2 x^2 = -\frac{1}{2} \omega^2 r^2 \cos^2 \lambda = -\frac{1}{2} \omega^2 r^2 \sin^2 \theta \quad (2.21)$$

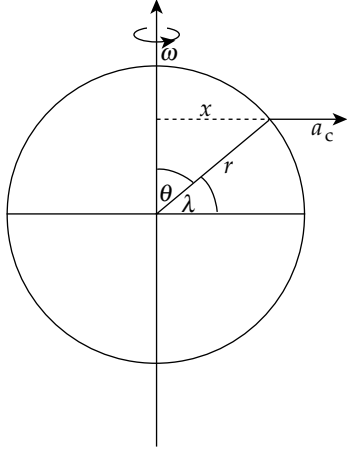


Fig. 2.8 The outwardly directed centrifugal acceleration a_c at latitude λ on a sphere rotating at angular velocity ω .

2.3.2.3 Kepler's third law of planetary motion

By comparing the centripetal acceleration of a planet about the Sun with the gravitational acceleration of the Sun, the third of Kepler's laws of planetary motion can be explained. Let S be the mass of the Sun, r_p the distance of a planet from the Sun, and T_p the period of orbital rotation of the planet around the Sun. Equating the gravitational and centripetal accelerations gives

$$G \frac{S}{r_p^2} = \omega_p^2 r_p = \left(\frac{2\pi}{T_p} \right)^2 r_p \quad (2.22)$$

Rearranging this equation we get Kepler's third law of planetary motion, which states that the square of the period of the planet is proportional to the cube of the radius of its orbit, or:

$$\frac{r_p^3}{T_p^2} = \frac{GS}{4\pi^2} = \text{constant} \quad (2.23)$$

2.3.2.4 Verification of the inverse square law of gravitation

Newton realized that the centripetal acceleration of the Moon in its orbit was supplied by the gravitational attraction of the Earth, and tried to use this knowledge to confirm the inverse square dependence on distance in his law of gravitation. The sidereal period (T_L) of the Moon about the Earth, a sidereal month, is equal to 27.3 days. Let the corresponding angular rate of rotation be ω_L . We can equate the gravitational acceleration of the Earth at the Moon with the centripetal acceleration due to ω_L :

$$G \frac{E}{r_L^2} = \omega_L^2 r_L \quad (2.24)$$

This equation can be rearranged as follows

$$\left(G \frac{E}{R^2} \right) \left(\frac{R}{r_L} \right)^2 = \omega_L^2 R \left(\frac{r_L}{R} \right) \quad (2.25)$$

Comparison with Eq. (2.15) shows that the first quantity in parentheses is the mean gravitational acceleration on the Earth's surface, a_G . Therefore, we can write

$$a_G = G \frac{E}{R^2} = \omega_L^2 R \left(\frac{r_L}{R} \right)^3 \quad (2.26)$$

In Newton's time little was known about the physical dimensions of our planet. The distance of the Moon was known to be approximately 60 times the radius of the Earth (see Section 1.1.3.2) and its sidereal period was known to be 27.3 days. At first Newton used the accepted value 5500 km for the Earth's radius. This gave a value of only 8.4 m s^{-2} for gravity, well below the known value of 9.8 m s^{-2} . However, in 1671 Picard determined the Earth's radius to be 6372 km. With this value, the inverse square character of Newton's law of gravitation was confirmed.

2.3.3 The tides

The gravitational forces of Sun and Moon deform the Earth's shape, causing tides in the oceans, atmosphere and solid body of the Earth. The most visible tidal effects are the displacements of the ocean surface, which is a hydrostatic equipotential surface. The Earth does not react rigidly to the tidal forces. The solid body of the Earth deforms in a like manner to the free surface, giving rise to so-called *bodily Earth-tides*. These can be observed with specially designed instruments, which operate on a similar principle to the long-period seismometer.

The height of the marine equilibrium tide amounts to only half a meter or so over the free ocean. In coastal areas the tidal height is significantly increased by the shallowing of the continental shelf and the confining shapes of bays and harbors. Accordingly, the height and variation of the tide at any place is influenced strongly by complex local factors. Subsequent subsections deal with the tidal deformations of the Earth's hydrostatic figure.

2.3.3.1 Lunar tidal periodicity

The Earth and Moon are coupled together by gravitational attraction. Their common motion is like that of a pair of ballroom dancers. Each partner moves around the center of mass (or *barycenter*) of the pair. For the Earth–Moon pair the location of the center of mass is easily found. Let E be the mass of the Earth, and m that of the Moon; let the separation of the centers of the Earth and Moon be r_L and let the distance of their common center of mass be d from the center of the Earth. The moment of the Earth about the center of mass is Ed and the moment of the Moon is $m(r_L - d)$. Setting these moments equal we get

$$d = \frac{m}{E + m} r_L \quad (2.27)$$

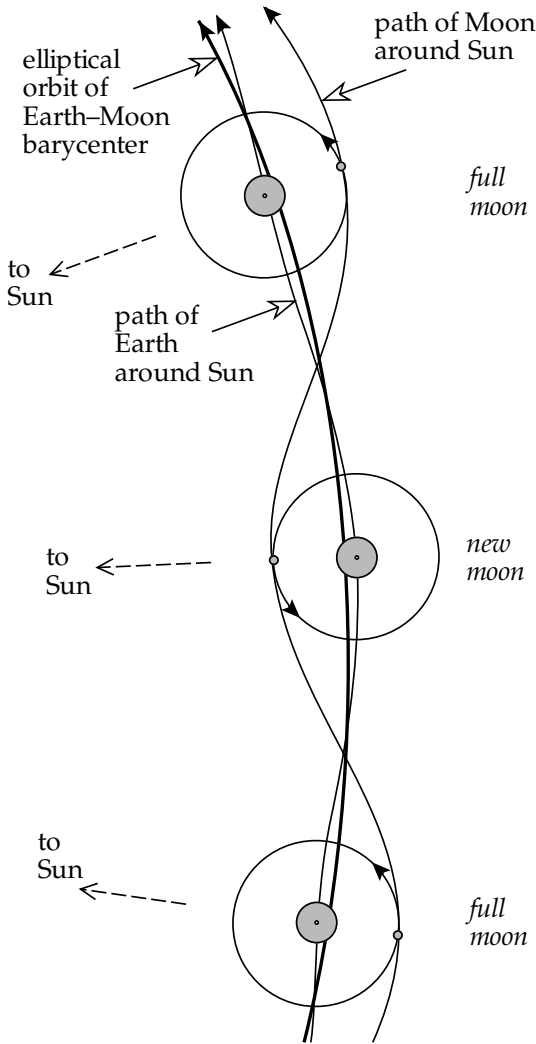


Fig. 2.9 Paths of the Earth and Moon, and their barycenter, around the Sun.

The mass of the Moon is 0.0123 that of the Earth and the distance between the centers is 384,100 km. These figures give $d=4600$ km, i.e., the center of revolution of the Earth–Moon pair lies within the Earth.

It follows that the paths of the Earth and the Moon around the Sun are more complicated than at first appears. The elliptical orbit is traced out by the *barycenter* of the pair (Fig. 2.9). The Earth and Moon follow wobbly paths, which, while always concave towards the Sun, bring each body at different times of the month alternately inside and outside the elliptical orbit.

To understand the common revolution of the Earth–Moon pair we have to exclude the rotation of the Earth about its axis. The “revolution without rotation” is illustrated in Fig. 2.10. The Earth–Moon pair revolves about S, the center of mass. Let the starting positions be as shown in Fig. 2.10a. Approximately one week later the Moon has advanced in its path by one-quarter of a revolution and the center of the Earth has moved so as to keep the center of mass fixed (Fig. 2.10b). The relationship is

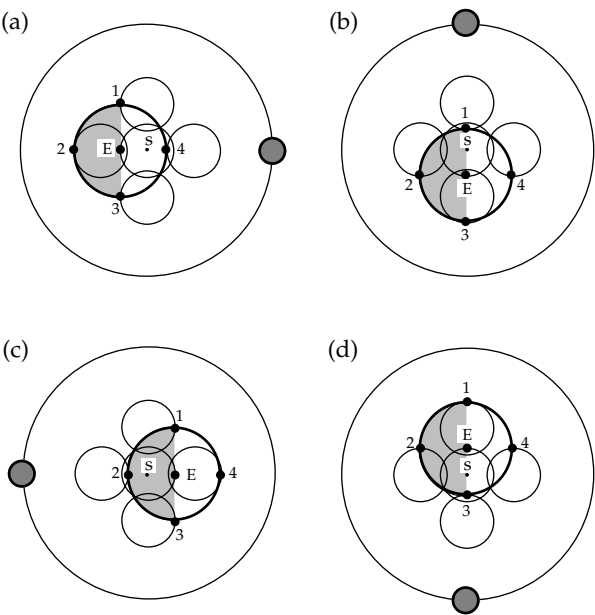


Fig. 2.10 Illustration of the “revolution without rotation” of the Earth–Moon pair about their common center of mass at S.

maintained in the following weeks (Fig. 2.10c, d) so that during one month the center of the Earth describes a circle about S. Now consider the motion of point number 2 on the left-hand side of the Earth in Fig. 2.10. If the Earth revolves as a rigid body and the rotation about its own axis is omitted, after one week point 2 will have moved to a new position but will still be the furthest point on the left. Subsequently, during one month point 2 will describe a small circle with the same radius as the circle described by the Earth’s center. Similarly points 1, 3 and 4 will also describe circles of exactly the same size. A simple illustration of this point can be made by chalking the tip of each finger on one hand with a different color, then moving your hand in a circular motion while touching a blackboard; your fingers will draw a set of identical circles.

The “revolution without rotation” causes each point in the body of the Earth to describe a circular path with identical radius. The centrifugal acceleration of this motion has therefore the same magnitude at all points in the Earth and, as can be seen by inspection of Fig. 2.10(a–d), it is directed away from the Moon parallel to the Earth–Moon line of centers. At C, the center of the Earth (Fig. 2.11a), this centrifugal acceleration exactly balances the gravitational attraction of the Moon. Its magnitude is given by

$$a_L = G \frac{m}{r_L^2} \tag{2.28}$$

At B, on the side of the Earth nearest to the Moon, the gravitational acceleration of the Moon is larger than at the center of the Earth and exceeds the centrifugal acceleration a_L . There is a residual acceleration *toward* the Moon, which raises a tide on this side of the Earth. The magnitude of the tidal acceleration at B is

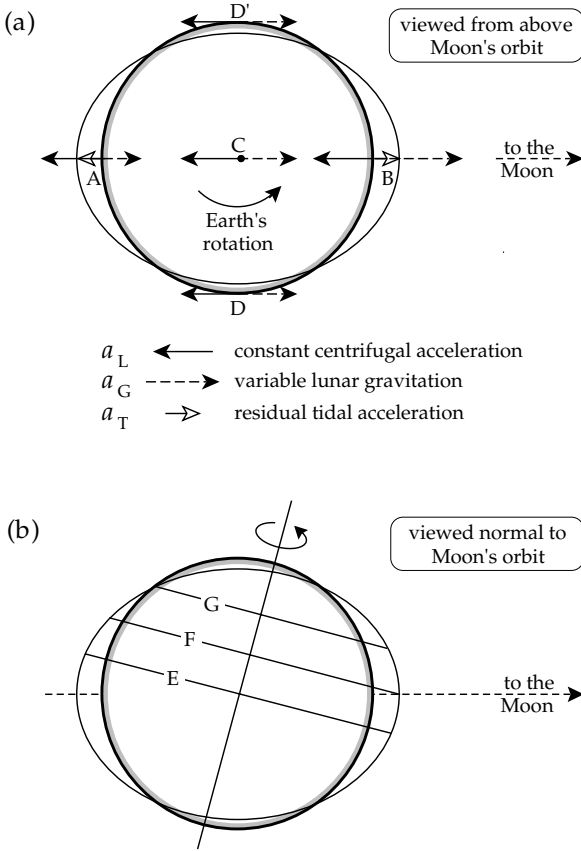


Fig. 2.11 (a) The relationships of the centrifugal, gravitational and residual tidal accelerations at selected points in the Earth. (b) Latitude effect that causes diurnal inequality of the tidal height.

$$a_T = Gm \left(\frac{1}{(r_L - R)^2} - \frac{1}{r_L^2} \right) \quad (2.29)$$

$$a_T = G \frac{m}{r_L^2} \left(\left(1 - \frac{R}{r_L} \right)^{-2} - 1 \right) \quad (2.30)$$

Expanding this equation with the binomial theorem and simplifying gives

$$a_T = G \frac{m}{r_L^2} \left(2 \frac{R}{r_L} + 3 \left(\frac{R}{r_L} \right)^2 + \dots \right) \quad (2.31)$$

At A, on the far side of the Earth, the gravitational acceleration of the Moon is less than the centrifugal acceleration a_L . The residual acceleration (Fig. 2.11a) is away from the Moon, and raises a tide on the far side of the Earth. The magnitude of the tidal acceleration at A is

$$a_T = Gm \left(\frac{1}{r_L^2} - \frac{1}{(r_L + R)^2} \right) \quad (2.32)$$

which reduces to

$$a_T = G \frac{m}{r_L^2} \left(2 \frac{R}{r_L} - 3 \left(\frac{R}{r_L} \right)^2 + \dots \right) \quad (2.33)$$

At points D and D' the direction of the gravitational acceleration due to the Moon is not exactly parallel to the line of centers of the Earth–Moon pair. The residual tidal acceleration is almost along the direction toward the center of the Earth. Its effect is to lower the free surface in this direction.

The free hydrostatic surface of the Earth is an equipotential surface (Section 2.2.3), which in the absence of the Earth's rotation and tidal effects would be a sphere. The lunar tidal accelerations perturb the equipotential surface, raising it at A and B while lowering it at D and D', as in Fig. 2.11a. The tidal deformation of the Earth produced by the Moon thus has an almost prolate ellipsoidal shape, like a rugby football, along the Earth–Moon line of centers. The daily tides are caused by superposing the Earth's rotation on this deformation. In the course of one day a point rotates past the points A, D, B and D' and an observer experiences two full tidal cycles, called the *semi-diurnal* tides. The extreme tides are not equal at every latitude, because of the varying angle between the Earth's rotational axis and the Moon's orbit (Fig. 2.11b). At the equator E the semi-diurnal tides are equal; at an intermediate latitude F one tide is higher than the other; and at latitude G and higher there is only one (diurnal) tide per day. The difference in height between two successive high or low tides is called the *diurnal inequality*.

In the same way that the Moon deforms the Earth, so the Earth causes a tidal deformation of the Moon. In fact, the tidal relationship between any planet and one of its moons, or between the Sun and a planet or comet, can be treated analogously to the Earth–Moon pair. A tidal acceleration similar to Eq. (2.31) deforms the smaller body; its self-gravitation acts to counteract the deformation. However, if a moon or comet comes too close to the planet, the tidal forces deforming it may overwhelm the gravitational forces holding it together, so that the moon or comet is torn apart. The separation at which this occurs is called the *Roche limit* (Box 2.1). The material of a disintegrated moon or comet enters orbit around the planet, forming a system of concentric rings, as around the great planets (Section 1.1.3.3).

2.3.3.2 Tidal effect of the Sun

The Sun also has an influence on the tides. The theory of the solar tides can be followed in identical manner to the lunar tides by again applying the principle of “revolution without rotation.” The Sun's mass is 333,000 times greater than that of the Earth, so the common center of mass is close to the center of the Sun at a radial distance of about 450 km from its center. The period of the revolution is one year. As for the lunar tide, the imbalance between gravitational acceleration of the Sun and centrifugal acceleration due to the common revolution leads to a prolate ellipsoidal tidal deformation. The solar effect is smaller than that of the Moon. Although the mass of the Sun is vastly

Box 2.1: The Roche limit

Suppose that a moon with mass M and radius R_M is in orbit at a distance d from a planet with mass P and radius R_P . The Roche limit is the distance at which the tidal attraction exerted by the planet on the moon overcomes the moon's self-gravitation (Fig. B2.1.1). If the moon is treated as an elastic body, its deformation to an elongate form complicates the calculation of the Roche limit. However, for a rigid body, the computation is simple because the moon maintains its shape as it approaches the planet.

Consider the forces acting on a small mass m forming part of the rigid moon's surface closest to the planet (Fig. B2.1.2). The tidal acceleration a_T caused by the planet can be written by adapting the first term of Eq. (2.31), and so the deforming force F_T on the small mass is

$$F_T = ma_T = G \frac{mP}{d^2} \left(2 \frac{R_M}{d} \right) = 2G \frac{mPR_M}{d^3} \quad (1)$$

This disrupting force is counteracted by the gravitational force F_G of the moon, which is

$$F_G = ma_G = G \frac{mM}{(R_M)^2} \quad (2)$$

The Roche limit d_R for a rigid solid body is determined by equating these forces:

$$2G \frac{mPR_M}{(d_R)^3} = G \frac{mM}{(R_M)^2} \quad (3)$$

$$(d_R)^3 = 2 \frac{P}{M} (R_M)^3 \quad (4)$$

If the densities of the planet, ρ_P , and moon, ρ_M , are known, Eq. (4) can be rewritten

$$(d_R)^3 = 2 \frac{\left(\frac{4}{3} \pi \rho_P (R_P)^3 \right)}{\left(\frac{4}{3} \pi \rho_M (R_M)^3 \right)} (R_M)^3 = 2 \left(\frac{\rho_P}{\rho_M} \right) (R_P)^3 \quad (5)$$

$$d_R = R_P \left(2 \frac{\rho_P}{\rho_M} \right)^{1/3} = 1.26 R_P \left(\frac{\rho_P}{\rho_M} \right)^{1/3} \quad (6)$$

If the moon is fluid, tidal attraction causes it to elongate progressively as it approaches the planet. This complicates the exact calculation of the Roche limit, but it is given approximately by

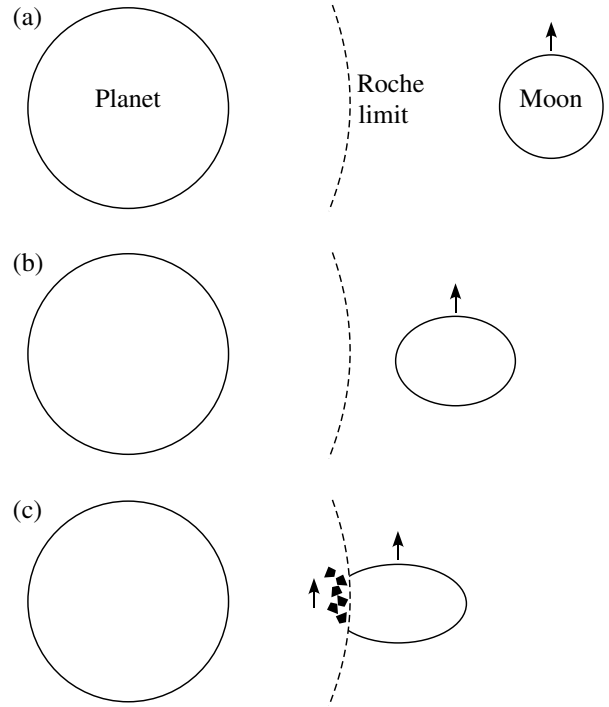


Fig. B2.1.1 (a) Far from its parent planet, a moon is spherical in shape, but (b) as it comes closer, tidal forces deform it into an ellipsoidal shape, until (c) within the Roche limit the moon breaks up. The disrupted material forms a ring of small objects orbiting the planet in the same sense as the moon's orbital motion.

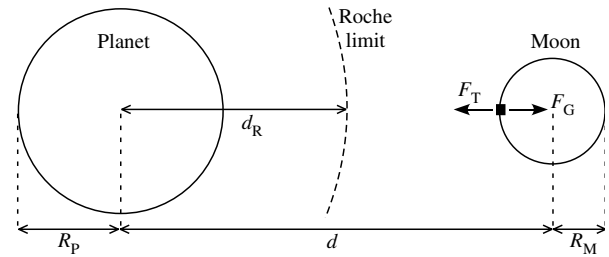


Fig. B2.1.2 Parameters for computation of the Roche limit.

$$d_R = 2.42 R_P \left(\frac{\rho_P}{\rho_M} \right)^{1/3} \quad (7)$$

Comparison of Eq. (6) and Eq. (7) shows that a fluid or gaseous moon disintegrates about twice as far from the planet as a rigid moon. In practice, the Roche limit for a moon about its parent planet (and the planet about the Sun) depends on the rigidity of the satellite and lies between the two extremes.

greater than that of the Moon, its distance from the Earth is also much greater and, because gravitational acceleration varies inversely with the square of distance, the maximum tidal effect of the Sun is only about 45% that of the Moon.

2.3.3.3 Spring and neap tides

The superposition of the lunar and solar tides causes a modulation of the tidal amplitude. The ecliptic plane is defined by the Earth's orbit around the Sun. The Moon's orbit around the Earth is not exactly in the ecliptic but is

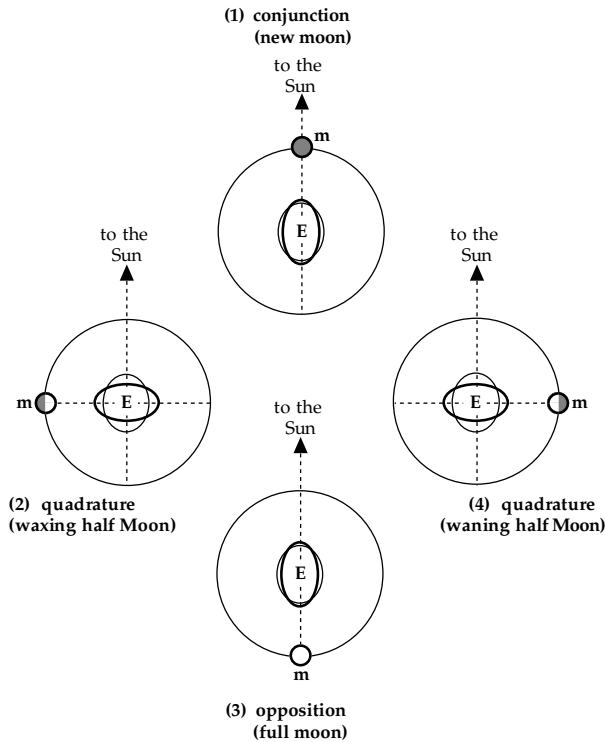


Fig. 2.12 The orientations of the solar and lunar tidal deformations of the Earth at different lunar phases.

inclined at a very small angle of about 5° to it. For discussion of the combination of lunar and solar tides we can assume the orbits to be coplanar. The Moon and Sun each produce a prolate tidal deformation of the Earth, but the relative orientations of these ellipsoids vary during one month (Fig. 2.12). At conjunction the (new) Moon is on the same side of the Earth as the Sun, and the ellipsoidal deformations augment each other. The same is the case half a month later at opposition, when the (full) Moon is on the opposite side of the Earth from the Sun. The unusually high tides at opposition and conjunction are called *spring tides*. In contrast, at the times of quadrature the waxing or waning half Moon causes a prolate ellipsoidal deformation out of phase with the solar deformation. The maximum lunar tide coincides with the minimum solar tide, and the effects partially cancel each other. The unusually low tides at quadrature are called *neap tides*. The superposition of the lunar and solar tides causes modulation of the tidal amplitude during a month (Fig. 2.13).

2.3.3.4 Effect of the tides on gravity measurements

The tides have an effect on gravity measurements made on the Earth. The combined effects of Sun and Moon cause an acceleration at the Earth's surface of approximately 0.3 mgal, of which about two-thirds are due to the Moon and one-third to the Sun. The sensitive modern instruments used for gravity exploration can readily detect gravity differences of 0.01 mgal. It is necessary to compen-

sate gravity measurements for the tidal effects, which vary with location, date and time of day. Fortunately, tidal theory is so well established that the gravity effect can be calculated and tabulated for any place and time before beginning a survey.

2.3.3.5 Bodily Earth-tides

A simple way to measure the height of the marine tide might be to fix a stake to the sea-bottom at a suitably sheltered location and to record continuously the measured water level (assuming that confusion introduced by wave motion can be eliminated or taken into account). The observed amplitude of the marine tide, defined by the displacement of the free water surface, is found to be about 70% of the theoretical value. The difference is explained by the elasticity of the Earth. The tidal deformation corresponds to a redistribution of mass, which modifies the gravitational potential of the Earth and augments the elevation of the free surface. This is partially counteracted by a bodily tide in the solid Earth, which deforms elastically in response to the attraction of the Sun and Moon. The free water surface is raised by the tidal attraction, but the sea-bottom in which the measuring rod is implanted is also raised. The measured tide is the difference between the marine tide and the bodily Earth-tide.

In practice, the displacement of the equipotential surface is measured with a horizontal pendulum, which reacts to the tilt of the surface. The bodily Earth-tides also affect gravity measurements and can be observed with sensitive gravimeters. The effects of the bodily Earth-tides are incorporated into the predicted tidal corrections to gravity measurements.

2.3.4 Changes in Earth's rotation

The Earth's rotational vector is affected by the gravitational attractions of the Sun, Moon and the planets. The rate of rotation and the orientation of the rotational axis change with time. The orbital motion around the Sun is also affected. The orbit rotates about the pole to the plane of the ecliptic and its ellipticity changes over long periods of time.

2.3.4.1 Effect of lunar tidal friction on the length of the day

If the Earth reacted perfectly elastically to the lunar tidal forces, the prolate tidal bulge would be aligned along the line of centers of the Earth–Moon pair (Fig. 2.14a). However, the motion of the seas is not instantaneous and the tidal response of the solid part of the Earth is partly anelastic. These features cause a slight delay in the time when high tide is reached, amounting to about 12 minutes. In this short interval the Earth's rotation carries the line of the maximum tides past the line of centers by a small angle of approximately 2.9° (Fig. 2.14b). A point on the rotating Earth passes under the line of maximum

Fig. 2.13 Schematic representation of the modulation of the tidal amplitude as a result of superposition of the lunar and solar tides.

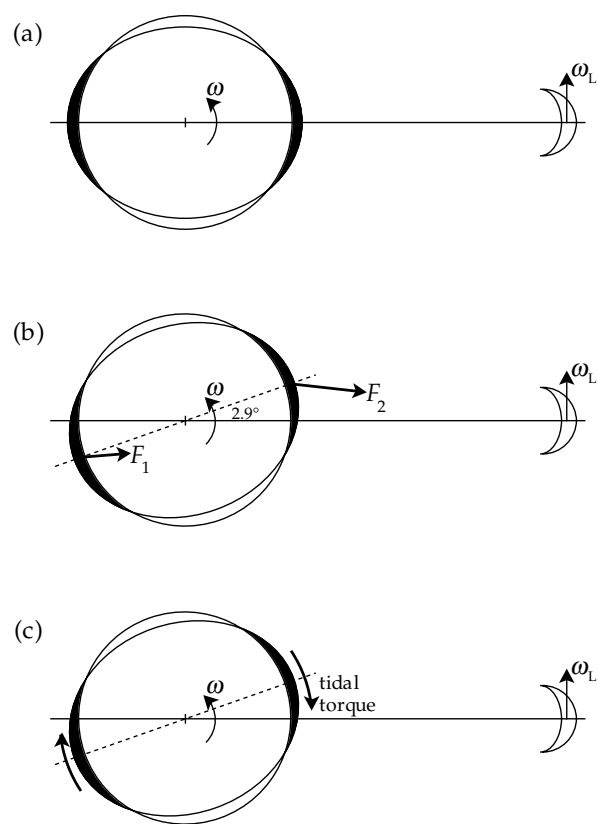
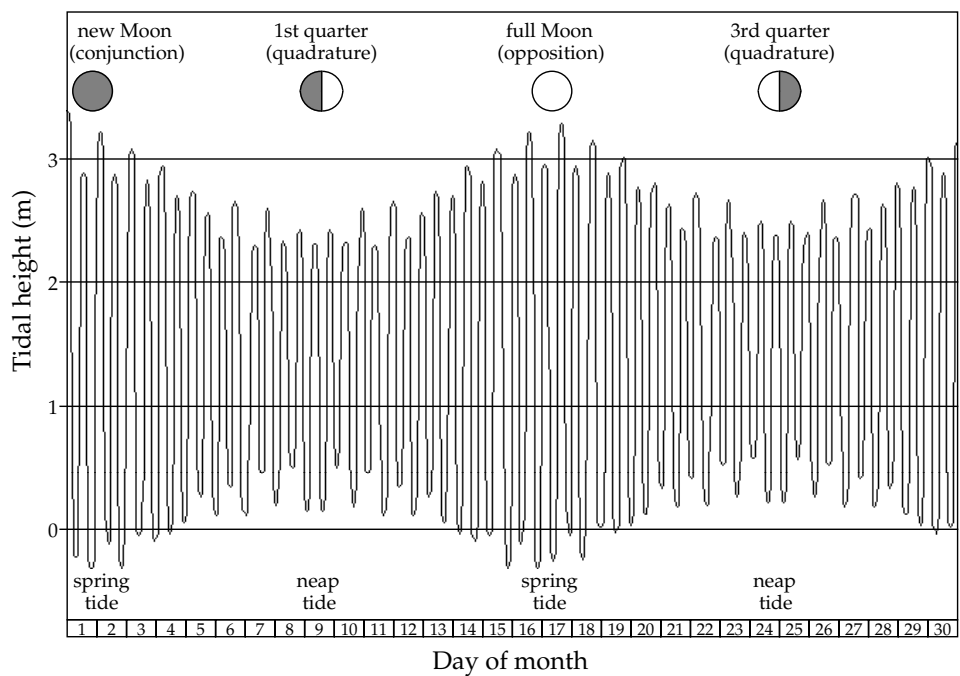


Fig. 2.14 (a) Alignment of the prolate tidal bulge of a perfectly elastic Earth along the line of centers of the Earth–Moon pair. (b) Tidal phase lag of 2.9° relative to the line of centers due to the Earth’s partially anelastic response. (c) Tidal decelerating torque due to unequal gravitational attractions of the Moon on the far- and near-sided tidal bulges.

tides 12 minutes after it passes under the Moon. The small phase difference is called the tidal lag.

Because of the tidal lag the gravitational attraction of the Moon on the tidal bulges on the far side and near side of the Earth (F_1 and F_2 , respectively) are not collinear (Fig. 2.14b). F_2 is stronger than F_1 so a torque is produced in the opposite sense to the Earth’s rotation (Fig. 2.14c). The tidal torque acts as a brake on the Earth’s rate of rotation, which is gradually slowing down.

The tidal deceleration of the Earth is manifested in a gradual increase in the length of the day. The effect is very small. Tidal theory predicts an increase in the length of the day of only 2.4 milliseconds per century. Observations of the phenomenon are based on ancient historical records of lunar and solar eclipses and on telescopically observed occultations of stars by the Moon. The current rate of rotation of the Earth can be measured with very accurate atomic clocks. Telescopic observations of the daily times of passage of stars past the local zenith are recorded with a camera controlled by an atomic clock. These observations give precise measures of the mean value and fluctuations of the length of the day.

The occurrence of a lunar or solar eclipse was a momentous event for ancient peoples, and was duly recorded in scientific and non-scientific chronicles. Untimed observations are found in non-astronomical works. They record, with variable reliability, the degree of totality and the time and place of observation. The unaided human eye is able to decide quite precisely just when an eclipse becomes total. Timed observations of both lunar and solar eclipses made by Arab astronomers around 800–1000 AD and Babylonian astronomers a thousand years earlier give two important groups of data (Fig. 2.15). By comparing the observed times of alignment

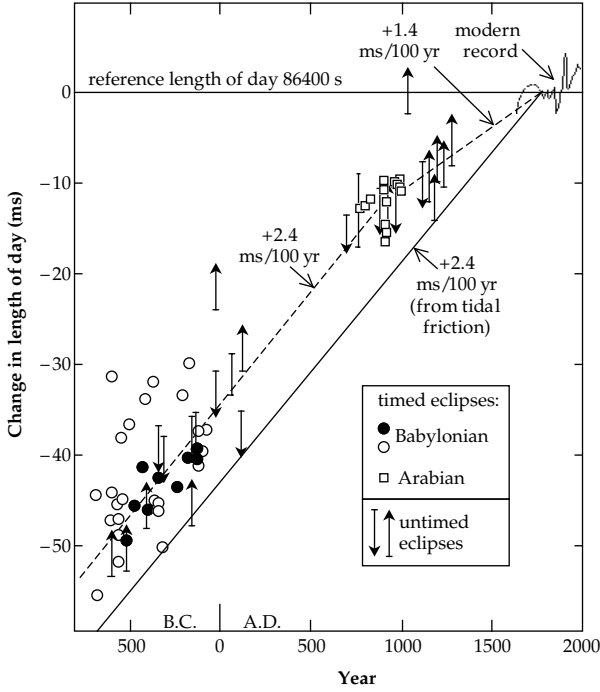


Fig. 2.15 Long-term changes in the length of the day deduced from observations of solar and lunar eclipses between 700 BC and 1980 AD (after Stephenson and Morrison, 1984).

of Sun, Moon and Earth with times predicted from the theory of celestial mechanics, the differences due to change in length of the day may be computed. A straight line with slope equal to the rate of increase of the length of the day inferred from tidal theory, 2.4 ms per century, connects the Babylonian and Arab data sets. Since the medieval observations of Arab astronomers the length of the day has increased on average by about 1.4 ms per century. The data set based on telescopic observations covers the time from 1620 to 1980 AD. It gives a more detailed picture and shows that the length of the day fluctuates about the long-term trend of 1.4 ms per century. A possible interpretation of the difference between the two slopes is that non-tidal causes have opposed the deceleration of the Earth's rotation since about 950 AD. It would be wrong to infer that some sudden event at that epoch caused an abrupt change, because the data are equally compatible with a smoothly changing polynomial. The observations confirm the importance of tidal braking, but they also indicate that tidal friction is not the only mechanism affecting the Earth's rotation.

The short-term fluctuations in rotation rate are due to exchanges of angular momentum with the Earth's atmosphere and core. The atmosphere is tightly coupled to the solid Earth. An increase in average global wind speed corresponds to an increase in the angular momentum of the atmosphere and corresponding decrease in angular momentum of the solid Earth. Accurate observations by very long baseline interferometry (see Section 2.4.6.6) confirm that rapid fluctuations in the length of the day are directly related to changes in the angular momentum

of the atmosphere. On a longer timescale of decades, the changes in length of the day may be related to changes in the angular momentum of the core. The fluid in the outer core has a speed of the order of 0.1 mm s^{-1} relative to the overlying mantle. The mechanism for exchange of angular momentum between the fluid core and the rest of the Earth depends on the way the core and mantle are coupled. The coupling may be mechanical if topographic irregularities obstruct the flow of the core fluid along the core-mantle interface. The core fluid is a good electrical conductor so, if the lower mantle also has an appreciable electrical conductivity, it is possible that the core and mantle are coupled electromagnetically.

2.3.4.2 Increase of the Earth-Moon distance

Further consequences of lunar tidal friction can be seen by applying the law of conservation of momentum to the Earth-Moon pair. Let the Earth's mass be E , its rate of rotation be ω and its moment of inertia about the rotation axis be C ; let the corresponding parameters for the Moon be m , Ω_L , and C_L , and let the Earth-Moon distance be r_L . Further, let the distance of the common center of revolution be d from the center of the Earth, as given by Eq. (2.27). The angular momentum of the system is given by

$$C\omega + E\Omega_L d^2 + m\Omega_L (r_L - d)^2 + C_L\Omega_L = \text{constant} \quad (2.34)$$

The fourth term is the angular momentum of the Moon about its own axis. Tidal deceleration due to the Earth's attraction has slowed down the Moon's rotation until it equals its rate of revolution about the Earth. Both Ω_L and C_L are very small and the fourth term can be neglected. The second and third terms can be combined so that we get

$$C\omega + \left(\frac{E}{E+M} \right) m\Omega_L r_L^2 = \text{constant} \quad (2.35)$$

The gravitational attraction of the Earth on the Moon is equal to the centripetal acceleration of the Moon about the common center of revolution, thus

$$G \frac{E}{r_L^2} = \Omega_L^2 (r_L - d) = \Omega_L^2 r_L \left(\frac{E}{E+M} \right) \quad (2.36)$$

from which

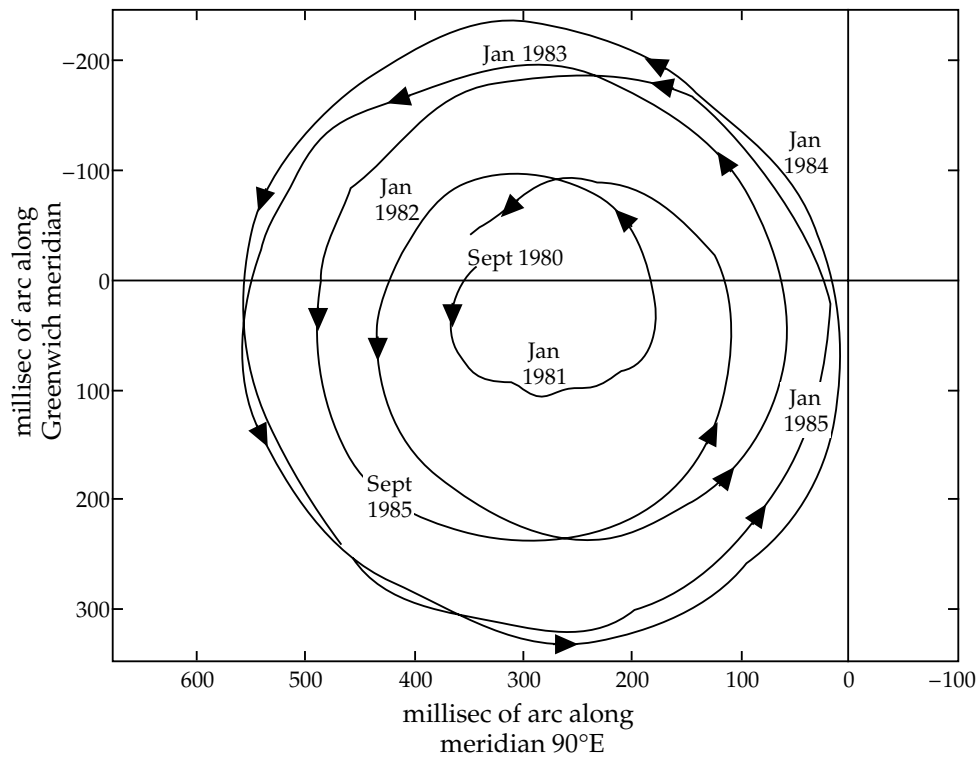
$$\Omega_L r_L^2 = \sqrt{G(E+m)r_L} \quad (2.37)$$

Inserting this in Eq. (2.35) gives

$$C\omega + \frac{Em}{\sqrt{(E+m)}} \sqrt{Gr_L} = \text{constant} \quad (2.38)$$

The first term in this equation decreases, because tidal friction reduces ω . To conserve angular momentum the second term must increase. Thus, lunar tidal braking of the

Fig. 2.16 Variation of latitude due to superposition of the 435 day Chandler wobble period and an annual seasonal component (after Carter, 1989).



Earth’s rotation causes an increase in the Earth–Moon distance, r_L . At present this distance is increasing at about 3.7 cm yr^{-1} . As a further consequence Eq. (2.37) shows that the Moon’s rate of revolution about the Earth (Ω_L) – and consequently also its synchronous rotation about its own axis – must decrease when r_L increases. Thus, tidal friction slows down the rates of Earth rotation, lunar rotation, and lunar orbital revolution and increases the Earth–Moon distance.

Eventually a situation will evolve in which the Earth’s rotation has slowed until it is synchronous with the Moon’s own rotation and its orbital revolution about the Earth. All three rotations will then be synchronous and equivalent to about 48 present Earth days. This will happen when the Moon’s distance from Earth is about 88 times the Earth’s radius ($r_L = 88R$; it is presently equal to about $60R$). The Moon will then be stationary over the Earth, and Earth and Moon will constantly present the same face to each other. This configuration already exists between the planet Pluto and its satellite Charon.

2.3.4.3 The Chandler wobble

The Earth’s rotation gives it the shape of a spheroid, or ellipsoid of revolution. This figure is symmetric with respect to the mean axis of rotation, about which the moment of inertia is greatest; this is also called the axis of figure (see Section 2.4). However, at any moment the instantaneous rotational axis is displaced by a few meters from the axis of figure. The orientation of the total angular momentum vector remains nearly constant but

the axis of figure changes location with time and appears to meander around the rotation axis (Fig. 2.16).

The theory of this motion was described by Leonhard Euler (1707–1783), a Swiss mathematician. He showed that the displaced rotational axis of a rigid spheroid would execute a circular motion about its mean position, now called the *Euler nutation*. Because it occurs in the absence of an external driving torque, it is also called the *free nutation*. It is due to differences in the way mass is distributed about the axis of rotational symmetry and an axis at right angles to it in the equatorial plane. The mass distributions are represented by the moments of inertia about these axes. If C and A are the moments of inertia about the rotational axis and an axis in the equatorial plane, respectively, Euler’s theory shows that the period of free nutation is $A/(C - A)$ days, or approximately 305 days.

Astronomers were unsuccessful in detecting a polar motion with this period. In 1891 an American geodesist and astronomer, S. C. Chandler, reported that the polar motion of the Earth’s axis contained two important components. An annual component with amplitude about 0.10 seconds of arc is due to the transfer of mass between atmosphere and hydrosphere accompanying the changing of the seasons. A slightly larger component with amplitude 0.15 seconds of arc has a period of 435 days. This polar motion is now called the *Chandler wobble*. It corresponds to the Euler nutation in an elastic Earth. The increase in period from 305 days to 435 days is a consequence of the elastic yielding of the Earth. The superposition of the annual and Chandler frequencies results in a beat effect, in which the amplitude of the

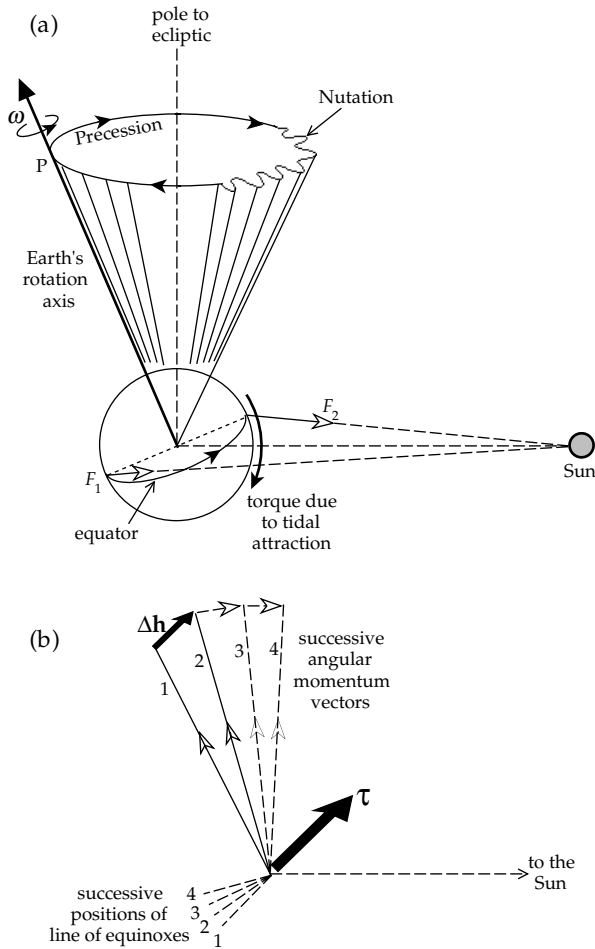


Fig. 2.17 (a) The precession and forced nutation (greatly exaggerated) of the rotation axis due to the lunar torque on the spinning Earth (after Strahler, 1963). (b) Torque and incremental angular momentum changes resulting in precession.

latitude variation is modulated with a period of 6–7 years (Fig. 2.16).

2.3.4.4 Precession and nutation of the rotation axis

During its orbital motion around the Sun the Earth's axis maintains an (almost) constant tilt of about 23.5° to the pole to the ecliptic. The line of intersection of the plane of the ecliptic with the equatorial plane is called the line of equinoxes. Two times a year, when this line points directly at the Sun, day and night have equal duration over the entire globe.

In the theory of the tides the unequal lunar attractions on the near and far side tidal bulges cause a torque about the rotation axis, which has a braking effect on the Earth's rotation. The attractions of the Moon (and Sun) on the equatorial bulge due to rotational flattening also produce torques on the spinning Earth. On the side of the Earth nearer to the Moon (or Sun) the gravitational attraction F_2 on the equatorial bulge is greater than the force F_1 on the distant side (Fig. 2.17a). Due to the tilt of the rotation axis to the ecliptic plane (23.5°), the forces are not

collinear. A torque results, which acts about a line in the equatorial plane, normal to the Earth–Sun line and normal to the spin axis. The magnitude of the torque changes as the Earth orbits around the Sun. It is minimum (and zero) at the spring and autumn equinoxes and maximum at the summer and winter solstices.

The response of a rotating system to an applied torque is to acquire an additional component of angular momentum parallel to the torque. In our example this will be perpendicular to the angular momentum (h) of the spinning Earth. The torque has a component (τ) parallel to the line of equinoxes (Fig. 2.17b) and a component normal to this line in the equatorial plane. The torque τ causes an increment Δh in angular momentum and shifts the angular momentum vector to a new position. If this exercise is repeated incrementally, the rotation axis moves around the surface of a cone whose axis is the pole to the ecliptic (Fig. 2.17a). The geographic pole P moves around a circle in the opposite sense from the Earth's spin. This motion is called *retrograde precession*. It is not a steady motion, but pulsates in sympathy with the driving torque. A change in orientation of the rotation axis affects the location of the line of equinoxes and causes the timing of the equinoxes to change slowly. The rate of change is only 50.4 seconds of arc per year, but it has been recognized during centuries of observation. For example, the Earth's rotation axis now points at Polaris in the constellation Ursa Minor, but in the time of the Egyptians around 3000 BC the pole star was Alpha Draconis, the brightest star in the constellation Draco. Hipparchus is credited with discovering the precession of the equinoxes in 120 BC by comparing his own observations with those of earlier astronomers.

The theory of the phenomenon is well understood. The Moon also exerts a torque on the spinning Earth and contributes to the precession of the rotation axis (and equinoxes). As in the theory of the tides, the small size of the Moon compared to the Sun is more than compensated by its nearness, so that the precessional contribution of the Moon is about double the effect of the Sun. The theory of precession shows that the period of 25,700 yr is proportional to the Earth's *dynamical ellipticity*, H (see Eq. (2.45)). This ratio (equal to $1/305.457$) is an important indicator of the internal distribution of mass in the Earth.

The component of the torque in the equatorial plane adds an additional motion to the axis, called *nutation*, because it causes the axis to nod up and down (Fig. 2.17a). The solar torque causes a semi-annual nutation, the lunar torque a semi-monthly one. In fact the motion of the axis exhibits many *forced nutations*, so-called because they respond to external torques. All are tiny perturbations on the precessional motion, the largest having an amplitude of only about 9 seconds of arc and a period of 18.6 yr. This nutation results from the fact that the plane of the lunar orbit is inclined at 5.145° to the plane of the ecliptic and (like the motion of artificial Earth

satellites) precesses retrogradely. This causes the inclination of the lunar orbit to the equatorial plane to vary between about 18.4° and 28.6° , modulating the torque and forcing a nutation with a period of 18.6 yr.

It is important to note that the Euler nutation and Chandler wobble are polar motions about the rotation axis, but the precession and forced nutations are displacements of the rotation axis itself.

2.3.4.5 Milankovitch climatic cycles

Solar energy can be imagined as flowing equally from the Sun in all directions. At distance r it floods a sphere with surface area $4\pi r^2$. The amount of solar energy falling per second on a square meter (the *insolation*) therefore decreases as the inverse square of the distance from the Sun. The gravitational attractions of the Moon, Sun, and the other planets – especially Jupiter – cause cyclical changes of the orientation of the rotation axis and variations in the shape and orientation of Earth's orbit. These variations modify the insolation of the Earth and result in long-term periodic changes in Earth's climate.

The angle between the rotational axis and the pole to the ecliptic is called the *obliquity*. It is the main factor determining the seasonal difference between summer and winter in each hemisphere. In the northern hemisphere, the insolation is maximum at the summer solstice (currently June 21) and minimum at the winter solstice (December 21–22). The exact dates change with the precession of the equinoxes, and also depend on the occurrence of leap years. The solstices do not coincide with extreme positions in Earth's orbit. The Earth currently reaches aphelion, its furthest distance from the Sun, around July 4–6, shortly after the summer solstice, and passes perihelion around January 2–4. About 13,000 yr from now, as a result of precession, the summer solstice will occur when Earth is close to perihelion. In this way, precession causes long-term changes in climate with a period related to the precession.

The gravitational attraction of the other planets causes the obliquity to change cyclically with time. It is currently equal to $23^\circ 26' 21.4''$ but varies slowly between a minimum of $21^\circ 55'$ and a maximum of $24^\circ 18'$. When the obliquity increases, the seasonal differences in temperature become more pronounced, while the opposite effect ensues if obliquity decreases. Thus, the variation in obliquity causes a modulation in the seasonal contrast between summer and winter on a global scale. This effect is manifest as a cyclical change in climate with a period of about 41 kyr.

A further effect of planetary attraction is to cause the eccentricity of the Earth's orbit, at present 0.017, to change cyclically (Fig. 2.18). At one extreme of the cycle, the orbit is almost circular, with an eccentricity of only 0.005. The closest distance from the Sun at perihelion is then 99% of the furthest distance at aphelion. At the other extreme, the orbit is more elongate, although with an eccentricity of 0.058 it is only slightly elliptical. The perihelion distance is then 89% of the aphelion distance.

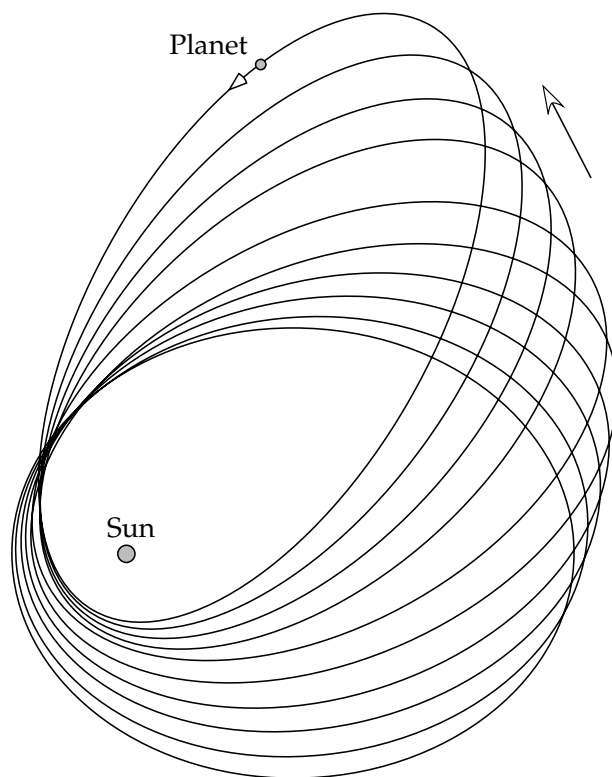


Fig. 2.18 Schematic illustration of the 100,000 yr variations in eccentricity and rotation of the axis of the Earth's elliptical orbit. The effects are greatly exaggerated for ease of visualization.

These slight differences have climatic effects. When the orbit is almost circular, the difference in insolation between summer and winter is negligible. However, when the orbit is most elongate, the insolation in winter is only 78% of the summer insolation. The cyclical variation in eccentricity has a dominant period of 404 kyr and lesser periodicities of 95 kyr, 99 kyr, 124 kyr and 131 kyr that together give a roughly 100 kyr period. The eccentricity variations generate fluctuations in paleoclimatic records with periods around 100 kyr and 400 kyr.

Not only does planetary attraction cause the shape of the orbit to change, it also causes the perihelion–aphelion axis of the orbit to precess. The orbital ellipse is not truly closed, and the path of the Earth describes a rosette with a period that is also around 100 kyr (Fig. 2.18). The precession of perihelion interacts with the axial precession and modifies the observed period of the equinoxes. The 26 kyr axial precession is retrograde with a rate of 0.038 cycles/kyr; the 100 kyr orbital precession is prograde, which speeds up the effective precession rate to 0.048 cycles/kyr. This is equivalent to a retrograde precession with a period of about 21 kyr. A corresponding climatic fluctuation has been interpreted in many sedimentary deposits.

Climatic effects related to cyclical changes in the Earth's rotational and orbital parameters were first studied between 1920 and 1938 by a Yugoslavian astronomer, Milutin Milanković (anglicized to Milankovitch). Periodicities of 21 kyr, 41 kyr, 100 kyr and 400 kyr – called the

Milankovitch climatic cycles – have been described in various sedimentary records ranging in age from Quaternary to Mesozoic. Caution must be used in interpreting the cyclicities in older records, as the characteristic Milankovitch periods are dependent on astronomical parameters that may have changed appreciably during the geological ages.

2.3.5 Coriolis and Eötvös accelerations

Every object on the Earth experiences the centrifugal acceleration due to the Earth's rotation. Moving objects on the rotating Earth experience additional accelerations related to the velocity at which they are moving. At latitude λ the distance d of a point on the Earth's surface from the rotational axis is equal to $R \cos \lambda$, and the rotational spin ω translates to an eastwards linear velocity v equal to $\omega R \cos \lambda$. Consider an object (e.g., a vehicle or projectile) that is moving at velocity v across the Earth's surface. In general v has a northward component v_N and an eastward component v_E . Consider first the effects related to the eastward velocity, which is added to the linear velocity of the rotation. The centrifugal acceleration increases by an amount Δa_c , which can be obtained by differentiating a_c in Eq. (2.19) with respect to ω

$$\Delta a_c = 2\omega(R \cos \lambda) \Delta \omega = 2\omega v_E \quad (2.39)$$

The extra centrifugal acceleration Δa_c can be resolved into a vertical component and a horizontal component (Fig. 2.19a). The vertical component, equal to $2\omega v_E \cos \lambda$, acts upward, opposite to gravity. It is called the *Eötvös acceleration*. Its effect is to *decrease* the measured gravity by a small amount. If the moving object has a westward component of velocity the Eötvös acceleration *increases* the measured gravity. If gravity measurements are made on a moving platform (for example, on a research ship or in an airplane), the measured gravity must be corrected to allow for the Eötvös effect. For a ship sailing eastward at 10 km h^{-1} at latitude 45° the Eötvös correction is 28.6 mgal ; in an airplane flying eastward at 300 km h^{-1} the correction is 856 mgal . These corrections are far greater than the sizes of many important gravity anomalies. However, the Eötvös correction can be made satisfactorily in marine gravity surveys, and recent technical advances now make it feasible in aerogravimetry.

The horizontal component of the extra centrifugal acceleration due to v_E is equal to $2\omega v_E \sin \lambda$. In the northern hemisphere it acts to the south. If the object moves westward, the acceleration is northward. In each case it acts horizontally *to the right* of the direction of motion. In the southern hemisphere the sense of this acceleration is reversed; it acts to the left of the direction of motion. This acceleration is a component of the *Coriolis acceleration*, another component of which derives from the northward motion of the object.

Consider an object moving northward along a meridian of longitude (Fig. 2.19b, point 1). The linear velocity of a

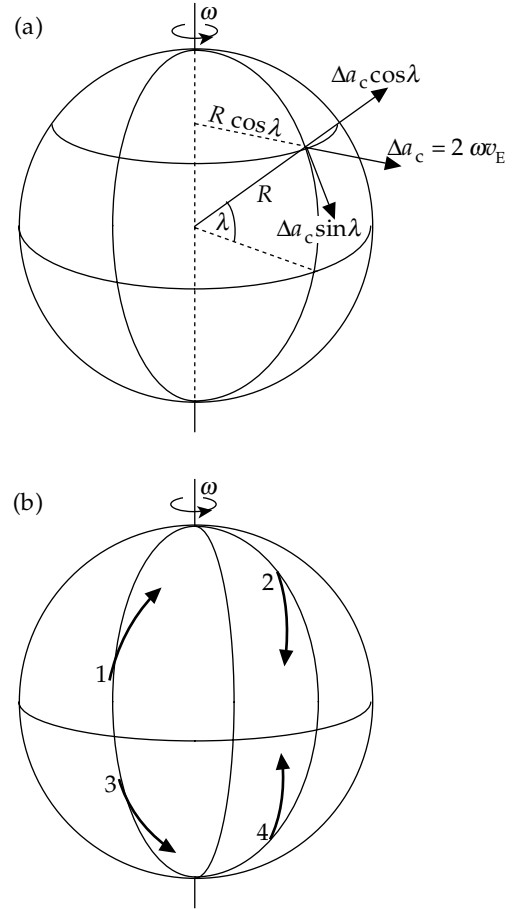


Fig. 2.19 (a) Resolution of the additional centrifugal acceleration Δa_c due to eastward velocity into vertical and horizontal components. (b) The horizontal deviations of the northward or southward trajectory of an object due to conservation of its angular momentum.

point on the Earth's surface decreases poleward, because the distance from the axis of rotation ($d = R \cos \lambda$) decreases. The angular momentum of the moving object must be conserved, so the eastward velocity v_E must increase. As the object moves to the north its eastward velocity is faster than the circles of latitude it crosses and its trajectory deviates to the right. If the motion is to the south (Fig. 2.19b, point 2), the inverse argument applies. The body crosses circles of latitude with faster eastward velocity than its own and, in order to maintain angular momentum, its trajectory must deviate to the west. In each case the deviation is *to the right* of the direction of motion. A similar argument applied to the southern hemisphere gives a Coriolis effect *to the left* of the direction of motion (Fig. 2.19b, points 3 and 4).

The magnitude of the Coriolis acceleration is easily evaluated quantitatively. The angular momentum h of a mass m at latitude λ is equal to $m\omega R^2 \cos^2 \lambda$. Conservation of angular momentum gives

$$\frac{\partial h}{\partial t} = mR^2 \cos^2 \lambda \frac{\partial \omega}{\partial t} + m\omega R^2 (-2 \cos \lambda \sin \lambda) \frac{\partial \lambda}{\partial t} = 0 \quad (2.40)$$

Rearranging and simplifying, we get

$$(R \cos \lambda) \frac{\partial \omega}{\partial t} = 2\omega \sin \lambda (R \frac{\partial \lambda}{\partial t}) \quad (2.41)$$

The expression on the left of the equation is an acceleration, a_E , equal to the rate of change of the eastward velocity. The expression in brackets on the right is the northward velocity component v_N . We can write this component of the Coriolis acceleration as $2\omega v_N \sin \lambda$. The north and east components of the Coriolis acceleration are therefore:

$$\begin{aligned} a_N &= 2\omega v_E \sin \lambda \\ a_E &= 2\omega v_N \sin \lambda \end{aligned} \quad (2.42)$$

The Coriolis acceleration deflects the horizontal path of any object moving on the Earth's surface. It affects the directions of wind and ocean currents, eventually constraining them to form circulatory patterns about centers of high or low pressure, and thereby plays an important role in determining the weather.

2.4 THE EARTH'S FIGURE AND GRAVITY

2.4.1 The figure of the Earth

The true surface of the Earth is uneven and irregular, partly land and partly water. For geophysical purposes the Earth's shape is represented by a smooth closed surface, which is called the figure of the Earth. Early concepts of the figure were governed by religion, superstition and non-scientific beliefs. The first circumnavigation of the Earth, completed in 1522 by Magellan's crew, established that the Earth was probably round. Before the era of scientific awakening the Earth's shape was believed to be a sphere. As confirmed by numerous photographs from spacecraft, this is in fact an excellent first approximation to Earth's shape that is adequate for solving many problems. The original suggestion that the Earth is a spheroid flattened at the poles is credited to Newton, who used a hydrostatic argument to account for the polar flattening. The slightly flattened shape permitted an explanation of why a clock that was precise in Paris lost time near to the equator (see Section 2.1).

Earth's shape and gravity are intimately associated. The figure of the Earth is the shape of an equipotential surface of gravity, in particular the one that coincides with mean sea level. The best mathematical approximation to the figure is an oblate ellipsoid, or spheroid (Fig. 2.20). The precise determination of the dimensions of the Earth (e.g., its polar and equatorial radii) is the main objective of the science of *geodesy*. It requires an exact knowledge of the Earth's gravity field, the description of which is the goal of *gravimetry*.

Modern analyses of the Earth's shape are based on precise observations of the orbits of artificial Earth satellites. These data are used to define a best-fitting oblate ellipsoid, called the *International Reference Ellipsoid*. In 1930 geodesists and geophysicists defined an optimum

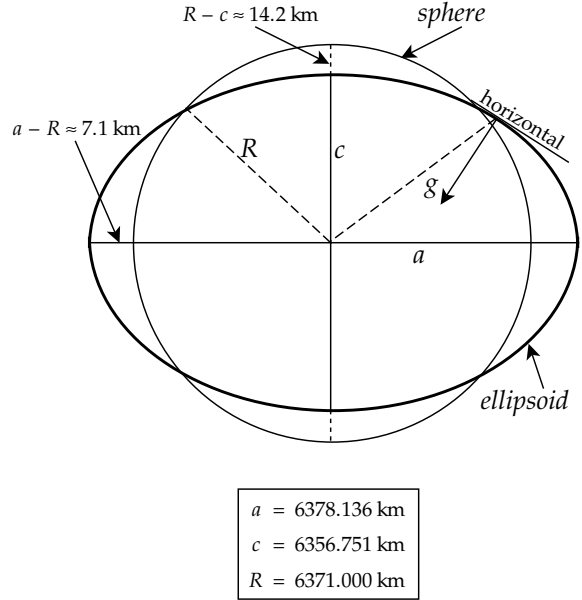


Fig. 2.20 Comparison of the dimensions of the *International Reference Ellipsoid* with a sphere of equal volume.

reference ellipsoid based on the best available data at the time. The dimensions of this figure have been subsequently refined as more exact data have become available. In 1980 the *International Association of Geodesy* adopted a *Geodetic Reference System* (GRS80) in which the reference ellipsoid has an equatorial radius (a) equal to 6378.137 km and a polar radius (c) equal to 6356.752 km. Subsequent determinations have resulted in only minor differences in the most important geodetic parameters. Some current values are listed in Table 2.1. The radius of the equivalent sphere (R) is found from $R = (a^2 c)^{1/3}$ to be 6371.000 km. Compared to the best-fitting sphere the spheroid is flattened by about 14.2 km at each pole and the equator bulges by about 7.1 km. The polar flattening f is defined as the ratio

$$f = \frac{a - c}{a} \quad (2.43)$$

The flattening of the optimum reference ellipsoid defined in 1930 was exactly 1/297. This ellipsoid, and the variation of gravity on its surface, served as the basis of gravimetric surveying for many years, until the era of satellite geodesy and highly sensitive gravimeters showed it to be too inexact. A recent best estimate of the flattening is $f = 3.35287 \times 10^{-3}$ (i.e., $f = 1/298.252$).

If the Earth is assumed to be a rotating fluid in perfect hydrostatic equilibrium (as assumed by Newton's theory), the flattening should be 1/299.5, slightly smaller than the observed value. The hydrostatic condition assumes that the Earth has no internal strength. A possible explanation for the tiny discrepancy in f is that the Earth has sufficient strength to maintain a non-hydrostatic figure, and the present figure is inherited from a time of more rapid rotation. Alternatively, the slightly more flattened form of the

Table 2.1 Some fundamental parameters relevant to the shape, rotation and orbit of the Earth. Sources: [1] Mohr and Taylor, 2005; [2] McCarthy and Petit, 2004; [3] Groten, 2004

Parameter	Symbol	Value	Units	Reference
<i>Terrestrial parameters (2004)</i>				
Gravitational constant	G	6.673×10^{-11}	$\text{m}^3 \text{kg}^{-1} \text{s}^{-2}$	[1]
Geocentric gravitational constant	GE	3.9860044×10^{14}	$\text{m}^3 \text{s}^{-2}$	[2]
Mass of the Earth: $E = (GE)/G$	E	5.9737×10^{24}	kg	
Earth's equatorial radius	a	6 378.137	km	[3]
Earth's polar radius: $c = a(1 - f)$	c	6 356.752	km	
Radius of equivalent sphere: $R_0 = (a^2c)^{1/3}$	R_0	6 371.000	km	
Mean equatorial gravity	g_e	9.7803278	m s^{-2}	[2]
Mean angular velocity of rotation	Ω	7.292115×10^{-5}	rad s^{-1}	[2]
Dynamical form-factor	J_2	1.0826359×10^{-3}		[3]
Flattening	f	1 : 298.252		[3]
Equatorial acceleration ratio	m	1 : 288.901		[3]
Dynamical ellipticity	H	1 : 305.457		[3]
<i>Orbital parameters (2003)</i>				
Astronomical unit	AU	149,597,870.691	km	[2]
Solar mass ratio	μ_S	332,946.0		[2]
Lunar mass ratio	μ_L	0.012300038		[2]
Obliquity of the ecliptic	ε_0	23° 26' 21.4"		[2]
Obliquity of lunar orbit to ecliptic		5° 0.9'		[3]
Eccentricity of solar orbit of barycenter		0.01671		[3]
Eccentricity of lunar orbit		0.05490		[3]

Earth may be due to internal density contrasts, which could be the consequence of slow convection in the Earth's mantle. This would take place over long time intervals and could result in a non-hydrostatic mass distribution.

The cause of the polar flattening is the deforming effect of the centrifugal acceleration. This is maximum at the equator where the gravitational acceleration is smallest. The parameter m is defined as the ratio of the equatorial centrifugal acceleration to the equatorial gravitational acceleration:

$$m = \frac{\omega^2 a}{GE/a^2} = \frac{\omega^2 a^3}{GE}$$

(2.44)

The value of m based on current geodetic values (Table 2.1) is $3.461\,39 \times 10^{-3}$ (i.e., $m = 1/288.901$).

As a result of the flattening, the distribution of mass within the Earth is not simply dependent on radius. The moments of inertia of the Earth about the rotation axis (C) and any axis in the equatorial plane (A) are unequal. As noted in the previous section the inequality affects the way the Earth responds to external gravitational torques and is a determining factor in perturbations of the Earth's rotation. The principal moments of inertia define the *dynamical ellipticity*:

$$H = \frac{C - \frac{1}{2}(A + B)}{C} \approx \frac{C - A}{C}$$

(2.45)

The dynamical ellipticity is obtained from precise observations of the orbits of artificial satellites of the Earth (see Section 2.4.5.1). The current optimum value for H is $3.273\,787\,5 \times 10^{-3}$ (i.e., $H = 1/305.457$).

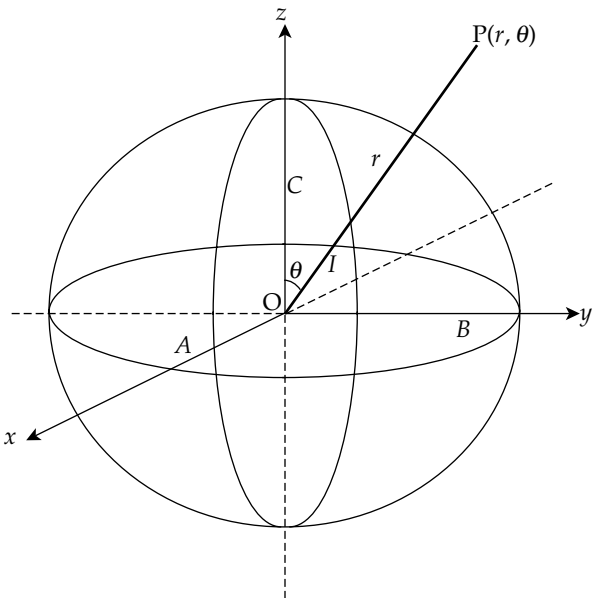


Fig. 2.21 Parameters of the ellipsoid used in MacCullagh's formula. A, B, and C are moments of inertia about the x-, y- and z-axes, respectively, and I is the moment of inertia about the line OP.

2.4.2 Gravitational potential of the spheroidal Earth

The ellipsoidal shape changes the gravitational potential of the Earth from that of an undeformed sphere. In 1849 J. MacCullagh developed the following formula for the gravitational potential of any body at large distance from its center of mass:

$$U_G = -G \frac{E}{r} - G \frac{(A + B + C - 3I)}{2r^3} - \dots$$

(2.46)

The first term, of order r^{-1} , is the gravitational potential of a point mass or sphere with mass E (Eqs. (2.10) and (2.14)); for the Earth it describes the potential of the undeformed globe. If the reference axes are centered on the body's center of mass, there is no term in r^{-2} . The second term, of order r^{-3} , is due to deviations from the spherical shape. For the flattened Earth it results from the mass displacements due to the rotational deformation. The parameters A , B , and C are the principal moments of inertia of the body and I is the moment of inertia about the line OP joining the center of mass to the point of observation (Fig. 2.21). In order to express the potential accurately an infinite number of terms of higher order in r are needed. In the case of the Earth these can be neglected, because the next term is about 1000 times smaller than the second term.

For a body with planes of symmetry, I is a simple combination of the principal moments of inertia. Setting A equal to B for rotational symmetry, and defining the angle between OP and the rotation axis to be θ , the expression for I is

$$I = A \sin^2 \theta + C \cos^2 \theta \quad (2.47)$$

MacCullagh's formula for the ellipsoidal Earth then becomes

$$U_G = -G \frac{E}{r} - G \frac{(C-A)(3\cos^2 \theta - 1)}{r^3} \quad (2.48)$$

The function $(3\cos^2 \theta - 1)/2$ is a second-order polynomial in $\cos \theta$, written as $P_2(\cos \theta)$. It belongs to a family of functions called Legendre polynomials (Box 2.2). Using this notation MacCullagh's formula for the gravitational potential of the oblate ellipsoid becomes

$$U_G = -G \frac{E}{r} - G \frac{(C-A)}{r^3} P_2(\cos \theta) \quad (2.49)$$

This can be written in the alternative form

$$U_G = -G \frac{E}{r} \left(1 - \left(\frac{C-A}{ER^2} \right) \left(\frac{R}{r} \right)^2 P_2(\cos \theta) \right) \quad (2.50)$$

Potential theory requires that the gravitational potential of the spheroidal Earth must satisfy an important equation, the Laplace equation (Box 2.3). The solution of this equation is the sum of an infinite number of terms of increasing order in $1/r$, each involving an appropriate Legendre polynomial:

$$U_G = -G \frac{E}{r} \left(1 - \sum_{n=2}^{\infty} \left(\frac{R}{r} \right)^n J_n P_n(\cos \theta) \right) \quad (2.51)$$

In this equation the coefficients J_n multiplying $P_n(\cos \theta)$ determine the relative importance of the term of n th order. The values of J_n are obtained from satellite geodesy: $J_2 = 1082.6 \times 10^{-6}$; $J_3 = -2.54 \times 10^{-6}$; $J_4 = -1.59 \times 10^{-6}$; higher orders are insignificant. The most important coefficient is the second order one, the

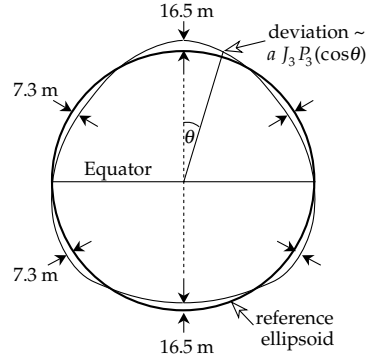


Fig. 2.22 The third-order term in the gravitational potential describes a pear-shaped Earth. The deviations from the reference ellipsoid are of the order of 10–20 m, much smaller than the deviations of the ellipsoid from a sphere, which are of the order of 10–20 km.

dynamical form-factor J_2 , which describes the effect of the polar flattening on the Earth's gravitational potential. Comparison of terms in Eqs. (2.48) and (2.51) gives the result

$$J_2 = \frac{C-A}{ER^2} \quad (2.52)$$

The term of next higher order ($n=3$) in Eq. (2.51) describes the deviations from the reference ellipsoid which correspond to a pear-shaped Earth (Fig. 2.22). These deviations are of the order of 7–17 m, a thousand times smaller than the deviations of the ellipsoid from a sphere, which are of the order of 7–14 km.

2.4.3 Gravity and its potential

The potential of gravity (U_g) is the sum of the gravitational and centrifugal potentials. It is often called the *geopotential*. At a point on the surface of the rotating spheroid it can be written

$$U_g = U_G - \frac{1}{2} \omega^2 r^2 \sin^2 \theta \quad (2.53)$$

If the free surface is an equipotential surface of gravity, then U_g is everywhere constant on it. The shape of the equipotential surface is constrained to be that of the spheroid with flattening f . Under these conditions a simple relation is found between the constants f , m and J_2 :

$$J_2 = \frac{1}{3}(2f - m) \quad (2.54)$$

By equating Eqs. (2.52) and (2.54) and re-ordering terms slightly we obtain the following relationship

$$\frac{C-A}{ER^2} = \frac{1}{3}(2f - m) \quad (2.55)$$

This yields useful information about the variation of density within the Earth. The quantities f , m and

Box 2.2: Legendre polynomials

In the triangle depicted in Fig. B2.2 the side u is related to the other two sides r and R and the angle θ they enclose by the cosine law. The expression for $1/u$ can then be written:

$$\frac{1}{u} = \frac{1}{(R^2 + r^2 - 2rR\cos\theta)^{1/2}} = \frac{1}{R} \left[1 + \left(\frac{r^2}{R^2} - 2\frac{r}{R}\cos\theta \right) \right]^{-1/2} \quad (1)$$

which on expanding becomes

$$\frac{1}{u} = \frac{1}{R} \left[1 + \frac{r}{R}\cos\theta + \frac{r^2}{R^2} \left(\frac{3\cos^2\theta - 1}{2} \right) + \frac{r^3}{R^3} \left(\frac{5\cos^3\theta - 3\cos\theta}{2} \right) + \dots \right] \quad (2)$$

This infinitely long series of terms in (r/R) is called the reciprocal distance formula. It can be written in short-hand form as

$$\frac{1}{u} = \frac{1}{R} \sum_{n=1}^{\infty} \left(\frac{r}{R} \right)^n P_n(\cos\theta) \quad (3)$$

The angle θ in this expression describes the angular deviation between the side r and the reference side R . The functions $P_n(\cos\theta)$ in the sum are called the *ordinary Legendre polynomials* of order n in $\cos\theta$. They are named after a French mathematician Adrien Marie Legendre (1752–1833). Each polynomial is a coefficient of $(r/R)^n$ in the infinite sum of terms for $(1/u)$, and so has order n . Writing $\cos\theta = x$, and $P_n(\cos\theta) = P_n(x)$, the first few polynomials, for $n = 0, 1, 2$, and 3 , respectively, are as follows

$$\begin{aligned} P_0(x) &= 1 & P_2(x) &= \frac{1}{2}(3x^2 - 1) \\ P_1(x) &= x & P_3(x) &= \frac{1}{2}(5x^3 - 3x) \end{aligned} \quad (4)$$

By substituting $\cos\theta$ for x these expressions can be converted into functions of $\cos\theta$. Legendre discovered that the polynomials satisfied the following second-order differential equation, in which n is an integer and $y = P_n(x)$:

$$\frac{\partial}{\partial x} (1 - x^2) \frac{\partial y}{\partial x} + n(n+1)y = 0 \quad (5)$$

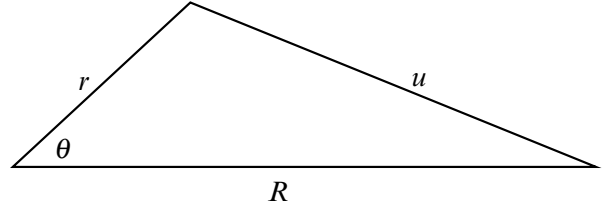


Fig. B2.2 Reference triangle for derivation of Legendre polynomials.

This, named in his honor, is the *Legendre equation*. It plays an important role in geophysical potential theory for situations expressed in spherical coordinates that have rotational symmetry about an axis. This is, for example, the case for the gravitational attraction of a spheroid, the simplified form of the Earth's shape.

The derivation of an individual polynomial of order n is rather tedious if the expanded expression for $(1/u)$ is used. A simple formula for calculating the Legendre polynomials for any order n was developed by another French mathematician, Olinde Rodrigues (1794–1851). The *Rodrigues formula* is

$$P_n(x) = \frac{1}{2^n n!} \frac{\partial^n}{\partial x^n} (x^2 - 1)^n \quad (6)$$

A relative of this equation encountered in many problems of potential theory is the *associated Legendre equation*, which written as a function of x is

$$\frac{\partial}{\partial x} (1 - x^2) \frac{\partial y}{\partial x} + \left(n(n+1) - \frac{m^2}{(1 - x^2)} \right) y = 0 \quad (7)$$

The solutions of this equation involve two integers, the order n and degree m . As in the case of the ordinary Legendre equation the solutions are polynomials in x , which are called the *associated Legendre polynomials* and written $P_n^m(x)$. A modification of the Rodrigues formula allows easy computation of these functions from the ordinary Legendre polynomials:

$$P_n^m(x) = (1 - x^2)^{m/2} \frac{\partial^m}{\partial x^m} P_n(x) \quad (8)$$

To express the associated Legendre polynomials as functions of θ , i.e. as $P_n^m(\cos\theta)$, it is again only necessary to substitute $\cos\theta$ for x .

$(C-A)/C$ are each equal to approximately $1/300$. Inserting their values in the equation gives $C \approx 0.33ER^2$. Compare this value with the principal moments of inertia of a hollow spherical shell ($0.66ER^2$) and a solid sphere with uniform density ($0.4ER^2$). The concentration of mass near the center causes a reduction in the multiplying factor from 0.66 to 0.4. The value of 0.33 for the Earth

implies that, in comparison with a uniform solid sphere, the density must increase towards the center of the Earth.

2.4.4 Normal gravity

The direction of gravity at a point is defined as perpendicular to the equipotential surface through the point. This

Box 2.3: Spherical harmonics

Many natural forces are directed towards a central point. Examples are the electrical field of a point charge, the magnetic field of a single magnetic pole, and the gravitational acceleration toward a center of mass. The French astronomer and mathematician Pierre Simon, marquis de Laplace (1749–1827) showed that, in order to fulfil this basic physical condition, the potential of the field must satisfy a second-order differential equation, the Laplace equation. This is one of the most famous and important equations in physics and geophysics, as it applies to many situations in potential theory. For the gravitational potential U_G the Laplace equation is written in Cartesian coordinates (x, y, z) as

$$\frac{\partial^2 U_G}{\partial x^2} + \frac{\partial^2 U_G}{\partial y^2} + \frac{\partial^2 U_G}{\partial z^2} = 0 \quad (1)$$

In spherical polar coordinates (r, θ, ϕ) the Laplace equation becomes

$$\frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial U_G}{\partial r} + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial U_G}{\partial \theta} + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 U_G}{\partial \phi^2} = 0 \quad (2)$$

The variation with azimuth ϕ disappears for symmetry about the rotational axis. The general solution of the Laplace equation for rotational symmetry (e.g., for a spheroidal Earth) is

$$U_G = \sum_{n=0}^{\infty} \left(A_n r^n + \frac{B_n}{r^{n+1}} \right) P_n(\cos \theta) \quad (3)$$

where $P_n(\cos \theta)$ is an ordinary Legendre polynomial of order n and the coordinate θ is the angular deviation of

the point of observation from the reference axis (see Box 2.1). In geographic coordinates θ is the co-latitude.

If the potential field is not rotationally symmetric – as is the case, for example, for the geoid and the Earth's magnetic field – the solution of the Laplace equation varies with azimuth ϕ as well as with radius r and axial angle θ and is given by

$$U_G = \sum_{n=0}^{\infty} \left(A_n r^n + \frac{B_n}{r^{n+1}} \right) \sum_{m=0}^n (a_n^m \cos m\phi + b_n^m \sin m\phi) P_n^m(\cos \theta) \quad (4)$$

where in this case $P_n^m(\cos \theta)$ is an associated Legendre polynomial of order n and degree m as described in Box 2.2. This equation can in turn be written in modified form as

$$U_G = \sum_{n=0}^{\infty} \left(A_n r^n + \frac{B_n}{r^{n+1}} \right) \sum_{m=0}^n Y_n^m(\theta, \phi) \quad (5)$$

The function

$$Y_n^m(\theta, \phi) = (a_n^m \cos m\phi + b_n^m \sin m\phi) P_n^m(\cos \theta) \quad (6)$$

is called a *spherical harmonic function*, because it has the same value when θ or ϕ is increased by an integral multiple of 2π . It describes the variation of the potential with the coordinates θ and ϕ on a spherical surface (i.e., for which r is a constant). Spherical harmonic functions are used, for example, for describing the variations of the gravitational and magnetic potentials, geoid height, and global heat flow with latitude and longitude on the surface of the Earth.

defines the *vertical* at the point, while the plane tangential to the equipotential surface defines the *horizontal* (Fig. 2.20). A consequence of the spheroidal shape of the Earth is that the vertical direction is generally not radial, except on the equator and at the poles.

On a spherical Earth there is no ambiguity in how we define latitude. It is the angle at the center of the Earth between the radius and the equator, the complement to the polar angle θ . This defines the geocentric latitude λ' . However, the geographic latitude in common use is not defined in this way. It is found by geodetic measurement of the angle of elevation of a fixed star above the horizon. But the horizontal plane is tangential to the ellipsoid, not to a sphere (Fig. 2.20), and the vertical direction (i.e., the local direction of gravity) intersects the equator at an angle λ that is slightly larger than the geocentric latitude λ' (Fig. 2.23). The difference $(\lambda - \lambda')$ is zero at the equator and poles and reaches a maximum

at a latitude of 45° , where it amounts to only 0.19° (about $12'$).

The *International Reference Ellipsoid* is the standardized reference figure of the Earth. The theoretical value of gravity on the rotating ellipsoid can be computed by differentiating the gravity potential (Eq. (2.53)). This yields the radial and transverse components of gravity, which are then combined to give the following formula for gravity normal to the ellipsoid:

$$g_n = g_e (1 + \beta_1 \sin^2 \lambda + \beta_2 \sin^2 2\lambda) \quad (2.56)$$

Where, to second order in f and m ,

$$\begin{aligned} g_n &= g_e \left(1 + f - \frac{3}{2}m + f^2 - \frac{27}{14}fm \right) \\ \beta_1 &= \frac{5}{2}m - f + \frac{15}{4}m^2 - \frac{17}{14}fm \\ \beta_2 &= \frac{1}{8}f^2 - \frac{5}{8}fm \end{aligned} \quad (2.57)$$

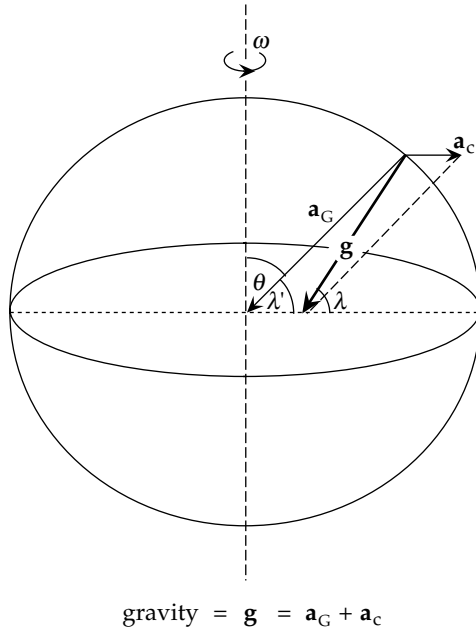


Fig. 2.23 Gravity on the ellipsoidal Earth is the vector sum of the gravitational and centrifugal accelerations and is not radial; consequently, geographic latitude (λ) is slightly larger than geocentric latitude (λ').

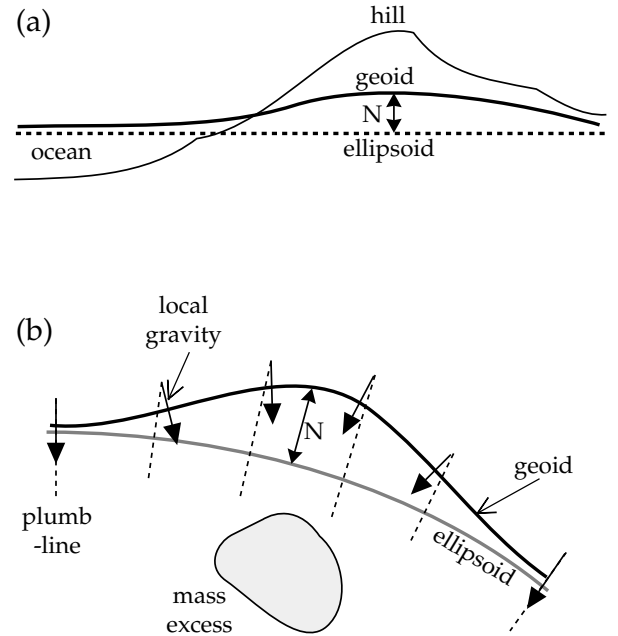


Fig. 2.24 (a) A mass outside the ellipsoid or (b) a mass excess below the ellipsoid elevates the geoid above the ellipsoid. N is the geoid undulation.

Equation (2.56) is known as the *normal gravity formula*. The constants in the formula, defined in 1980 for the Geodetic Reference System (GRS80) still in common use, are: $g_e = 9.780\,327\,\text{m s}^{-2}$; $\beta_1 = 5.30244 \times 10^{-3}$; $\beta_2 = -5.8 \times 10^{-6}$. They allow calculation of normal gravity at any latitude with an accuracy of 0.1 mgal. Modern instruments can measure gravity differences with even greater precision, in which case a more exact formula, accurate to 0.0001 mgal, can be used. The normal gravity formula is very important in the analysis of gravity measurements on the Earth, because it gives the theoretical variation of normal gravity (g_n) with latitude on the surface of the reference ellipsoid.

The normal gravity is expressed in terms of g_e , the value of gravity on the equator. The second-order terms f^2 , m^2 and fm are about 300 times smaller than the first-order terms f and m . The constant β_2 is about 1000 times smaller than β_1 . If we drop second-order terms and use $\lambda = 90^\circ$, the value of normal gravity at the pole is $g_p = g_e (1 + \beta_1)$, so by rearranging and retaining only first-order terms, we get

$$\frac{g_p - g_e}{g_e} = \frac{5}{2}m - f \quad (2.58)$$

This expression is called Clairaut's theorem. It was developed in 1743 by a French mathematician, Alexis-Claude Clairaut, who was the first to relate the variation of gravity on the rotating Earth with the flattening of the spheroid. The normal gravity formula gives $g_p = 9.832\,186\,\text{m s}^{-2}$. Numerically, this gives an increase in gravity from equator to pole of approximately $5.186 \times 10^{-2}\,\text{m s}^{-2}$, or 5186 mgal.

There are two obvious reasons for the poleward increase in gravity. The distance to the center of mass of the Earth is shorter at the poles than at the equator. This gives a stronger gravitational acceleration (a_G) at the poles. The difference is

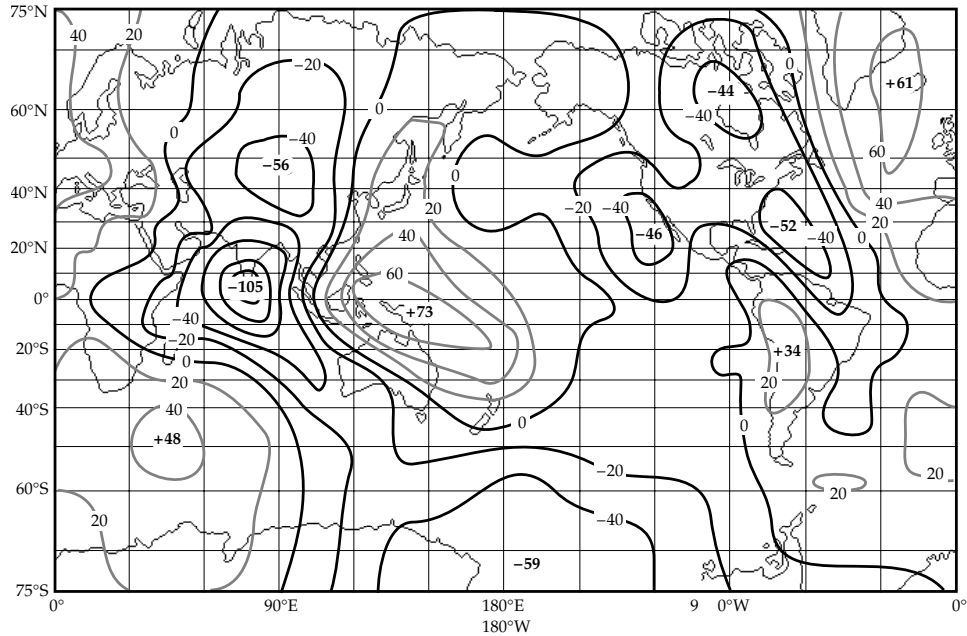
$$\Delta a_G = \left(\frac{GE}{c^2} - \frac{GE}{a^2} \right) \quad (2.59)$$

This gives an excess gravity of approximately 6600 mgal at the poles. The effect of the centrifugal force in diminishing gravity is largest at the equator, where it equals (ma_G), and is zero at the poles. This also results in a poleward increase of gravity, amounting to about 3375 mgal. These figures indicate that gravity should increase by a total of 9975 mgal from equator to pole, instead of the observed difference of 5186 mgal. The discrepancy can be resolved by taking into account a third factor. The computation of the difference in gravitational attraction is not so simple as indicated by Eq. (2.59). The equatorial bulge places an excess of mass under the equator, increasing the equatorial gravitational attraction and thereby reducing the gravity decrease from equator to pole.

2.4.5 The geoid

The international reference ellipsoid is a close approximation to the equipotential surface of gravity, but it is really a mathematical convenience. The physical equipotential surface of gravity is called the *geoid*. It reflects the true distribution of mass inside the Earth and differs from the theoretical ellipsoid by small amounts. Far from

Fig. 2.25 World map of geoid undulations relative to a reference ellipsoid of flattening $f = 1/298.257$ (after Lerch *et al.*, 1979).



land the geoid agrees with the free ocean surface, excluding the temporary perturbing effects of tides and winds. Over the continents the geoid is affected by the mass of land above mean sea level (Fig. 2.24a). The mass within the ellipsoid causes a downward gravitational attraction toward the center of the Earth, but a hill or mountain whose center of gravity is outside the ellipsoid causes an upward attraction. This causes a local elevation of the geoid above the ellipsoid. The displacement between the geoid and the ellipsoid is called a *geoid undulation*; the elevation caused by the mass above the ellipsoid is a positive undulation.

2.4.5.1 Geoid undulations

In computing the theoretical figure of the Earth the distribution of mass beneath the ellipsoid is assumed to be homogeneous. A local excess of mass under the ellipsoid will deflect and strengthen gravity locally. The potential of the ellipsoid is achieved further from the center of the Earth. The equipotential surface is forced to warp upward while remaining normal to gravity. This gives a positive geoid undulation over a mass excess under the ellipsoid (Fig. 2.24b). Conversely, a mass deficit beneath the ellipsoid will deflect the geoid below the ellipsoid, causing a negative geoid undulation. As a result of the uneven topography and heterogeneous internal mass distribution of the Earth, the geoid is a bumpy equipotential surface.

The potential of the geoid is represented mathematically by *spherical harmonic functions* that involve the *associated Legendre polynomials* (Box 2.3). These are more complicated than the ordinary Legendre polynomials used to describe the gravitational potential of the ellipsoid (Eqs. (2.49)–(2.51)). Until now we have only considered

variation of the potential with distance r and with the co-latitude angle θ . This is an oversimplification, because density variations within the Earth are not symmetrical about the rotation axis. The geoid is an equipotential surface for the real density distribution in the Earth, and so the potential of the geoid varies with longitude as well as co-latitude. These variations are taken into account by expressing the potential as a sum of spherical harmonic functions, as described in Box 2.3. This representation of the geopotential is analogous to the simpler expression for the gravitational potential of the rotationally symmetric Earth using a series of Legendre polynomials (Eq. (2.51)).

In modern analyses the coefficient of each term in the geopotential – similar to the coefficients J_n in Eq. (2.51) – can be calculated up to a high harmonic degree. The terms up to a selected degree are then used to compute a model of the geoid and the Earth’s gravity field. A combination of satellite data and surface gravity measurements was used to construct Goddard Earth Model (GEM) 10. A global comparison between a reference ellipsoid with flattening 1/298.257 and the geoid surface computed from the GEM 10 model shows long-wavelength geoid undulations (Fig. 2.25). The largest negative undulation (–105 m) is in the Indian Ocean south of India, and the largest positive undulation (+73 m) is in the equatorial Pacific Ocean north of Australia. These large-scale features are too broad to be ascribed to shallow crustal or lithospheric mass anomalies. They are thought to be due to heterogeneities that extend deep into the lower mantle, but their origin is not yet understood.

2.4.6 Satellite geodesy

Since the early 1960s knowledge of the geoid has been dramatically enhanced by the science of satellite geodesy.

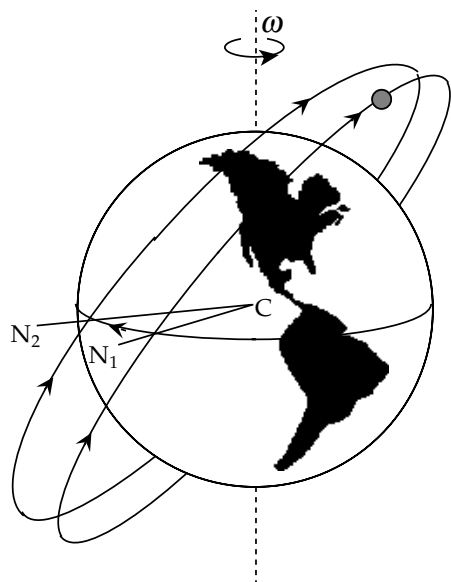


Fig. 2.26 The retrograde precession of a satellite orbit causes the line of nodes (CN₁, CN₂) to change position on successive equatorial crossings.

The motions of artificial satellites in Earth orbits are influenced by the Earth’s mass distribution. The most important interaction is the simple balance between the centrifugal force and the gravitational attraction of the Earth’s mass, which determines the radius of the satellite’s orbit. Analysis of the precession of the Earth’s rotation axis (Section 2.3.4.4) shows that it is determined by the dynamical ellipticity H , which depends on the difference between the principal moments of inertia resulting from the rotational flattening. In principle, the gravitational attraction of an artificial satellite on the Earth’s equatorial bulge also contributes to the precession, but the effect is too tiny to be measurable. However, the inverse attraction of the equatorial bulge on the satellite causes the orbit of the satellite to precess around the rotation axis. The plane of the orbit intersects the equatorial plane in the *line of nodes*. Let this be represented by the line CN₁ in Fig. 2.26. On the next passage of the satellite around the Earth the precession of the orbit has moved the nodal line to a new position CN₂. The orbital precession in this case is *retrograde*; the nodal line *regresses*. For a satellite orbiting in the same sense as the Earth’s rotation the longitude of the nodal line shifts gradually westward; if the orbital sense is opposite to the Earth’s rotation the longitude of the nodal line shifts gradually eastward. Because of the precession of its orbit the path of a satellite eventually covers the entire Earth between the north and south circles of latitude defined by the inclination of the orbit. The profusion of high-quality satellite data is the best source for calculating the dynamical ellipticity or the related parameter J_2 in the gravity potential. Observations of satellite orbits are so precise that small perturbations of the orbit can be related to the gravitational field and to the geoid.

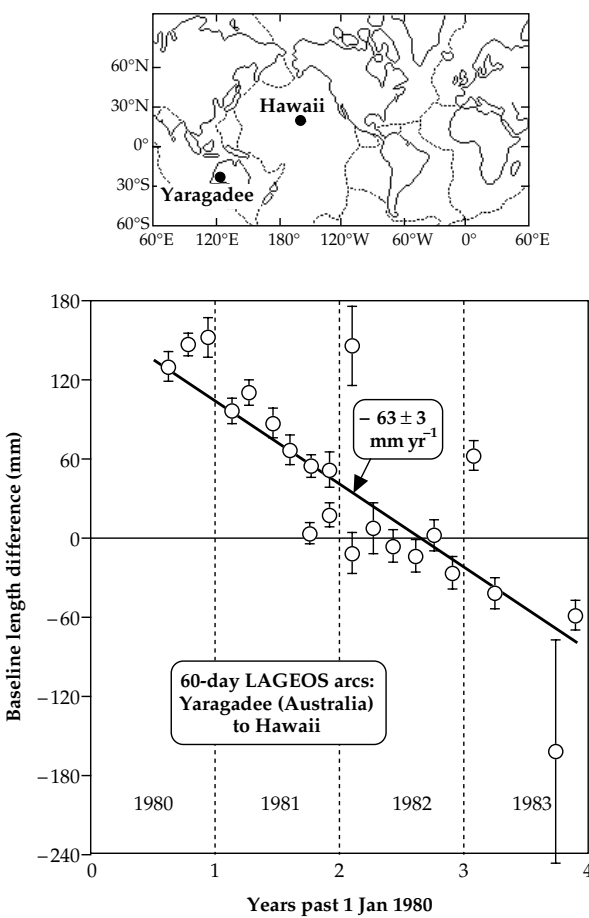


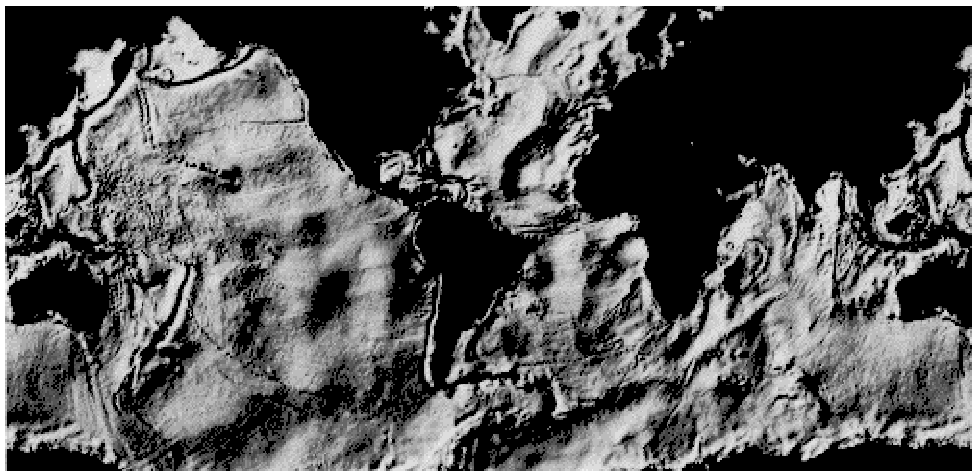
Fig. 2.27 Changes in the arc distance between satellite laser-ranging (SLR) stations in Australia and Hawaii determined from LAGEOS observations over a period of four years. The mean rate of convergence, $63 \pm 3 \text{ mm yr}^{-1}$, agrees well with the rate of 67 mm yr^{-1} deduced from plate tectonics (after Tapley *et al.*, 1985).

2.4.6.1 Satellite laser-ranging

The accurate tracking of a satellite orbit is achieved by satellite laser-ranging (SLR). The spherical surface of the target satellite is covered with numerous retro-reflectors. A retro-reflector consists of three orthogonal mirrors that form the corner of a cube; it reflects an incident beam of light back along its path. A brief pulse of laser light with a wavelength of 532 nm is sent from the tracking station on Earth to the satellite, and the two-way travel-time of the reflected pulse is measured. Knowing the speed of light, the distance of the satellite from the tracking station is obtained. The accuracy of a single range measurement is about 1 cm.

America’s *Laser Geodynamics Satellite* (LAGEOS 1) and France’s *Starlette* satellite have been tracked for many years. LAGEOS 1 flies at 5858–5958 km altitude, the inclination of its orbit is 110° (i.e., its orbital sense is opposite to the Earth’s rotation), and the nodal line of the orbit advances at 0.343° per day. Starlette flies at an altitude of 806–1108 km, its orbit is inclined at 50° , and its nodal line regresses at 3.95° per day.

Fig. 2.28 The mean sea surface as determined from SEASAT and GEOS-3 satellite altimetry, after removal of long-wavelength features of the GEM-10B geoid up to order and degree 12 (from Marsh *et al.*, 1992). The surface is portrayed as though illuminated from the northwest.



The track of a satellite is perturbed by many factors, including the Earth's gravity field, solar and lunar tidal effects, and atmospheric drag. The perturbing influences of these factors can be computed and allowed for. For the very high accuracy that has now been achieved in SLR results, variations in the coordinates of the tracking stations become detectable. The motion of the pole of rotation of the Earth can be deduced and the history of changes in position of the tracking station can be obtained. LAGEOS 1 was launched in 1976 and has been tracked by more than twenty laser-tracking stations on five tectonic plates. The relative changes in position between pairs of stations can be compared with the rates of plate tectonic motion deduced from marine geophysical data. For example, a profile from the Yarragadee tracking station in Australia and the tracking station in Hawaii crosses the converging plate boundary between the Indo-Australian and Pacific plates (Fig. 2.27). The results of four years of measurement show a decrease of the arc distance between the two stations at a rate of $63 \pm 3 \text{ mm yr}^{-1}$. This is in good agreement with the corresponding rate of 67 mm yr^{-1} inferred from the relative rotation of the tectonic plates.

2.4.6.2 Satellite altimetry

From satellite laser-ranging measurements the altitude of a spacecraft can be determined relative to the reference ellipsoid with a precision in the centimeter range. In satellite altimetry the tracked satellite carries a transmitter and receiver of microwave (radar) signals. A brief electromagnetic pulse is emitted from the spacecraft and reflected from the surface of the Earth. The two-way travel-time is converted using the speed of light to an estimate of the height of the satellite above the Earth's surface. The difference between the satellite's height above the ellipsoid and above the Earth's surface gives the height of the topography relative to the reference ellipsoid. The precision over land areas is poorer than over the oceans, but over smooth land features like deserts and inland water bodies an accuracy of better than a meter is achievable.

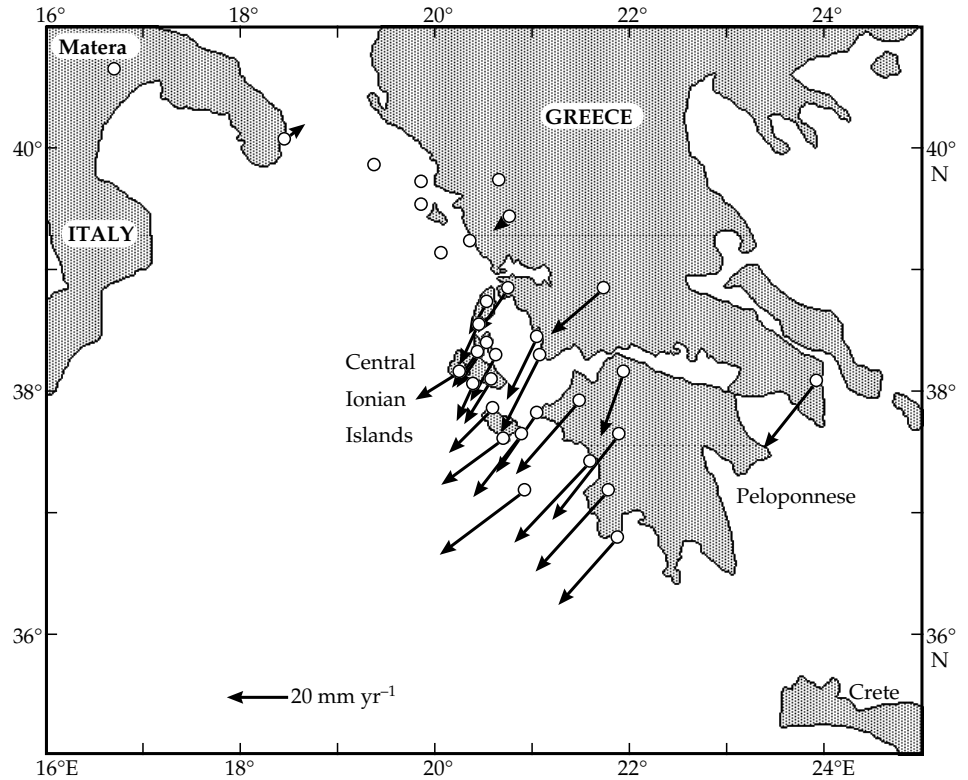
Satellite altimeters are best suited for marine surveys, where sub-meter accuracy is possible. The satellite GEOS-3 flew from 1975–1978, SEASAT was launched in 1978, and GEOSAT was launched in 1985. Specifically designed for marine geophysical studies, these satellite altimeters revealed remarkable aspects of the marine geoid. The long-wavelength geoid undulations (Fig. 2.25) have large amplitudes up to several tens of meters and are maintained by mantle-wide convection. The short-wavelength features are accentuated by removing the computed geoid elevation up to a known order and degree. The data are presented in a way that emphasizes the elevated and depressed areas of the sea surface (Fig. 2.28).

There is a strong correlation between the short-wavelength anomalies in elevation of the mean sea surface and features of the sea-floor topography. Over the ocean ridge systems and seamount chains the mean sea surface (geoid) is raised. The locations of fracture zones, in which one side is elevated relative to the other, are clearly discernible. Very dark areas mark the locations of deep ocean trenches, because the mass deficiency in a trench depresses the geoid. Seaward of the deep ocean trenches the mean sea surface is raised as a result of the upward flexure of the lithosphere before it plunges downward in a subduction zone.

2.4.6.3 Satellite-based global positioning systems (GPS)

Geodesy, the science of determining the three-dimensional coordinates of a position on the surface of the Earth, received an important boost with the advent of the satellite era. The first global satellite navigation system, the US Navy Navigation Satellite System known as TRANSIT consisted of six satellites in polar orbits about 1100 km above the surface of the Earth. Signals transmitted from these satellites were combined in a receiver on Earth with a signal generated at the same frequency in the receiver. Because of the motion of the satellite the frequency of its signal was modified by the Doppler effect and was thus slightly different from the receiver-generated signal, producing a beat frequency. Using the speed of light, the beat signal was

Fig. 2.29 Annual displacement rates in southeastern Italy, the Ionian Islands and western Greece relative to Matera (Italy), determined from GPS surveys in 1989 and 1992. The displacement arrows are much larger than the measurement errors, and indicate a distinct southwestward movement of western Greece relative to Italy (after Kahle *et al.*, 1995).



converted to the oblique distance between the satellite and receiver. By integrating the beat signal over a chosen time interval the change in range to the satellite in the interval was obtained. This was repeated several times. The orbit of the satellite was known precisely from tracking stations on the ground, and so the position of the receiver could be calculated. Originally developed to support ballistic missile submarines in the 1960s, the system was extended to civilian navigation purposes, especially for fixing the position of a ship at sea. The TRANSIT program was terminated in 1996, and succeeded by the more precise GPS program.

The Navigation Satellite Timing and Ranging Global Positioning System (NAVSTAR GPS, or, more commonly, just GPS) utilizes satellites in much higher orbits, at an altitude of around 20,200 km (i.e., a radial distance of 26,570 km), with an orbital period of half a sidereal day. The GPS system consists of 24 satellites. There are four satellites in each of six orbital planes, equally spaced at 60° intervals around the equator and inclined to the equator at about 55°. Between five and eight GPS satellites are visible at any time and location on Earth. Each satellite broadcasts its own predetermined position and reference signal every six seconds. The time difference between emission and reception on Earth gives the “pseudo-range” of the satellite, so-called because it must be corrected for errors in the clock of the receiver and for tropospheric refraction. Pseudo-range measurements to four or more satellites with known positions allows computation of the clock error and the exact position of the receiver. The precision with which a point can be located

depends on the quality of the receiver and signal processing. Low-cost single civilian-quality receivers have about 100 m positioning accuracy. In scientific and military missions a roving receiver is used in conjunction with a base station (a fixed receiver), and differential signal processing improves the accuracy of location to around 1 cm.

The GPS system allows very precise determination of changes in the distance between observation points. For example, a dense network of GPS measurements was made in southeastern Italy, the Ionian Islands and western Greece in 1989 and 1993. The differences between the two measuring campaigns show that southwestern Greece moved systematically to the southwest relative to Matera in southeastern Italy at mean annual rates of 20–40 mm yr⁻¹ (Fig. 2.29).

2.4.6.4 Measurement of gravity and the geoid from orbiting satellites

The equipotential surface of gravity, the geoid (Section 2.4.5), is characterized by undulations caused by inhomogeneous distribution of mass in the Earth. Until recently, construction of a global model of the geoid was very laborious, as it required combining data from many different sources of variable precision. Surface gravity measurements made on land or at sea were augmented by data from a large number of Earth-orbiting satellites. The resulting figure showed large-scale features (Fig. 2.25), but fine details were impossible to define accurately. Satellites in comparatively low orbits, a few hundreds of

kilometers above the Earth's surface, can now be used in conjunction with the GPS satellites orbiting at high altitudes (20,200 km) to measure the global gravity field and geoid with a precision that is several orders of magnitude better than was previously possible.

In 2000 the German CHAMP (Challenging Mini-satellite Payload) satellite was inserted into a nearly circular, almost polar orbit with an initial altitude of 450 km. At this altitude the thin atmosphere is still capable of exerting drag, which lowers the altitude of the satellite to about 300 km over a 5 year interval. Sensitive accelerometers on board the satellite allow correction for non-gravitational forces, such as atmospheric drag or the pressure of solar radiation. A highly precise GPS receiver on board the CHAMP satellite, using position data from up to 12 GPS satellites simultaneously, allows retrieval of CHAMP's position with an accuracy of a few centimeters. Whereas the orbits of earlier satellites were compiled from many comparatively short tracks measured when the satellite was in view of different ground stations, the CHAMP orbit is continuously tracked by the GPS satellites. Small perturbations of CHAMP's orbit may be tracked and modelled. The models of the Earth's gravity field and of the global geoid derived from CHAMP data were greatly improved in accuracy and definition over previous models.

Building on the experience gained from CHAMP, a joint American–German project, the Gravity Recovery and Climate Experiment (GRACE), was launched in 2002. The GRACE mission uses two nearly identical satellites in near-circular polar orbits (inclination 89.5° to the equator), initially about 500 km above Earth's surface. The twin satellites each carry on-board GPS receivers, which allow precise determination of their absolute positions over the Earth at any time. The satellites travel in tandem in the same orbital plane, separated by approximately 220 km along their track. Changes in gravity along the orbit are determined by observing small differences in the separation of the two satellites. This is achieved by using a highly accurate microwave ranging system. Each satellite carries a microwave antenna transmitting in the K-band frequency range (wavelength ~ 1 cm) and directed accurately at the other satellite. With this ranging system the separation of the two satellites can be measured with a precision of one micrometer ($1\ \mu\text{m}$).

As the satellite-pair orbits the Earth, it traverses variations in the gravity field due to the inhomogeneous mass distribution in the Earth. If there is a mass excess, the equipotential surface bulges upward, and gravity is enhanced locally. The leading satellite encounters this anomaly first and is accelerated away from the trailing satellite. Tiny changes in separation between the two satellites as they move along-track are detected by the accurate microwave ranging system. In conjunction with exact location of the satellite by the on-board GPS devices, the GRACE satellites provide fine-scale definition of the gravity field, and determination of the geoid from a single source. Moreover, the satellites measure the gravity field

completely in about 30 days. Thus, comparison of data from selected surveys of a region can reveal very small, time-dependent changes in gravity resulting, for example, from transient effects such as changes in groundwater level, or the melting of glaciers, in the observed region. Other instruments on board the GRACE satellites make further observations for atmospheric and ionospheric research.

2.4.6.5 Observation of crustal deformation with satellite-borne radar

Among the many satellites in Earth orbit, some (identified by acronyms such as ERS1, ERS2, JERS, IRS, RADARSAT, Envisat, etc.) are specifically designed to direct beams of radar waves at the Earth and record the reflections from the Earth's surface. Synthetic aperture radar (SAR) is a remote sensing technique that has made it possible to record features of the Earth's surface in remarkable detail based on these radar reflections. In a typical SAR investigation enormous amounts of radar data are gathered and subjected to complex data-processing. This requires massive computational power, and so is usually performed on the ground after the survey has been carried out.

Radar signals, like visible light, are subject to reflection, refraction and diffraction (these phenomena are explained in Section 3.6.2 for seismic waves). Diffraction (see Fig. 3.55) bends light passing through a lens in such a way that a point source becomes diffuse. When two adjacent point sources are observed with the lens, their diffuse images overlap and if they are very close, they may not be seen as distinct points. The resolving power of an optical instrument, such as a lens, is defined by the smallest angular separation (θ) of two points that the instrument can distinguish clearly. For a given lens this angle is dependent inversely on the diameter (d) of the aperture that allows light to pass through the lens, and directly on the wavelength (λ) of the light. It is given by the approximate relationship $\theta \sim \lambda/d$. High resolution requires that closely spaced details of the object can be distinguished, i.e., the angular resolution θ should be a small number. Thus, the larger the aperture of the lens, the higher is the optical resolution.

The same principle applies to radar. Instead of being dependent on the diameter of an optical lens, the resolution of a radar system is determined by the antenna length. When mounted in a satellite, the physical dimensions of the antenna are limited to just a few meters. SAR makes use of the motion of the antenna and powerful data-processing to get around this limitation.

The radar antenna is mounted so that it directs its beam at right angles to the direction of motion of the host spacecraft. The beam “illuminates” a swathe of ground surface, each particle of which reflects a signal to the antenna. Hundreds of radar pulses are sent out per second (e.g., the European Radar Satellites (ERS) emit 1700 pulses per

second); this results in huge amounts of reflected signals. As the craft moves forward, the illuminated swathe moves across the target surface. Each particle of the target reflects hundreds of radar pulses from the time when it is first energized until it is no longer covered by the beam. During this time the craft (and real antenna) move some distance along the track. In subsequent data-processing the signals reflected from the target are combined and corrected for the changing position of the antenna in such a way that they appear to have been gathered by an antenna as long as the distance moved along the track. This distance is called the *synthetic aperture* of the radar. For example, a SAR investigation with an ERS1 satellite in orbit 800 km above the Earth's surface created a synthetic aperture of about 4 km. The high resolving power achieved with this large aperture produced SAR images of ground features with a resolution of about 30 m.

An important aspect of the data reduction is the ability to reconstruct the path of each reflection precisely. This is achieved using the Doppler effect, the principle of which is described in Box 1.2. Reflections from target features ahead of the moving spacecraft have elevated frequencies; those from behind have lowered frequencies. Correcting the frequency of each signal for its Doppler shift is necessary to obtain the true geometry of the reflections.

A further development of the SAR method is *Interferometric SAR* (InSAR). This technique analyzes the phase of the reflected radar signal to determine small changes in topography between repeated passages of the satellite over an area. The phase of the wave is a measure of the time-delay the wave experiences in transit between transmitter and receiver. To illustrate this point, picture the shape of a waveform as an alternation of crests and troughs, which leaves the satellite at the instant its amplitude is maximum (i.e., at a crest). If the reflected signal returns to the satellite as a crest, it has the same phase as the transmitted signal. Its amplitude could be expressed by the equation $y = A \cos \omega t$. This will be the case if the path of the reflected wave to and from the target is an exact number of complete wavelengths. On the other hand, if the reflection arrives back at the satellite as a trough, it is exactly out-of-phase with the original wave. This happens when the length of its path is an odd number of half-wavelengths. More generally, the path length is not an exact even or odd number of half-wavelengths, and the equation of the reflected wave must be written $y = A \cos(\omega t + \Delta)$, where the phase difference Δ depends on the path length. The InSAR technique developed in the 1990s is based on analysis of the phases inherent in each reflection recorded by the satellite.

If a SAR image is made of a target area during one orbit, it should be reproduced exactly on a later orbit that revisits the same location (this is not exactly possible, but paths that repeat within a few hundred meters can be corrected for geometric differences). In particular, because each point of the target is the same distance from the trans-

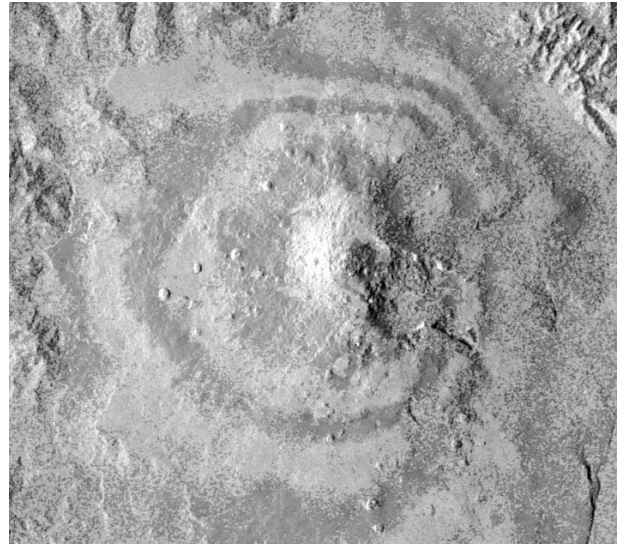


Fig. 2.30 Interferometric Synthetic Aperture Radar (InSAR) pattern of interference fringes showing changes in elevation of Mount Etna, Sicily, following the 1992–1993 eruptive cycle. The four pairs of light and dark fringes correspond to subsidence of the mountaintop by about 11 cm as magma drains out of the volcano (after Massonnet, 1997).

mitter, the phases of the imaged signals should be identical. However, if geological events have caused surface displacements between the times of the two images there will be phase differences between the two images. These are made visible by combining the two images so that they interfere with each other.

When harmonic signals with different phases are mixed, they interfere with each other. Constructive interference occurs when the signals have the same phase; if they are superposed, the combined signal is strengthened. Destructive interference occurs when out-of-phase signals are mixed; the combined signal is weakened. The interference pattern that results from mixing the two waveforms consists of alternating zones of reinforcement and reduction of the signal, forming a sequence of so-called “interference fringes.” The use of color greatly enhances the visual impact of the interference fringes.

When this procedure is carried out with SAR images, the resulting interference pattern makes it possible to interpret ground motion of large areas in much greater detail than would be possible from ground-based observations. The method has been used to record various large-scale ground displacements related, for example, to earthquakes, tectonic faulting, and volcanism. Figure 2.30 shows an interference pattern superposed on the background topography of Mount Etna, in Sicily, following a cycle of eruptions in 1992 and 1993. Successive radar images from a common vantage point were obtained 13 months apart by the ERS1 satellite, which transmitted radar signals with wavelength 5.66 cm. In order to change the along-path distance to and from the target by a full wavelength, the ground must move by a half-wavelength perpendicular to the path, in this case by 2.83 cm. The concentric dark and light fringes around the

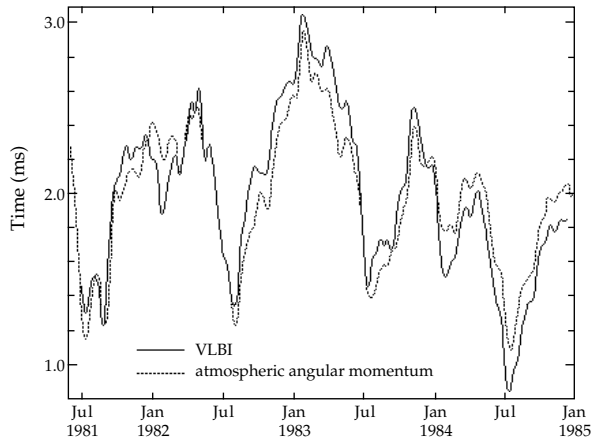


Fig. 2.31 Fine-scale fluctuations in the LOD observed by VLBI, and LOD variations expected from changes in the angular momentum of the atmosphere (after Carter, 1989).

crater show four cycles of interference, corresponding to a change in elevation of the mountaintop of about 11 cm. The fringes result from the subsidence of the crater as magma drained out of it following the eruptive cycle.

2.4.6.6 Very long baseline interferometry

Extra-galactic radio sources (quasars) form the most stable inertial coordinate system yet known for geodetic measurements. The extra-galactic radio signals are detected almost simultaneously by radio-astronomy antennas at observatories on different continents. Knowing the direction of the incoming signal, the small differences in times of arrival of the signal wavefronts at the various stations are processed to give the lengths of the baselines between pairs of stations. This highly precise geodetic technique, called Very Long Baseline Interferometry (VLBI), allows determination of the separation of observatories several thousand kilometers apart with an accuracy of a few centimeters. Although not strictly a satellite-based technique, it is included in this section because of its use of non-terrestrial signals for high resolution geodetic measurements.

By combining VLBI observations from different stations the orientation of the Earth to the extra-galactic inertial coordinate system of the radio sources is obtained. Repeated determinations yield a record of the Earth's orientation and rotational rate with unprecedented accuracy. Motion of the rotation axis (e.g., the Chandler wobble, Section 2.3.4.3) can be described optically with a resolution of 0.5–1 m; the VLBI data have an accuracy of 3–5 cm. The period of angular rotation can be determined to better than 0.1 millisecond. This has enabled very accurate observation of irregularities in the rotational rate of the Earth, which are manifest as changes in the length of the day (LOD).

The most important, first-order changes in the LOD are due to the braking of the Earth's rotation by the lunar and solar marine tides (Section 2.3.4.1). The most significant

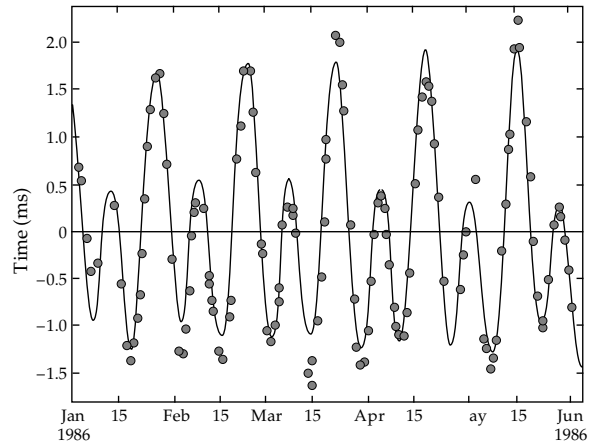


Fig. 2.32 High-frequency changes in the LOD after correction for the effects due to atmospheric angular momentum (points) and the theoretical variations expected from the solid bodily Earth-tides (after Carter, 1989).

non-tidal LOD variations are associated with changes in the angular momentum of the atmosphere due to shifts in the east–west component of the wind patterns. To conserve the total angular momentum of the Earth a change in the angular momentum of the atmosphere must be compensated by an equal and opposite change in the angular momentum of the crust and mantle. The largely seasonal transfers of angular momentum correlate well with high-frequency variations in the LOD obtained from VLBI results (Fig. 2.31).

If the effects of marine tidal braking and non-tidal transfers of atmospheric angular momentum variations are taken into account, small residual deviations in the LOD remain. These are related to the tides in the solid Earth (Section 2.3.3.5). The lunar and solar tidal forces deform the Earth elastically and change its ellipticity slightly. The readjustment of the mass distribution necessitates a corresponding change in the Earth's rate of rotation in order to conserve angular momentum. The expected changes in LOD due to the influence of tides in the solid Earth can be computed. The discrepancies in LOD values determined from VLBI results agree well with the fluctuations predicted by the theory of the bodily Earth-tides (Fig. 2.32).

2.5 GRAVITY ANOMALIES

2.5.1 Introduction

The mean value of gravity at the surface of the Earth is approximately 9.80 m s^{-2} , or 980,000 mgal. The Earth's rotation and flattening cause gravity to increase by roughly 5300 mgal from equator to pole, which is a variation of only about 0.5%. Accordingly, measurements of gravity are of two types. The first corresponds to determination of the absolute magnitude of gravity at any place; the second consists of measuring the change in gravity from one place to another. In geophysical studies, especially in gravity prospecting, it is necessary to measure accurately the small

changes in gravity caused by underground structures. These require an instrumental sensitivity of the order of 0.01 mgal. It is very difficult to design an instrument to measure the absolute value of gravity that has this high precision and that is also portable enough to be used easily in different places. Gravity surveying is usually carried out with a portable instrument called a *gravimeter*, which determines the variation of gravity relative to one or more reference locations. In national gravity surveys the relative variations determined with a gravimeter may be converted to absolute values by calibration with absolute measurements made at selected stations.

2.5.2 Absolute measurement of gravity

The classical method of measuring gravity is with a pendulum. A simple pendulum consists of a heavy weight suspended at the end of a thin fiber. The compound (or reversible) pendulum, first described by Henry Kater in 1818, allows more exact measurements. It consists of a stiff metal or quartz rod, about 50 cm long, to which is attached a movable mass. Near each end of the rod is fixed a pivot, which consists of a quartz knife-edge resting on a flat quartz plane. The period of the pendulum is measured for oscillations about one of the pivots. The pendulum is then inverted and its period about the other pivot is determined. The position of the movable mass is adjusted until the periods about the two pivots are equal. The distance L between the pivots is then measured accurately. The period of the instrument is given by

$$T = 2\pi\sqrt{\frac{I}{mgh}} = 2\pi\sqrt{\frac{L}{g}} \quad (2.60)$$

where I is the moment of inertia of the pendulum about a pivot, h is the distance of the center of mass from the pivot, and m is the mass of the pendulum. Knowing the length L from Kater's method obviates knowledge of I , m and h .

The sensitivity of the compound pendulum is found by differentiating Eq. (2.60). This gives

$$\frac{\Delta g}{g} = -2\frac{\Delta T}{T} \quad (2.61)$$

To obtain a sensitivity of about 1 mgal it is necessary to determine the period with an accuracy of about 0.5 μ s. This can be achieved easily today with precise atomic clocks. The compound pendulum was the main instrument for gravity prospecting in the 1930s, when timing the swings precisely was more difficult. It was necessary to time as accurately as possible a very large number of swings. As a result a single gravity measurement took about half an hour.

The performance of the instrument was handicapped by several factors. The inertial reaction of the housing to the swinging mass of the pendulum was compensated by mounting two pendulums on the same frame and swinging them in opposite phase. Air resistance was reduced by

housing the pendulum assemblage in an evacuated thermostatically controlled chamber. Friction in the pivot was minimized by the quartz knife-edge and plane, but due to minor unevenness the contact edge was not exactly repeatable if the assemblage was set up in a different location, which affected the reliability of the measurements. The apparatus was bulky but was used until the 1950s as the main method of making absolute gravity measurements.

2.5.2.1 Free-fall method

Modern methods of determining the acceleration of gravity are based on observations of falling objects. For an object that falls from a starting position z_0 with initial velocity u the equation of motion gives the position z at time t as

$$z = z_0 + ut + \frac{1}{2}gt^2 \quad (2.62)$$

The absolute value of gravity is obtained by fitting a quadratic to the record of position versus time.

An important element in modern experiments is the accurate measurement of the change of position with a Michelson interferometer. In this device a beam of monochromatic light passes through a beam splitter, consisting of a semi-silvered mirror, which reflects half of the light incident upon it and transmits the other half. This divides the incident ray into two subrays, which subsequently travel along different paths and are then recombined to give an interference pattern. If the path lengths differ by a full wavelength (or a number of full wavelengths) of the monochromatic light, the interference is constructive. The recombined light has maximum intensity, giving a bright interference fringe. If the path lengths differ by half a wavelength (or by an odd number of half-wavelengths) the recombined beams interfere destructively, giving a dark fringe. In modern experiments the monochromatic light source is a laser beam of accurately known wavelength.

In an absolute measurement of gravity a laser beam is split along two paths that form a Michelson interferometer (Fig. 2.33). The horizontal path is of fixed length, while the vertical path is reflected off a corner-cube retro-reflector that is released at a known instant and falls freely. The path of free-fall is about 0.5 m long. The cube falls in an evacuated chamber to minimize air resistance. A photo-multiplier and counter permit the fringes within any time interval to be recorded and counted. The intensity of the recombined light fluctuates sinusoidally with increasing frequency the further and faster the cube falls. The distance between each zero crossing corresponds to half the wavelength of the laser light, and so the distance travelled by the falling cube in any time interval may be obtained. The times of the zero crossings must be measured with an accuracy of 0.1 ns (10^{-10} s) to give an accuracy of 1 μ gal in the gravity measurement.

Although the apparatus is compact, it is not quite portable enough for gravity surveying. It gives measure-

Fig. 2.33 The free-fall method of measuring absolute gravity.

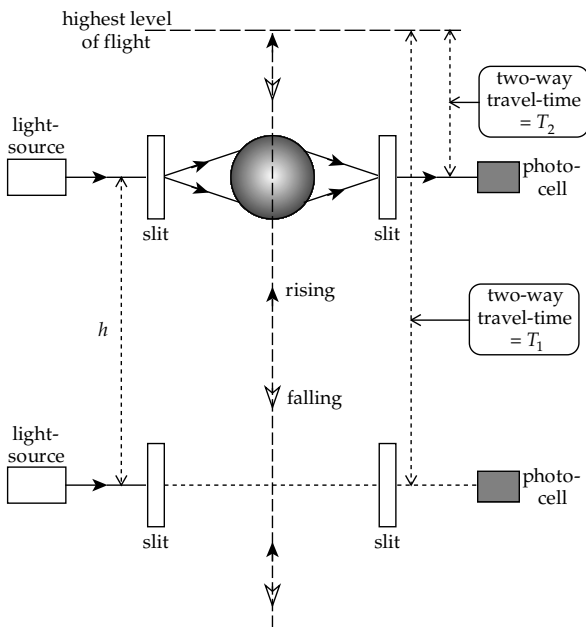
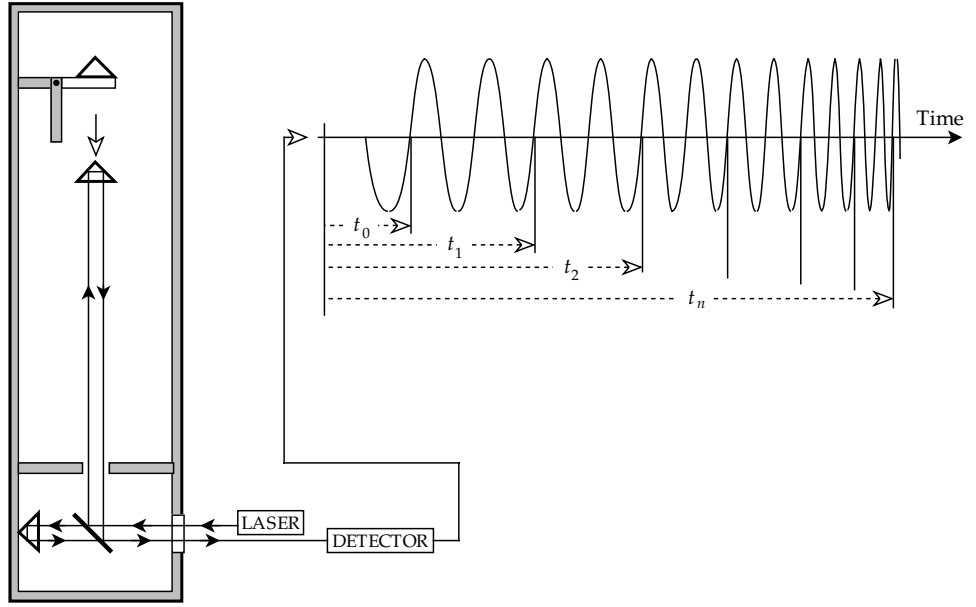


Fig. 2.34 The rise-and-fall method of measuring absolute gravity.

ments of the absolute value of gravity with an accuracy of about 0.005–0.010 mgal (5–10 μ gal). A disadvantage of the free-fall method is the resistance of the residual air molecules left in the evacuated chamber. This effect is reduced by placing the retro-reflector in a chamber that falls simultaneously with the cube, so that in effect the cube falls in still air. Air resistance is further reduced in the rise-and-fall method.

2.5.2.2 Rise-and-fall method

In the original version of the rise-and-fall method a glass sphere was fired vertically upward and fell back along the same path (Fig. 2.34). Timing devices at two different levels

registered the times of passage of the ball on the upward and downward paths. In each timer a light beam passed through a narrow slit. As the glass sphere passed the slit it acted as a lens and focussed one slit on the other. A photo-multiplier and detector registered the exact passage of the ball past the timing level on the upward and downward paths. The distance h between the two timing levels (around 1 m) was measured accurately by optical interferometry.

Let the time spent by the sphere above the first timing level be T_1 and the time above the second level be T_2 ; further, let the distances from the zenith level to the timing levels be z_1 and z_2 , respectively. The corresponding times of fall are $t_1 = T_1/2$ and $t_2 = T_2/2$. Then,

$$z_1 = \frac{1}{2} g \left(\frac{T_1}{2} \right)^2 \quad (2.63)$$

with a similar expression for the second timing level. Their separation is

$$h = z_1 - z_2 = \frac{1}{8} g (T_1^2 - T_2^2) \quad (2.64)$$

The following elegantly simple expression for the value of gravity is obtained:

$$g = \frac{8h}{(T_1^2 - T_2^2)} \quad (2.65)$$

Although the experiment is conducted in a high vacuum, the few remaining air molecules cause a drag that opposes the motion. On the upward path the air drag is downward, in the same direction as gravity; on the downward path the air drag is upward, opposite to the direction of gravity. This asymmetry helps to minimize the effects of air resistance.

In a modern variation Michelson interferometry is used as in the free-fall method. The projectile is a corner-cube

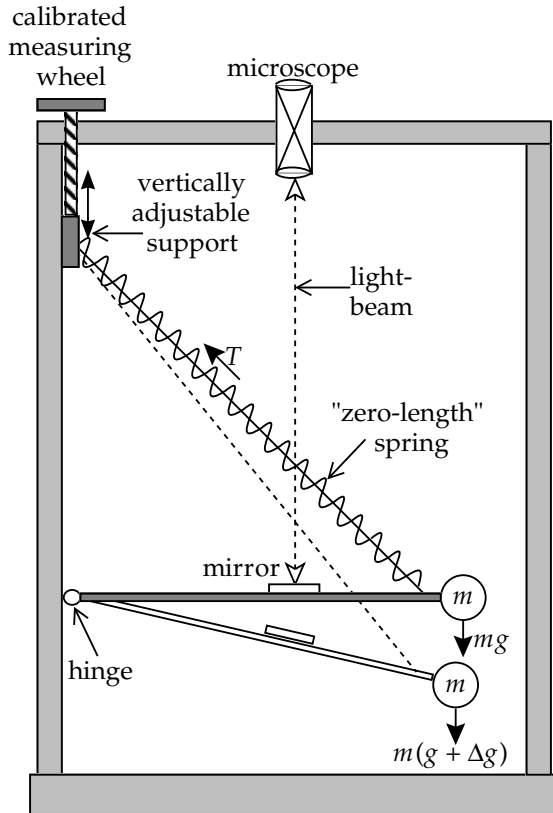


Fig. 2.35 The principle of operation of an unstable (astatic) type of gravimeter.

retro-reflector, and interference fringes are observed and counted during its upward and downward paths. Sensitivity and accuracy are comparable to those of the free-fall method.

2.5.3 Relative measurement of gravity: the gravimeter

In principle, a gravity meter, or *gravimeter*, is a very sensitive balance. The first gravimeters were based on the straightforward application of Hooke's law (Section 3.2.1). A mass m suspended from a spring of length s_0 causes it to stretch to a new length s . The extension, or change in length, of the spring is proportional to the restoring force of the spring and so to the value of gravity, according to:

$$F = mg = -k(s - s_0) \quad (2.66)$$

where k is the elastic constant of the spring. The gravimeter is calibrated at a known location. If gravity is different at another location, the extension of the spring changes, and from this the change in gravity can be computed.

This type of gravimeter, based directly on Hooke's law, is called a stable type. It has been replaced by more sensitive unstable or astaticized types, which are constructed so that an additional force acts in the same direction as gravity and opposes the restoring force of the spring. The instrument is then in a state of unstable equilibrium. This

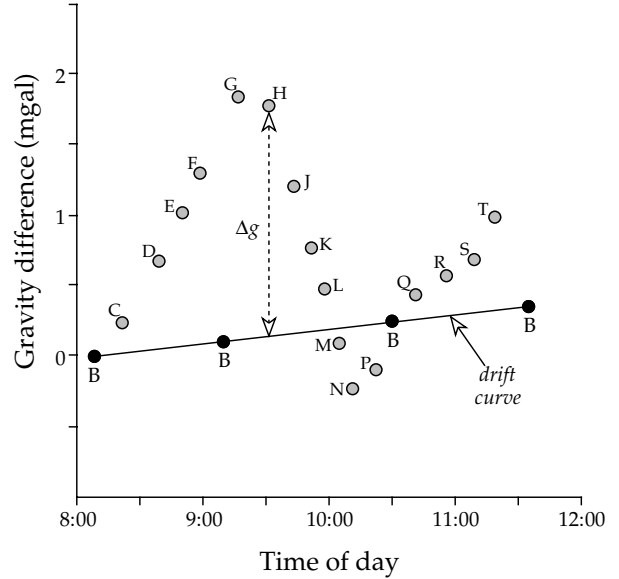


Fig. 2.36 Compensation of gravity readings for instrumental drift. The gravity stations B–T are occupied in sequence at known times. The repeated measurements at the base station B allow a drift correction to be made to the gravity readings at the other stations.

condition is realized through the design of the spring. If the natural length s_0 can be made as small as possible, ideally zero, Eq. (2.66) shows that the restoring force is then proportional to the physical length of the spring instead of its extension. The *zero-length spring*, first introduced in the LaCoste–Romberg gravimeter, is now a common element in modern gravimeters. The spring is usually of the helical type. When a helical spring is stretched, the fiber of the spring is twisted; the total twist along the length of the fiber equals the extension of the spring as a whole. During manufacture of a zero-length spring the helical spring is given an extra twist, so that its tendency is to uncoil. An increase in gravity stretches the spring against its restoring force, and the extension is augmented by the built-in pre-tension.

The operation of a gravimeter is illustrated in Fig. 2.35. A mass is supported by a horizontal rod to which a mirror is attached. The position of the rod is observed with a light-beam reflected into a microscope. If gravity changes, the zero-length spring is extended or shortened and the position of the rod is altered, which deflects the light-beam. The null-deflection principle is utilized. An adjusting screw changes the position of the upper attachment of the spring, which alters its tension and restores the rod to its original horizontal position as detected by the light-beam and microscope. The turns of the adjusting screw are calibrated in units of the change in gravity, usually in mgal.

The gravimeter is light, robust and portable. After initially levelling the instrument, an accurate measurement of a gravity difference can be made in a few minutes. The gravimeter has a sensitivity of about 0.01 mgal (10 μ gal). This high sensitivity makes it susceptible to small changes in its own properties.

2.5.3.1 Gravity surveying

If a gravimeter is set up at a given place and monitored for an hour or so, the repeated readings are found to vary smoothly with time. The changes amount to several hundredths of a mgal. The *instrumental drift* is partly due to thermally induced changes in the elastic properties of the gravimeter spring, which are minimized by housing the critical elements in an evacuated chamber. In addition, the elastic properties of the spring are not perfect, but creep slowly with time. The effect is small in modern gravimeters and can be compensated by making a *drift correction*. This is obtained by repeated occupation of some measurement stations at intervals during the day (Fig. 2.36). Gravity readings at other stations are adjusted by comparison with the drift curve. In order to make this correction the time of each measurement must be noted.

During the day, while measurements are being made, the gravimeter is subject to tidal attraction, including vertical displacement due to the bodily Earth-tides. The theory of the tides is known well (see Section 2.3.3) and their time-dependent effect on gravity can be computed precisely for any place on Earth at any time. Again, the *tidal correction* requires that the time of each measurement be known.

The goal of gravity surveying is to locate and describe subsurface structures from the gravity effects caused by their anomalous densities. Most commonly, gravimeter measurements are made at a network of stations, spaced according to the purpose of the survey. In environmental studies a detailed high-resolution investigation of the gravity expression of a small area requires small distances of a few meters between measurement stations. In regional gravity surveys, as used for the definition of hidden structures of prospective commercial interest, the distance between stations may be several kilometers. If the area surveyed is not too large, a suitable site is selected as base station (or reference site), and the gravity differences between the surveyed sites and this site are measured. In a gravity survey on a national scale, the gravity differences may be determined relative to a site where the absolute value of gravity is known.

2.5.4 Correction of gravity measurements

If the interior of the Earth were uniform, the value of gravity on the international reference ellipsoid would vary with latitude according to the normal gravity formula (Eq. (2.56)). This provides us with a reference value for gravity measurements. In practice, it is not possible to measure gravity on the ellipsoid at the place where the reference value is known. The elevation of a measurement station may be hundreds of meters above or below the ellipsoid. Moreover, the gravity station may be surrounded by mountains and valleys that perturb the measurement. For example, let P and Q represent gravity stations at different elevations in hilly terrain (Fig. 2.37a).

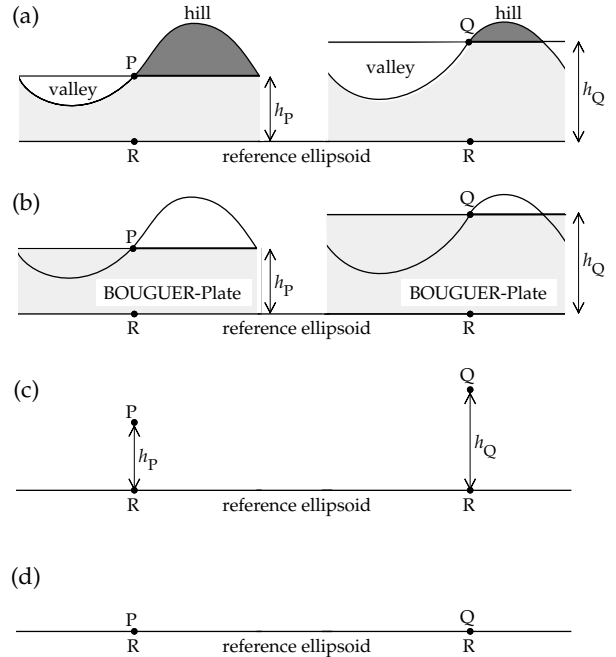


Fig. 2.37 After (a) terrain corrections, (b) the Bouguer plate correction, and (c) the free-air correction, the gravity measurements at stations P and Q can be compared to (d) the theoretical gravity at R on the reference ellipsoid.

The theoretical value of gravity is computed at the points R on the reference ellipsoid below P and Q. Thus, we must correct the measured gravity before it can be compared with the reference value.

The hill-top adjacent to stations P and Q has a center of mass that lies higher than the measurement elevation (Fig. 2.37a). The gravimeter measures gravity in the vertical direction, along the local plumb-line. The mass of the hill-top above P attracts the gravimeter and causes an acceleration with a vertically upward component at P. The measured gravity is reduced by the presence of the hill-top; to compensate for this a *terrain* (or *topographic*) *correction* is calculated and *added* to the measured gravity. A similar effect is observed at Q, but the hill-top above Q is smaller and the corresponding terrain correction is smaller. These corrections effectively level the topography to the same elevation as the gravity station. The presence of a valley next to each measurement station also requires a terrain correction. In this case, imagine that we could fill the valley up to the level of each station with rock of the same density ρ as under P and Q. The downward attraction on the gravimeter would be increased, so the terrain correction for a valley must also be *added* to the measured gravity, just as for a hill. Removing the effects of the topography around a gravity station requires making positive terrain corrections (Δg_T) for both hills and valleys.

After levelling the topography there is now a fictive uniform layer of rock with density ρ between the gravity station and the reference ellipsoid (Fig. 2.37b). The gravitational acceleration of this rock-mass is included in the

measured gravity and must be removed before we can compare with the theoretical gravity. The layer is taken to be a flat disk or plate of thickness h_p or h_Q under each station; it is called the *Bouguer plate*. Its gravitational acceleration can be computed for known thickness and density ρ , and gives a *Bouguer plate correction* (Δg_{BP}) that must be subtracted from the measured gravity, if the gravity station is above sea-level. Note that, if the gravity station is below sea-level, we have to fill the space above it up to sea-level with rock of density ρ ; this requires increasing the measured gravity correspondingly. The Bouguer plate correction (Δg_{BP}) is negative if the station is above sea-level but positive if it is below sea-level. Its size depends on the density of the local rocks, but typically amounts to about 0.1 mgal m^{-1} .

Finally, we must compensate the measured gravity for the elevation h_p or h_Q of the gravity station above the ellipsoid (Fig. 2.37c). The main part of gravity is due to gravitational attraction, which decreases proportionately to the inverse square of distance from the center of the Earth. The gravity measured at P or Q is smaller than it would be if measured on the ellipsoid at R. A *free-air correction* (Δg_{FA}) for the elevation of the station must be added to the measured gravity. This correction ignores the effects of material between the measurement and reference levels, as this is taken care of in Δg_{BP} . Note that, if the gravity station were below sea-level, the gravitational part of the measured gravity would be too large by comparison with the reference ellipsoid; we would need to subtract Δg_{FA} in this case. The free-air correction is positive if the station is above sea-level but negative if it is below sea-level (as might be the case in Death Valley or beside the Dead Sea). It amounts to about 0.3 mgal m^{-1} .

The free-air correction is always of opposite sense to the Bouguer plate correction. For convenience, the two are often combined in a single *elevation correction*, which amounts to about 0.2 mgal m^{-1} . This must be added for gravity stations above sea-level and subtracted if gravity is measured below sea-level. In addition, a *tidal correction* (Δg_{tide}) must be made (Section 2.3.3), and, if gravity is measured in a moving vehicle, the *Eötvös correction* (Section 2.3.5) is also necessary.

After correction the measured gravity can be compared with the theoretical gravity on the ellipsoid (Fig. 2.37d). Note that the above procedure reduces the measured gravity to the surface of the ellipsoid. In principle it is equally valid to correct the theoretical gravity from the ellipsoid upward to the level where the measurement was made. This method is preferred in more advanced types of analysis of gravity anomalies where the possibility of an anomalous mass between the ellipsoid and ground surface must be taken into account.

2.5.4.1 Latitude correction

The theoretical gravity at a given latitude is given by the normal gravity formula (Eq. 2.56). If the measured

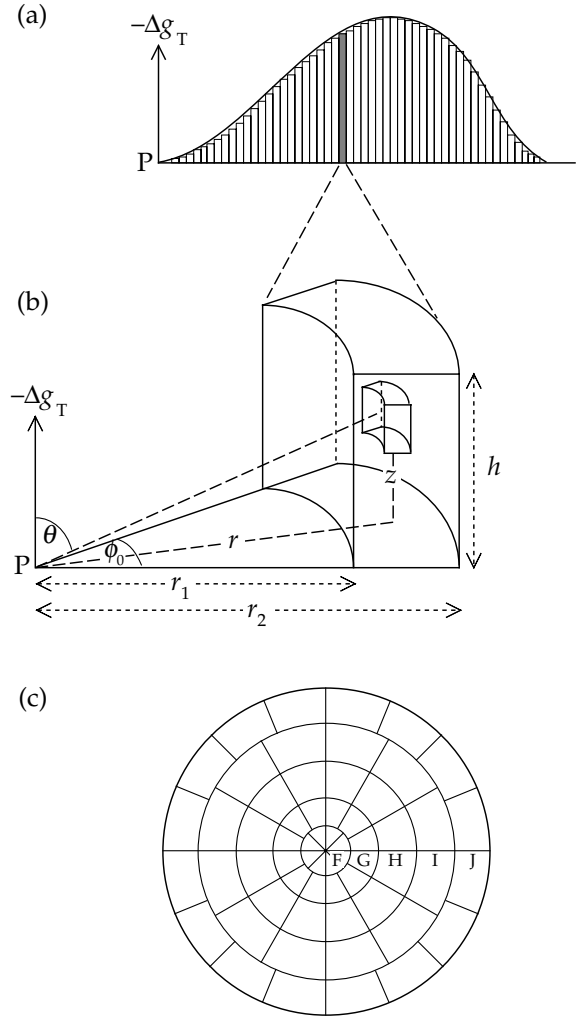


Fig. 2.38 Terrain corrections Δg_T are made by (a) dividing the topography into vertical elements, (b) computing the correction for each cylindrical element according to its height above or below the measurement station, and (c) adding up the contributions for all elements around the station with the aid of a transparent overlay on a topographic map.

gravity is an absolute value, the correction for latitude is made by subtracting the value predicted by this formula. Often, however, the gravity survey is made with a gravimeter, and the quantity measured, g_m , is the gravity difference relative to a base station. The normal reference gravity g_n may then be replaced by a latitude correction, obtained by differentiating Eq. (2.56):

$$\frac{\partial g_n}{\partial \lambda} = g_e(\beta_1 \sin 2\lambda + \beta_2 \sin 4\lambda) \quad (2.67)$$

After converting $\partial \lambda$ from radians to kilometers and neglecting the β_2 term, the latitude correction (Δg_{lat}) is $0.8140 \sin 2\lambda \text{ mgal}$ per kilometer of north–south displacement. Because gravity decreases towards the poles, the correction for stations closer to the pole than the base station must be added to the measured gravity.

2.5.4.2 Terrain corrections

The terrain correction (Δg_T) for a hill adjacent to a gravity station is computed by dividing the hill into a number of vertical prisms (Fig. 2.38a). The contribution of each vertical element to the vertical acceleration at the point of observation P is calculated by assuming cylindrical symmetry about P. The height of the prism is h , its inner and outer radii are r_1 and r_2 , respectively, the angle subtended at P is ϕ_0 , and the density of the hill is ρ (Fig. 2.38b). Let the sides of a small cylindrical element be dr , dz and $r d\phi$; its mass is $dm = \rho r d\phi dr dz$ and its contribution to the upward acceleration caused by the prism at P is

$$\Delta g = G \frac{dm}{(r^2 + z^2)^{3/2}} \cos \theta = G \frac{\rho r dr dz d\phi}{(r^2 + z^2)^{3/2}} \frac{z}{\sqrt{r^2 + z^2}} \quad (2.68)$$

Combining and rearranging terms and the order of integration gives the upward acceleration at P due to the cylindrical prism:

$$\Delta g_T = G\rho \int_{\phi=0}^{\phi_0} d\phi \int_{r=r_1}^{r_2} \left(\int_{z=0}^h \frac{z dz}{(r^2 + z^2)^{3/2}} \right) r dr \quad (2.69)$$

The integration over ϕ gives ϕ_0 ; after further integration over z we get:

$$\Delta g_T = G\rho\phi_0 \int_{r=r_1}^{r_2} \left(\frac{r}{\sqrt{r^2 + h^2}} - 1 \right) dr \quad (2.70)$$

Integration over r gives the upward acceleration produced at P by the cylinder:

$$\Delta g_T = G\rho\phi_0 \left((\sqrt{r^2 + h^2} - r_1) - (\sqrt{r^2 + h^2} - r_2) \right) \quad (2.71)$$

The direction of Δg_T in Fig. 2.38b is upward, opposite to gravity; the corresponding terrain correction must be *added* to the measured gravity.

In practice, terrain corrections can be made using a terrain chart (Fig. 2.38c) on which concentric circles and radial lines divide the area around the gravity station into sectors that have radial symmetry like the cross-section of the element of a vertical cylinder in Fig. 2.38b. The inner and outer radii of each sector correspond to r_1 and r_2 , and the angle subtended by the sector is ϕ . The terrain correction for each sector within each zone is pre-calculated using Eq. (2.71) and tabulated. The chart is drawn on a transparent sheet that is overlaid on a topographic map at the same scale and centered on the gravity station. The mean elevation within each sector is estimated as accurately as possible, and the elevation difference (i.e., h in Eq. (2.71)) of the sector relative to the station is computed. This is multiplied by the correction factor for the sector to give its contribution to the terrain correction. Finally, the terrain correction at the gravity station is obtained by summing up the contributions of all sectors. The procedure must be repeated for each gravity station. When the terrain chart is centered on a new station, the

mean topographic relief within each sector changes and must be computed anew. As a result, terrain corrections are time consuming and tedious. The most important effects come from the topography nearest to the station. However, terrain corrections are generally necessary if a topographic difference within a sector is more than about 5% of its distance from the station.

2.5.4.3 Bouguer plate correction

The Bouguer plate correction (Δg_{BP}) compensates for the effect of a layer of rock whose thickness corresponds to the elevation difference between the measurement and reference levels. This is modelled by a solid disk of density ρ and infinite radius centered at the gravity station P. The correction is computed by extension of the calculation for the terrain correction. An elemental cylindrical prism is defined as in Fig. 2.38b. Let the angle ϕ subtended by the prism increase to 2π and the inner radius decrease to zero; the first term in brackets in Eq. (2.71) reduces to h . The gravitational acceleration at the center of a solid disk of radius r is then

$$\Delta g_T = 2\pi G\rho(h - (\sqrt{r^2 + h^2} - r)) \quad (2.72)$$

Now let the radius r of the disk increase. The value of h gradually becomes insignificant compared to r ; in the limit, when r is infinite, the second term in Eq. (2.72) tends to zero. Thus, the Bouguer plate correction (Δg_{BP}) is given by

$$\Delta g_{BP} = 2\pi G\rho h \quad (2.73)$$

Inserting numerical values gives $0.0419 \times 10^{-3} \rho \text{ mgal m}^{-1}$ for Δg_{BP} , where the density ρ is in kg m^{-3} (see Section 2.5.5). The correct choice of density is very important in computing Δg_{BP} and Δg_T . Some methods of determining the optimum choice are described in detail below.

An additional consideration is necessary in marine gravity surveys. Δg_{BP} requires uniform density below the surface of the reference ellipsoid. To compute Δg_{BP} over an oceanic region we must in effect replace the sea-water with rock of density ρ . However, the measured gravity contains a component due to the attraction of the sea-water (density 1030 kg m^{-3}) in the ocean basin. The Bouguer plate correction in marine gravity surveys is therefore made by replacing the density ρ in Eq. (2.73) by $(\rho - 1030) \text{ kg m}^{-3}$. When a shipboard gravity survey is made over a large deep lake, a similar allowance must be made for the depth of water in the lake using an assumed density of $(\rho - 1000) \text{ kg m}^{-3}$.

2.5.4.4 Free-air correction

The free-air correction (Δg_{FA}) has a rather colorful, but slightly misleading title, giving the impression that the measurement station is floating in air above the ellipsoid. The density of air at standard temperature and pressure is around 1.3 kg m^{-3} and a mass of air between

the observation and reference levels would cause a detectable gravity effect of about 50 μgal at an elevation of 1000 m. In fact, the free-air correction pays no attention to the density of the material between the measurement elevation and the ellipsoid. It is a straightforward correction for the decrease of gravitational acceleration with distance from the center of the Earth:

$$\frac{\partial g}{\partial r} = \frac{\partial}{\partial r} \left(-G \frac{E}{r^2} \right) = +2G \frac{E}{r^3} = -\frac{2}{r}g \quad (2.74)$$

On substituting the Earth's radius (6371 km) for r and the mean value of gravity (981,000 mgal) for g , the value of Δg_{FA} is found to be 0.3086 mgal m^{-1} .

2.5.4.5 Combined elevation correction

The free-air and Bouguer plate corrections are often combined into a single elevation correction, which is $(0.3086 - (0.0419\rho \times 10^{-3})) \text{ mgal m}^{-1}$. Substituting a typical density for crustal rocks, usually taken to be 2670 kg m^{-3} , gives a combined elevation correction of 0.197 mgal m^{-1} . This must be added to the measured gravity if the gravity station is above the ellipsoid and subtracted if it is below.

The high sensitivity of modern gravimeters allows an achievable accuracy of 0.01–0.02 mgal in modern gravity surveys. To achieve this accuracy the corrections for the variations of gravity with latitude and elevation must be made very exactly. This requires that the precise coordinates of a gravity station must be determined by accurate geodetic surveying. The necessary precision of horizontal positioning is indicated by the latitude correction. This is maximum at 45° latitude, where, in order to achieve a survey accuracy of $\pm 0.01 \text{ mgal}$, the north–south positions of gravity stations must be known to about $\pm 10 \text{ m}$. The requisite precision in vertical positioning is indicated by the combined elevation correction of 0.2 mgal m^{-1} . To achieve a survey accuracy of $\pm 0.01 \text{ mgal}$ the elevation of the gravimeter above the reference ellipsoid must be known to about $\pm 5 \text{ cm}$.

The elevation of a site above the ellipsoid is often taken to be its altitude above mean sea-level. However, mean sea-level is equated with the geoid and not with the ellipsoid. Geoid undulations can amount to tens of meters (Section 2.4.5.1). They are long-wavelength features. Within a local survey the distance between geoid and ellipsoid is unlikely to vary much, and the gravity differences from the selected base station are unlikely to be strongly affected. In a national survey the discrepancies due to geoid undulations may be more serious. In the event that geoid undulations are large enough to affect a survey, the station altitudes must be corrected to true elevations above the ellipsoid.

2.5.5 Density determination

The density of rocks in the vicinity of a gravity profile is important for the calculation of the Bouguer plate and

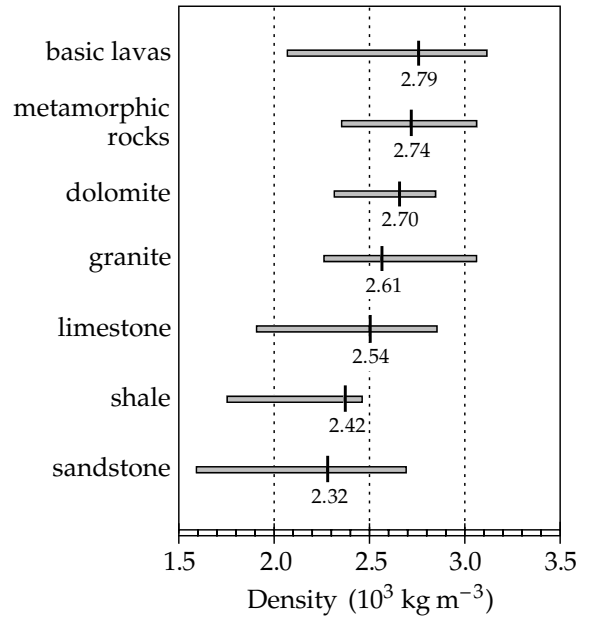


Fig. 2.39 Typical mean values and ranges of density for some common rock types (data source: Dobrin, 1976).

terrain corrections. Density is defined as the mass per unit of volume of a material. It has different units and different numerical values in the c.g.s. and SI systems. For example, the density of water is 1 g cm^{-3} in the c.g.s. system, but 1000 kg m^{-3} in the SI system. In gravity prospecting c.g.s. units are still in common use, but are slowly being replaced by SI units. The formulas given for Δg_{T} and Δg_{BP} in Eq. (2.71) and Eq. (2.73), respectively, require that density be given in kg m^{-3} .

A simple way of determining the appropriate density to use in a gravity study is to make a representative collection of rock samples with the aid of a geological map. The specific gravity of a sample may be found directly by weighing it first in air and then in water, and applying Archimedes' principle. This gives its density ρ_r relative to that of water:

$$\rho_r = \frac{W_a}{W_a - W_w} \quad (2.75)$$

Typically, the densities found for different rock types by this method show a large amount of scatter about their means, and the ranges of values for different rock types overlap (Fig. 2.39). The densities of igneous and metamorphic rocks are generally higher than those of sedimentary rocks. This method is adequate for reconnaissance of an area. Unfortunately, it is often difficult to ensure that the surface collection of rocks is representative of the rock types in subsurface structures, so alternative methods of determining the appropriate density are usually employed. Density can be measured in vertical boreholes, drilled to explore the nature of a presumed structure. The density determined in the borehole is used to refine the interpretation of the structure.

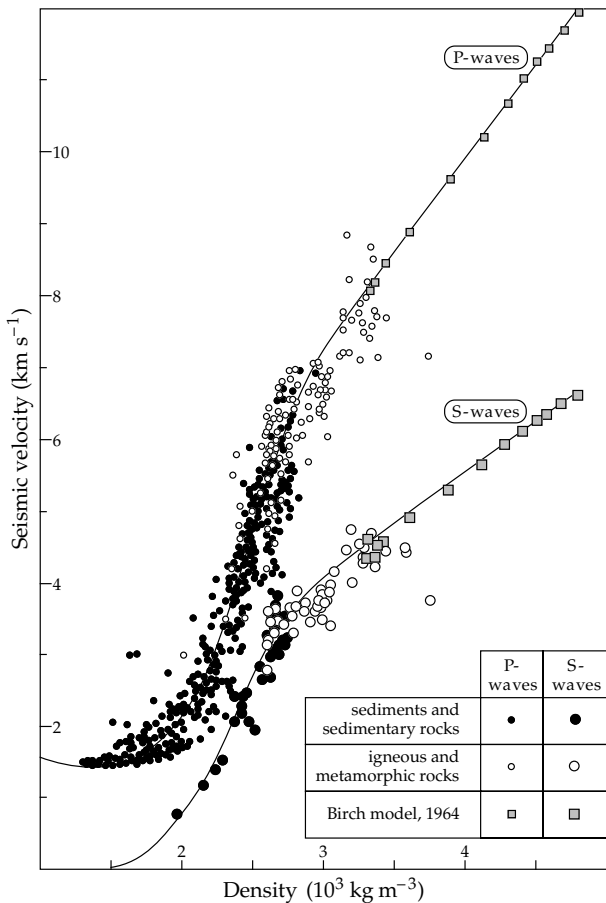


Fig. 2.40 The empirical relationships between density and the seismic P-wave and S-wave velocities in water-saturated sediments and sedimentary rocks, igneous and metamorphic rocks (after Ludwig *et al.*, 1970).

2.5.5.1 Density from seismic velocities

Measurements on samples of water-saturated sediments and sedimentary rocks, and on igneous and metamorphic rocks show that density and the seismic P-wave and S-wave velocities are related. The optimum fit to each data-set is a smooth curve (Fig. 2.40). Each curve is rather idealized, as the real data contain considerable scatter. For this reason the curves are best suited for computing the mean density of a large crustal body from its mean seismic velocity. Adjustments must be made for the higher temperatures and pressures at depth in the Earth, which affect both the density and the elastic parameters of rocks. However, the effects of high pressure and temperature can only be examined in laboratory experiments on small specimens. It is not known to what extent the results are representative of the *in situ* velocity–density relationship in large crustal blocks.

The velocity–density curves are empirical relationships that do not have a theoretical basis. The P-wave data are used most commonly. In conjunction with seismic refraction studies, they have been used for modelling the density distributions in the Earth’s crust and upper mantle responsible for large-scale, regional gravity anomalies (see Section 2.6.4).

2.5.5.2 Gamma–gamma logging

The density of rock formations adjacent to a borehole can be determined from an instrument in the borehole. The principle makes use of the Compton scattering of γ -rays by loosely bound electrons in the rock adjacent to a borehole. An American physicist, Arthur H. Compton, discovered in 1923 that radiation scattered by loosely bound electrons experienced an increase in wavelength. This simple observation cannot be explained at all if the radiation is treated as a wave; the scattered radiation would have the same wavelength as the incident radiation. The Compton effect is easily explained by regarding the radiation as particles or photons, i.e., particles of quantized energy, rather than as waves. The energy of a photon is inversely proportional to its wavelength. The collision of a γ -ray photon with an electron is like a collision between billiard balls; part of the photon’s energy is transferred to the electron. The scattered photon has lower energy and hence a longer wavelength than the incident photon. The Compton effect was an important verification of quantum theory.

The density logger, or *gamma–gamma* logger (Fig. 2.41), is a cylindrical device that contains a radioactive source of γ -rays, such as ^{137}Cs , which emits radiation through a narrow slit. The γ -ray photons collide with the loosely bound electrons of atoms near the hole, and are scattered. A scintillation counter to detect and measure the intensity of γ -rays is located in the tool about 45–60 cm above the emitter; the radiation reaching it also passes through a slit. Emitter and detector are shielded with lead, and the tool is pressed against the wall of the borehole by a strong spring, so that the only radiation registered is that resulting from the Compton scattering in the surrounding formation. The intensity of detected radiation is determined by the density of electrons, and so by the density of rock near to the logging tool. The γ -rays penetrate only about 15 cm into the rock.

Calibrated gamma–gamma logs give the bulk density of the rock surrounding a borehole. This information is also needed for calculating porosity, which is defined as the fractional volume of the rock represented by pore spaces. Most sedimentary rocks are porous, the amount depending on the amount of compaction experienced. Igneous and metamorphic rocks generally have low porosity, unless they have been fractured. Usually the pores are filled with air, gas or a fluid, such as water or oil. If the densities of the matrix rock and pore fluid are known, the bulk density obtained from gamma–gamma logging allows the porosity of the rock to be determined.

2.5.5.3 Borehole gravimetry

Modern instrumentation allows gravity to be measured accurately in boreholes. One type of borehole gravimeter is a modification of the LaCoste–Romberg instrument, adapted for use in the narrow borehole and under conditions of elevated temperature and pressure. Alternative

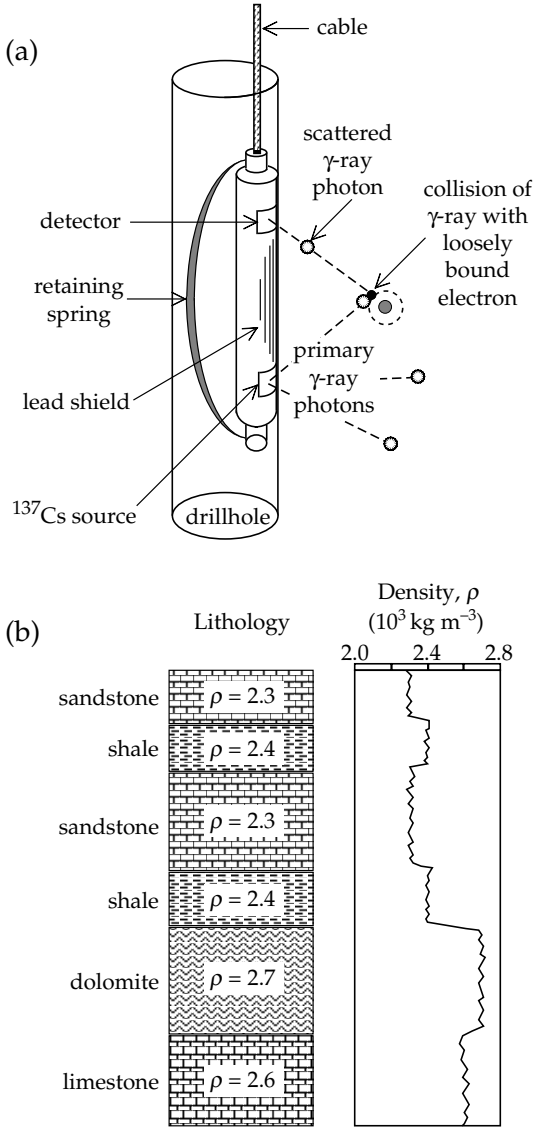


Fig. 2.41 (a) The design of a gamma–gamma logging device for determining density in a borehole (after Telford *et al.*, 1990), and (b) a schematic gamma–gamma log calibrated in terms of the rock density.

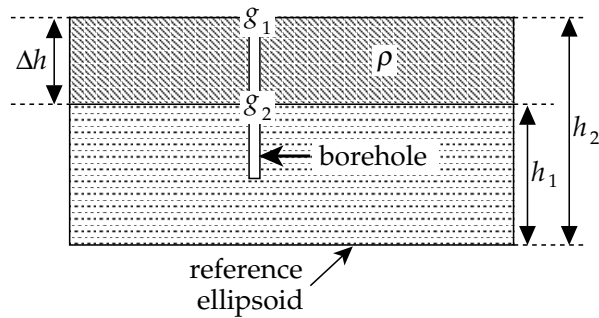


Fig. 2.42 Geometry for computation of the density of a rock layer from gravity measurements made in a vertical borehole.

instruments have been designed on different principles; they have a comparable sensitivity of about 0.01 mgal. Their usage for down-hole density determination is based on application of the free-air and Bouguer plate corrections.

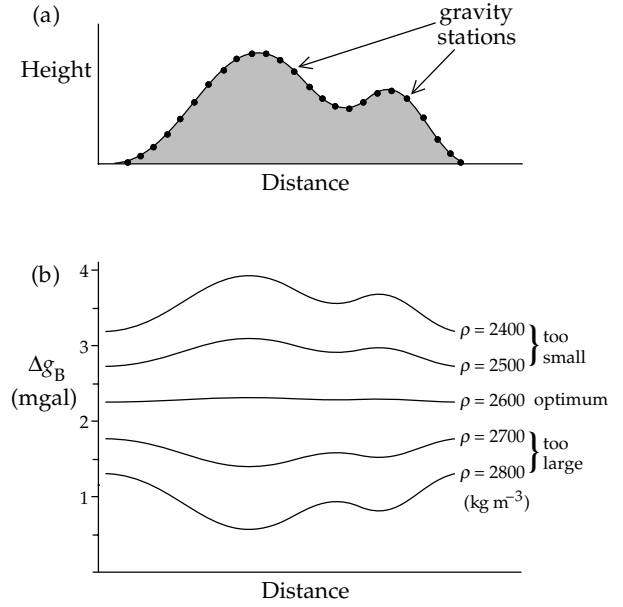


Fig. 2.43 Determination of the density of near-surface rocks by Nettleton's method. (a) Gravity measurements are made on a profile across a small hill. (b) The data are corrected for elevation with various test values of the density. The optimum density gives minimum correlation between the gravity anomaly (Δg_B) and the topography.

Let g_1 and g_2 be the values of gravity measured in a vertical borehole at heights h_1 and h_2 , respectively, above the reference ellipsoid (Fig. 2.42). The difference between g_1 and g_2 is due to the different heights and to the material between the two measurement levels in the borehole. The value g_2 will be larger than g_1 for two reasons. First, because the lower measurement level is closer to the Earth's center, g_2 will be greater than g_1 by the amount of the combined elevation correction, namely $(0.3086 - (0.0419\rho \times 10^{-3}))\Delta h$ mgal, where $\Delta h = h_1 - h_2$. Second, at the lower level h_2 the gravimeter experiences an upward Bouguer attraction due to the material between the two measurement levels. This reduces the measured gravity at h_2 and requires a compensating increase to g_2 of amount $(0.0419\rho \times 10^{-3})\Delta h$ mgal. The difference Δg between the *corrected* values of g_1 and g_2 after reduction to the level h_2 is then

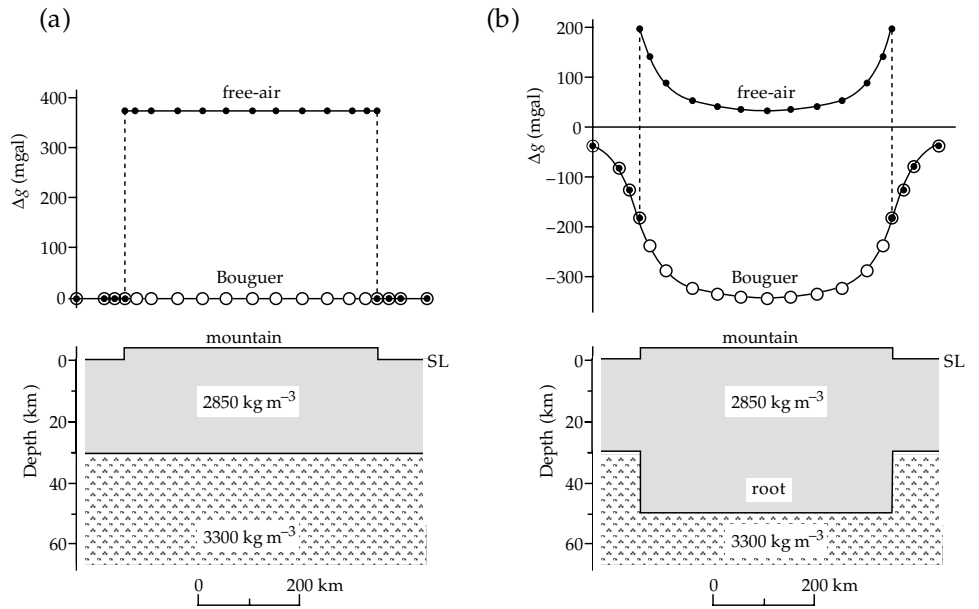
$$\begin{aligned} \Delta g &= (0.3086 - 0.0419\rho \times 10^{-3})\Delta h - 0.0419\rho \times 10^{-3}\Delta h \\ &= (0.3086 - 0.0838\rho \times 10^{-3})\Delta h \end{aligned} \quad (2.76)$$

Rearranging this equation gives the density ρ of the material between the measurement levels in the borehole:

$$\rho = \left(3.683 - 11.93 \frac{\Delta g}{\Delta h} \right) \times 10^3 \text{ kg m}^{-3} \quad (2.77)$$

If borehole gravity measurements are made with an accuracy of ± 0.01 mgal at a separation of about 10 m, the density of the material near the borehole can be determined with an accuracy of about $\pm 10 \text{ kg m}^{-3}$. More than 90% of the variation in gravity in the borehole is due to material within a radius of about $5\Delta h$ from the borehole (about 50 m

Fig. 2.44 Free-air and Bouguer anomalies across a mountain range. In (a) the mountain is modelled by a fully supported block, and in (b) the mass of the mountain above sea-level (SL) is compensated by a less-dense crustal root, which projects down into the denser mantle (based on Bott, 1982).



for a distance $\Delta h \approx 10 \text{ m}$ between measurement levels). This is much larger than the lateral range penetrated by gamma–gamma logging. As a result, effects related to the borehole itself are unimportant.

2.5.5.4 Nettleton's method for near-surface density

The near-surface density of the material under a hill can be determined by a method devised by L. Nettleton that compares the shape of a Bouguer gravity anomaly (see Section 2.5.6) with the shape of the topography along a profile. The method makes use of the combined elevation correction ($\Delta g_{\text{FA}} + \Delta g_{\text{BP}}$) and the terrain correction (Δg_{T}), which are density dependent. The terrain correction is less important than the Bouguer plate correction and can usually be neglected.

A profile of closely spaced gravity stations is measured across a small hill (Fig. 2.43). The combined elevation correction is applied to each measurement. Suppose that the true average density of the hill is 2600 kg m^{-3} . If the value assumed for ρ is too small (say, 2400 kg m^{-3}), Δg_{BP} at each station will be too small. The discrepancy is proportional to the elevation, so the Bouguer gravity anomaly is a *positive* image of the topography. If the value assumed for ρ is too large (say, 2800 kg m^{-3}), the opposite situation occurs. Too much is subtracted at each point, giving a computed anomaly that is a *negative* image of the topography. The optimum value for the density is found when the gravity anomaly has minimum correlation with the topography.

2.5.6 Free-air and Bouguer gravity anomalies

Suppose that we can measure gravity on the reference ellipsoid. If the distribution of density inside the Earth is homogeneous, the measured gravity should agree with the

theoretical gravity given by the normal gravity formula. The gravity corrections described in Section 2.5.4 compensate for the usual situation that the point of measurement is not on the ellipsoid. A discrepancy between the corrected, measured gravity and the theoretical gravity is called a *gravity anomaly*. It arises because the density of the Earth's interior is not homogeneous as assumed. The most common types of gravity anomaly are the *Bouguer anomaly* and the *free-air anomaly*.

The Bouguer gravity anomaly (Δg_{B}) is defined by applying all the corrections described individually in Section 2.5.4:

$$\Delta g_{\text{B}} = g_{\text{m}} + (\Delta g_{\text{FA}} - \Delta g_{\text{BP}} + \Delta g_{\text{T}} + \Delta g_{\text{tide}}) - g_{\text{n}} \quad (2.78)$$

In this formula g_{m} and g_{n} are the measured and normal gravity values; the corrections in parentheses are the free-air correction (Δg_{FA}), Bouguer plate correction (Δg_{BP}), terrain correction (Δg_{T}) and tidal correction (Δg_{tide}).

The free-air anomaly Δg_{F} is defined by applying only the free-air, terrain and tidal corrections to the measured gravity:

$$\Delta g_{\text{F}} = g_{\text{m}} + (\Delta g_{\text{FA}} + \Delta g_{\text{T}} + \Delta g_{\text{tide}}) - g_{\text{n}} \quad (2.79)$$

The Bouguer and free-air anomalies across the same structure can look quite different. Consider first the topographic block (representing a mountain range) shown in Fig. 2.44a. For this simple structure we neglect the terrain and tidal corrections. The difference between the Bouguer anomaly and the free-air anomaly arises from the Bouguer plate correction. In computing the Bouguer anomaly the simple elevation of the measurement station is taken into account together with the free-air correction. The measured gravity contains the attraction of the land-mass above the ellipsoid, which is compensated with the Bouguer plate correction. The underground structure

does not vary laterally, so the corrected measurement agrees with the theoretical value and the Bouguer anomaly is everywhere zero across the mountain range. In computing the free-air anomaly only the free-air correction is applied; the part of the measured gravity due to the attraction of the landmass above the ellipsoid is not taken into account. Away from the mountain-block the Bouguer and free-air anomalies are both equal to zero. Over the mountain the mass of the mountain-block increases the measured gravity compared to the reference value and results in a positive free-air anomaly across the mountain range.

In fact, seismic data show that the Earth's crust is usually much thicker than normal under a mountain range. This means that a block of less-dense crustal rock projects down into the denser mantle (Fig. 2.44b). After making the free-air and Bouguer plate corrections there remains a Bouguer anomaly due to a block that represents the "root-zone" of the mountain range. As this is less dense than the adjacent and underlying mantle it constitutes a mass deficit. The attraction on a gravimeter at stations on a profile across the mountain range will be less than in Fig. 2.44a, so the corrected measurement will be less than the reference value. A strongly negative Bouguer anomaly is observed along the profile. At some distance from the mountain-block the Bouguer and free-air anomalies are equal but they are no longer zero, because the Bouguer anomaly now contains the effect of the root-zone. Over the mountain-block the free-air anomaly has a constant positive offset from the Bouguer anomaly, as in the previous example. Note that, although the free-air anomaly is positive, it falls to a very low value over the center of the block. At this point the attraction of the mountain is partly cancelled by the missing attraction of the less-dense root-zone.

2.6 INTERPRETATION OF GRAVITY ANOMALIES

2.6.1 Regional and residual anomalies

A gravity anomaly results from the inhomogeneous distribution of density in the Earth. Suppose that the density of rocks in a subsurface body is ρ and the density of the rocks surrounding the body is ρ_0 . The difference $\Delta\rho = \rho - \rho_0$ is called the *density contrast* of the body with respect to the surrounding rocks. If the body has a higher density than the host rock, it has a positive density contrast; a body with lower density than the host rock has a negative density contrast. Over a high-density body the measured gravity is augmented; after reduction to the reference ellipsoid and subtraction of the normal gravity a positive gravity anomaly is obtained. Likewise a negative anomaly results over a region of low density. The presence of a gravity anomaly indicates a body or structure with anomalous density; the sign of the anomaly is the same as that of the density contrast and shows whether the density of the body is higher or lower than normal.

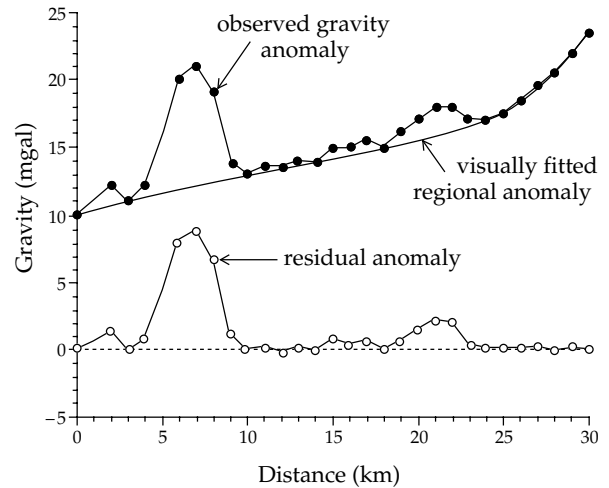


Fig. 2.45 Representation of the regional anomaly on a gravity profile by visually fitting the large-scale trend with a smooth curve.

The appearance of a gravity anomaly is affected by the dimensions, density contrast and depth of the anomalous body. The horizontal extent of an anomaly is often called its apparent "wavelength." The wavelength of an anomaly is a measure of the depth of the anomalous mass. Large, deep bodies give rise to broad (long-wavelength), low-amplitude anomalies, while small, shallow bodies cause narrow (short-wavelength), sharp anomalies.

Usually a map of Bouguer gravity anomalies contains superposed anomalies from several sources. The long-wavelength anomalies due to deep density contrasts are called *regional* anomalies. They are important for understanding the large-scale structure of the Earth's crust under major geographic features, such as mountain ranges, oceanic ridges and subduction zones. Short-wavelength *residual* anomalies are due to shallow anomalous masses that may be of interest for commercial exploitation. Geological knowledge is essential for interpreting the residual anomalies. In eroded shield areas, like Canada or Scandinavia, anomalies with very short wavelengths may be due to near-surface mineralized bodies. In sedimentary basins, short- or intermediate-wavelength anomalies may arise from structures related to reservoirs for petroleum or natural gas.

2.6.2 Separation of regional and residual anomalies

The separation of anomalies of regional and local origin is an important step in the interpretation of a gravity map. The analysis may be based on selected profiles across some structure, or it may involve the two-dimensional distribution of anomalies in a gravity map. Numerous techniques have been applied to the decomposition of a gravity anomaly into its constituent parts. They range in sophistication from simple visual inspection of the anomaly pattern to advanced mathematical analysis. A few examples of these methods are described below.

Fig. 2.46 Removal of regional trend from a gravity map by contour smoothing: (a) hand-drawn smoothing of contour lines on original Bouguer gravity map, (b) map of regional gravity variation, (c) residual gravity anomaly after subtracting the regional variation from the Bouguer gravity map (after Robinson and Çoruh, 1988). Values are in mgal.

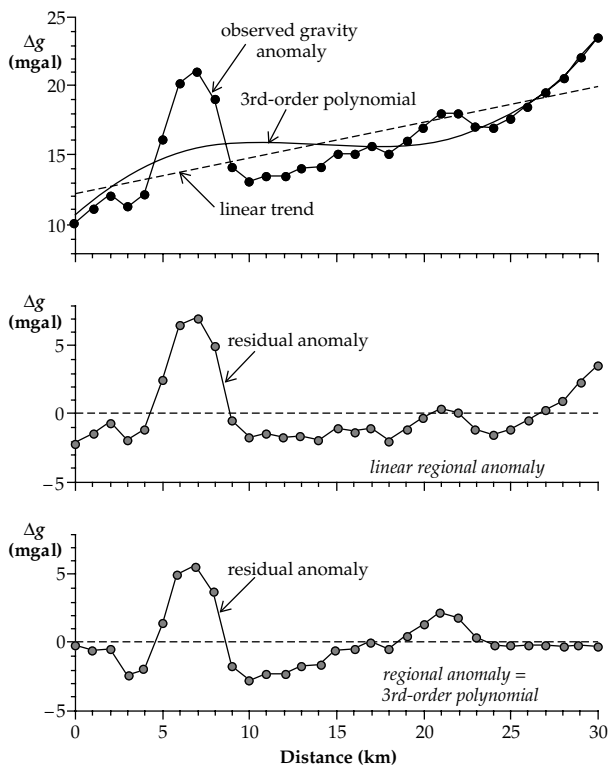
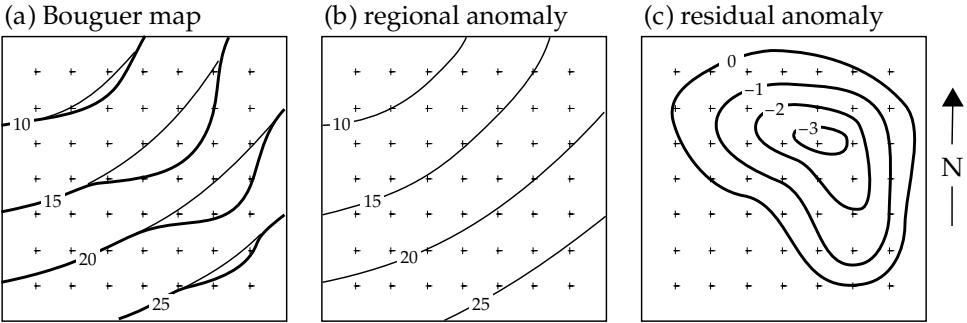


Fig. 2.47 Representation of the regional trend by a smooth polynomial curve fitted to the observed gravity profile by the method of least squares.

2.6.2.1 Visual analysis

The simplest way of representing the regional anomaly on a gravity profile is by visually fitting the large-scale trend with a smooth curve (Fig. 2.45). The value of the regional gravity given by this trend is subtracted point by point from the Bouguer gravity anomaly. This method allows the interpreter to fit curves that leave residual anomalies with a sign appropriate to the interpretation of the density distribution.

This approach may be adapted to the analysis of a gravity map by visually smoothing the contour lines. In Fig. 2.46a the contour lines of equal Bouguer gravity curve sharply around a local abnormality. The more gently curved contours have been continued smoothly as dotted lines. They indicate how the interpreter thinks the regional

gravity field (Fig. 2.46b) would continue in the absence of the local abnormality. The values of the regional and original Bouguer gravity are interpolated from the corresponding maps at points spaced on a regular grid. The regional value is subtracted from the Bouguer anomaly at each point and the computed residuals are contoured to give a map of the local gravity anomaly (Fig. 2.46c). The experience and skill of the interpreter are important factors in the success of visual methods.

2.6.2.2 Polynomial representation

In an alternative method the regional trend is represented by a straight line or, more generally, by a smooth polynomial curve. If x denotes the horizontal position on a gravity profile, the regional gravity Δg_R may be written

$$\Delta g_R = \Delta g_0 + \Delta g_1 x + \Delta g_2 x^2 + \Delta g_3 x^3 + \dots + \Delta g_n x^n \quad (2.80)$$

The polynomial is fitted by the method of least squares to the observed gravity profile. This gives optimum values for the coefficients Δg_n . The method also has drawbacks. The higher the order of the polynomial, the better it fits the observations (Fig. 2.47). The ludicrous extreme is when the order of the polynomial is one less than the number of observations; the curve then passes perfectly through all the data points, but the regional gravity anomaly has no meaning geologically. The interpreter's judgement is important in selecting the order of the polynomial, which is usually chosen to be the lowest possible order that represents most of the regional trend. Moreover, a curve fitted by least squares must pass through the mean of the gravity values, so that the residual anomalies are divided equally between positive and negative values. Each residual anomaly is flanked by anomalies of opposite sign (Fig. 2.47), which are due to the same anomalous mass that caused the central anomaly and so have no significance of their own.

Polynomial fitting can also be applied to gravity maps. It is assumed that the regional anomaly can be represented by a smooth surface, $\Delta g(x, y)$, which is a low order polynomial of the horizontal position coordinates x and y . In the simplest case the regional anomaly is expressed

Box 2.4: Fourier analysis

If integrated over a full cycle, the resulting value of a sine or cosine function is zero. The integrated value of the product of a sine function and a cosine function is also zero. Mathematically, this defines the sines and cosines as *orthogonal* functions. The squared values of sines and cosines do not integrate to zero over a full cycle; this property can be used to *normalize* functions that can be expressed in terms of sines and cosines. These observations may be summarized as follows for the functions $\sin(n\theta)$ and $\cos(n\theta)$

$$\begin{aligned}\int_0^{2\pi} \sin(n\theta) d\theta &= \int_0^{2\pi} \cos(n\theta) d\theta = 0 \\ \int_0^{2\pi} \sin^2(n\theta) d\theta &= \frac{1}{2} \int_0^{2\pi} (1 - \cos(2n\theta)) d\theta = \pi \\ \int_0^{2\pi} \cos^2(n\theta) d\theta &= \frac{1}{2} \int_0^{2\pi} (1 + \cos(2n\theta)) d\theta = \pi\end{aligned}\quad (1)$$

It follows by applying these results and invoking the formulas for the sums and differences of sines and cosines that

$$\begin{aligned}\int_0^{2\pi} \sin(n\theta) \cos(m\theta) d\theta &= \frac{1}{2} \int_0^{2\pi} \left\{ \sin\left(\frac{n+m}{2}\theta\right) \right. \\ &\quad \left. + \sin\left(\frac{n-m}{2}\theta\right) \right\} d\theta = 0 \\ \int_0^{2\pi} \cos(n\theta) \cos(m\theta) d\theta &= \int_0^{2\pi} \sin(n\theta) \sin(m\theta) d\theta \\ &= \frac{1}{2} \int_0^{2\pi} \left\{ \cos\left(\frac{n-m}{2}\theta\right) \pm \cos\left(\frac{n+m}{2}\theta\right) \right\} d\theta \\ &= \begin{cases} 0, & \text{if } m \neq n \\ \pi, & \text{if } m = n \end{cases}\end{aligned}\quad (2)$$

These relationships form the basis of Fourier analysis.

Geophysical signals that vary periodically in time (e.g., seismic waves) or space (e.g., gravity, magnetic or thermal anomalies) can be expressed as the superposition of harmonics of a fundamental frequency or wave number. For example, consider a gravity anomaly $\Delta g(x)$ along a profile of length L in the x -direction. The fundamental “wavelength” λ of the anomaly is equal to $2L$, i.e., twice the profile length. A corresponding fundamental wave number is defined for the profile as $k = (2\pi/\lambda)$, so that the argument θ in the sine and cosine functions in Eqs. (1) and (2) is replaced by (kx) . The observed anomaly is represented by adding together components corresponding to harmonics of the fundamental wavelength:

$$\Delta g(x) = a_1 \cos(kx) + b_1 \sin(kx) + a_2 \cos(2kx) + b_2 \sin(2kx) + a_3 \cos(3kx) + b_3 \sin(3kx) + \dots$$

$$\Delta g(x) = \sum_{n=1}^N (a_n \cos(nkx) + b_n \sin(nkx)) \quad (3)$$

This expression for $\Delta g(x)$ is called a *Fourier series*. In Eq. (3) the summation is truncated after N sine and cosine terms. The value of N is chosen to be as large as necessary to describe the gravity anomaly adequately. The importance of any individual term of order n is given by the values of the corresponding coefficients a_n and b_n , which act as weighting functions.

The coefficients a_n and b_n can be calculated using the orthogonal properties of sine and cosine functions summarized in Eqs. (1) and (2). If both sides of Eq. (3) for the gravity anomaly $\Delta g(x)$ are multiplied by $\cos(mkx)$, we get

$$\begin{aligned}\Delta g(x) \cos(mkx) &= \sum_{n=1}^N (a_n \cos(nkx) \cos(mkx) \\ &\quad + b_n \sin(nkx) \cos(mkx))\end{aligned}\quad (4)$$

Each product on the right-hand side of this equation can be written as the sum or difference of two sines or cosines. Thus,

$$\begin{aligned}\Delta g(x) \cos(mkx) &= \frac{1}{2} \sum_{n=1}^N \left(a_n [\cos((n+m)kx) \right. \\ &\quad \left. + \cos((n-m)kx)] + b_n [\sin((n-m)kx) \right. \\ &\quad \left. + \sin((n+m)kx)] \right)\end{aligned}\quad (5)$$

Integration of $\Delta g(x) \cos(2mkx)$ over a full wavelength of x causes all terms on the right side of Eq. (5) to vanish unless $n=m$, when the term $\cos((n-m)kx) = \cos(0) = 1$. This allows us to determine the coefficients a_n in Eq. (3):

$$\int_0^{\lambda} \Delta g(x) \cos(nkx) dx = \frac{1}{2} a_n \int_0^{\lambda} dx = \frac{\lambda}{2} a_n \quad (6)$$

$$a_n = \frac{2}{\lambda} \int_0^{\lambda} \Delta g(x) \cos(nkx) dx = \frac{1}{\pi} \int_0^{2\pi} \Delta g(x) \cos(n\theta) d\theta \quad (7)$$

The coefficients b_n in Eq. (3) are obtained similarly by multiplying $\Delta g(x)$ by $\sin(mkx)$ and integrating over a full cycle of the signal. This gives

$$b_n = \frac{2}{\lambda} \int_0^{\lambda} \Delta g(x) \sin(nkx) dx = \frac{1}{\pi} \int_0^{2\pi} \Delta g(x) \sin(n\theta) d\theta \quad (8)$$

Box 2.5: Double Fourier series

The two-dimensional variation of a mapped gravity anomaly can be analyzed with the aid of *double Fourier series*. In this case the gravity anomaly is a function of both the x - and y -coordinates and can be written

$$\Delta g(x, y) = \sum_{n=1}^N \sum_{m=1}^M (a_{nm} C_n C_m^* + b_{nm} C_n S_m^* + c_{nm} S_n C_m^* + d_{nm} S_n S_m^*) \quad (1)$$

where

$$\begin{aligned} C_n &= \cos\left(2n\pi \frac{x}{\lambda_x}\right) & S_n &= \sin\left(2n\pi \frac{x}{\lambda_x}\right) \\ C_m^* &= \cos\left(2m\pi \frac{y}{\lambda_y}\right) & S_m^* &= \sin\left(2m\pi \frac{y}{\lambda_y}\right) \end{aligned} \quad (2)$$

In these expressions the fundamental wavelengths λ_x and λ_y express the extent of the anomaly in the x - and y -directions, respectively. The derivation of the coefficients a_{nm} , b_{nm} , c_{nm} and d_{nm} is similar in principle to the one-dimensional case, relying on the orthogonality of the individual sine and cosine terms, and the property that the products of two sine terms, two cosine terms or a sine term with a cosine term, can be expressed as the sum or difference of other sine or cosine functions. As might be expected, the analysis of double Fourier series is somewhat more complicated than in the one-dimensional case, but it delivers results that characterize the two-dimensional variation of the regional gravity anomaly.

as a first-order polynomial, or plane. To express changes in the gradient of gravity a higher-order polynomial is needed. For example, the regional gravity given by a second-order polynomial is written

$$\begin{aligned} \Delta g(x, y) &= \Delta g_0 + \Delta g_{x1}x + \Delta g_{y1}y \\ &+ \Delta g_{x2}x^2 + \Delta g_{y2}y^2 + \Delta g_{xy}xy \end{aligned} \quad (2.81)$$

As in the analysis of a profile, the optimum values of the coefficients Δg_{x1} , Δg_{y1} , etc., are determined by least-squares fitting. The residual anomaly is again computed point by point by subtracting the regional from the original data.

2.6.2.3 Representation by Fourier series

The gravity anomaly along a profile can be analyzed with techniques developed for investigating time series. Instead of varying with time, as the seismic signal does in a seismometer, the gravity anomaly $\Delta g(x)$ varies with position

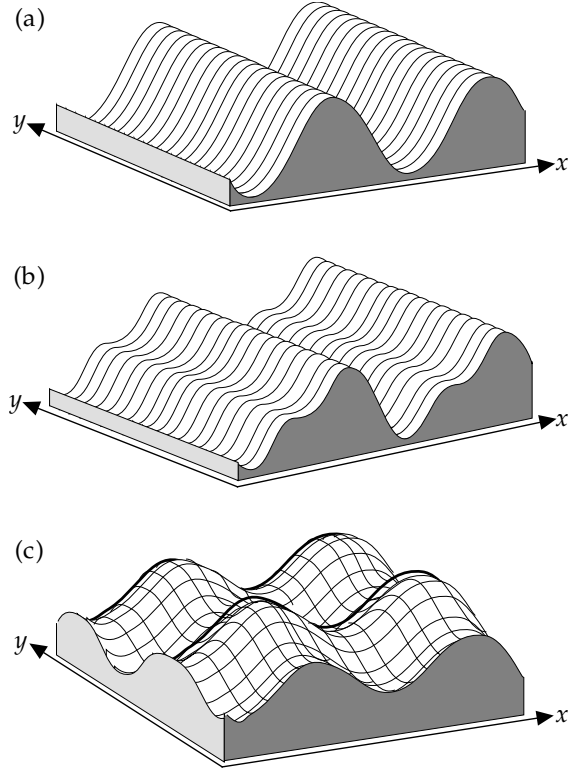


Fig. 2.48 Expression of the two-dimensional variation of a gravity anomaly using double Fourier series: (a) a single harmonic in the x -direction, (b) two harmonics in the x -direction, (c) superposed single harmonics in the x - and y -directions, respectively (after Davis, 1973).

x along the profile. For a spatial distribution the wave number, $k = 2\pi/\lambda$, is the counterpart of the frequency of a time series. If it can be assumed that its variation is periodic, the function $\Delta g(x)$ can be expressed as the sum of a series of discrete harmonics. Each harmonic is a sine or cosine function whose argument is a multiple of the fundamental wave number. The expression for $\Delta g(x)$ is called a *Fourier series* (Box 2.4).

The breakdown of a complex anomaly (or time series) in terms of simpler periodic variations of different wavelengths is called *Fourier analysis* and is a powerful method for resolving the most important components of the original signal.

The two-dimensional variation of a mapped gravity anomaly can be expressed in a similar way with the aid of *double Fourier series* (Box 2.5). As in the simpler one-dimensional case of a gravity anomaly on a profile, the expression of two-dimensional gravity anomalies by double Fourier series is analogous to summing weighted sinusoidal functions. These can be visualized as corrugations of the x - y plane (Fig. 2.48), with each corrugation weighted according to its importance to $\Delta g(x, y)$.

2.6.2.4 Anomaly enhancement and filtering

The above discussion shows how a function that is periodic can be expressed as a Fourier sum of harmonics of a

Box 2.6: Complex numbers

In determining the roots of cubic and higher order polynomials it is sometimes necessary to use the square roots of negative numbers. A negative number can be written as the positive number multiplied by (-1) , so the square root of a negative number contains the imaginary unit i , defined as the square root of (-1) . A complex number z consists of a real part x and an imaginary part y , and is written

$$z = x + iy \quad (1)$$

The number $z^* = x - iy$ is called the *complex conjugate* of z . The product $zz^* = x^2 + y^2$ gives the squared amplitude of the complex number.

The great Swiss mathematician Leonhard Euler showed in 1748 how the trigonometric functions, $\cos\theta$ and $\sin\theta$, are related to a complex exponential function:

$$\cos\theta + i \sin\theta = e^{i\theta} \quad (2)$$

where e is the base of the natural logarithms and i is the imaginary unit. This formula is fundamental to the study of functions of a complex variable, a branch of mathematics known as *complex analysis* that can be used to solve a wide range of practical and theoretical problems. In this expression, $\cos\theta$ is called the real part of $e^{i\theta}$ and $\sin\theta$ is called the imaginary part. It follows that

$$i = e^{i\pi/2} \\ \sqrt{i} = e^{i\pi/4} = \frac{1}{\sqrt{2}}(1 + i) \quad (3)$$

Complex numbers can be depicted geometrically on an *Argand diagram*, (also called the *complex plane*) invented in 1806 by another Swiss mathematician, Jean-Robert Argand, as an aid to their visualization. In this diagram (Fig. B2.6) the real part of the number, x in our case, is plotted on the horizontal axis and the imaginary part, y , on the vertical axis. If the distance of the point P at (x, y) from the origin of the plot at O is r , and the angle between OP and the x -axis is θ , then the real part of the complex number is $x = r \cos\theta$ and the imaginary part is $y = r \sin\theta$, so that, using Euler's formula

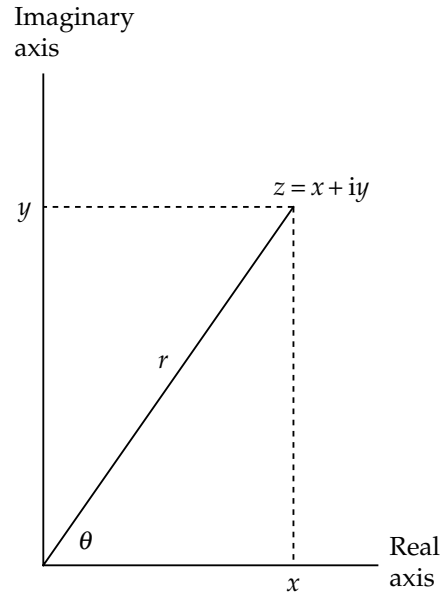


Fig. B2.6 The complex plane.

$$z = x + iy = r(\cos\theta + i \sin\theta) = re^{i\theta} \quad (4)$$

Complex numbers are combined in the same way as real numbers. Two complex numbers, z_1 and z_2 , combine to form a new complex number, z . A special case is when the complex number is combined with its *complex conjugate*; this results in a real number. Some examples of possible combinations of complex numbers are as follows:

$$z = z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2) \quad (5)$$

$$z = r_1 e^{i\theta_1} + r_2 e^{i\theta_2}$$

$$z = z_1 z_2 = (x_1 + iy_1)(x_2 + iy_2) \\ = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + y_1 x_2) \quad (6)$$

$$z = r_1 e^{i\theta_1} r_2 e^{i\theta_2} = r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

fundamental wavelength. By breaking down the observed signal into discrete components, it is possible to remove some of these and reconstruct a filtered version of the original anomaly. However, the requirements of periodic behavior and discreteness of harmonic content are often not met. For example, the variation of gravity from one point to another is usually not periodic. Moreover, if the harmonic content of a function is made up of distinct multiples of a fundamental frequency or wave number, the wavelength spectrum consists of a number of distinct values. Yet many functions of geophysical interest are best represented by a continuous spectrum of wavelengths.

To handle this kind of problem the spatial variation of gravity is represented by a *Fourier integral*, which consists of a continuous set of frequencies or wave numbers instead of a discrete set. The Fourier integral can be used to represent non-periodic functions. It uses complex numbers (Box 2.6), which are numbers that involve i , the square-root of -1 . If the gravity anomaly is analyzed in two dimensions (instead of on a single profile), a two-dimensional integral is needed, analogous to the double Fourier series representation described in Section 2.6.2.3 and Box 2.5. The observed gravity can then be manipulated using the techniques of *Fourier transforms* (Box 2.7). These techniques

Box 2.7: Fourier transforms

Functions that cannot be expressed as the Fourier series of individual terms, as in Boxes 2.4 and 2.5, may be replaced by a *Fourier integral*. This consists of a continuous set of frequencies or wave numbers instead of a discrete set. The Fourier integral can be used to represent non-periodic functions. The gravity anomaly $\Delta g(x)$ is now written as an integral instead of as a sum of discrete terms:

$$\Delta g(x) = \int_{-\infty}^{\infty} G(u) e^{iux} du \quad (1)$$

where, by using the properties of complex numbers, it can be shown that

$$G(u) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \Delta g(x) e^{-iux} dx \quad (2)$$

The complex function $G(u)$ is the *Fourier transform* of the real-valued function $\Delta g(x)$. An adequate treatment of Fourier transforms is beyond the scope of this book. However, the use of this powerful mathematical technique can be illustrated without delving deeply into the theory.

A map of gravity anomalies can be represented by a function $\Delta g(x, y)$ of the Cartesian map coordinates. The Fourier transform of $\Delta g(x, y)$ is a two-dimensional complex function that involves wave numbers k_x and k_y defined by wavelengths of the gravity field with respect to the x - and y -axes ($k_x = 2\pi/\lambda_x$, $k_y = 2\pi/\lambda_y$). It is

$$G(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \Delta g(x, y) \left\{ \cos(k_x x + k_y y) - i \sin(k_x x + k_y y) \right\} dx dy \quad (3)$$

This equation assumes that the observations $\Delta g(x, y)$ can be represented by a continuous function defined over an infinite x - y plane, whereas in fact the data are of finite extent and are known at discrete points of a measurement grid. In practice, these inconsistencies are usually not important. Efficient computer algorithms permit the rapid computation of the Fourier transform $G(x, y)$ of the gravity anomaly $\Delta g(x, y)$. The transformed signal can be readily manipulated in the Fourier domain by convolution with a filter function and the result deconvoluted back into the spatial domain. This allows the operator to examine the effects of low- and high-pass filtering on the observed anomaly, which can greatly help in its interpretation.

involve intensive computations and are ideally suited to digital data-processing with powerful computers.

The two-dimensional Fourier transform of a gravity map makes it possible to digitally filter the gravity anomalies. A filter is a spatial function of the coordinates x and y . When the function $\Delta g(x, y)$ representing the gravity data is multiplied by the filter function, a new function is produced. The process is called *convolution* and the output is a map of the filtered gravity data. The computation in the spatial domain defined by the x - and y -coordinates can be time consuming. It is often faster to compute the Fourier transforms of the gravity and filter functions, multiply these together in the Fourier domain, then perform an inverse Fourier transform on the product to convert it back to the spatial domain.

The nature of the filter applied in the Fourier domain can be chosen to eliminate certain wavelengths. For example, it can be designed to cut out all wavelengths shorter than a selected wavelength and to pass longer wavelengths. This is called a *low-pass filter*; it passes long wavelengths that have low wave numbers. The irregularities in a Bouguer gravity anomaly map (Fig. 2.49a) are removed by low-pass filtering, leaving a filtered map (Fig. 2.49b) that is much smoother than the original. Alternatively, the filter in the Fourier domain can be designed to eliminate wavelengths longer than a selected wavelength and to pass shorter wavelengths. The application of such a *high-pass filter* enhances the short-wavelength (high wave number) component of the gravity map (Fig. 2.49c).

Wavelength filtering can be used to emphasize selected anomalies. For example, in studying large-scale crustal structure the gravity anomalies due to local small bodies are of less interest than the regional anomalies, which can be enhanced by applying a low-pass filter. Conversely, in the investigation of anomalies due to shallow crustal sources the regional effect can be suppressed by high-pass filtering.

2.6.3 Modelling gravity anomalies

After removal of regional effects the residual gravity anomaly must be interpreted in terms of an anomalous density distribution. Modern analyses are based on iterative modelling using high-speed computers. Earlier methods of interpretation utilized comparison of the observed gravity anomalies with the computed anomalies of geometric shapes. The success of this simple approach is due to the insensitivity of the shape of a gravity anomaly to minor variations in the anomalous density distribution. Some fundamental problems of interpreting gravity anomalies can be learned from the computed effects of geometric models. In particular, it is important to realize that the interpretation of gravity anomalies is not unique; different density distributions can give the same anomaly.

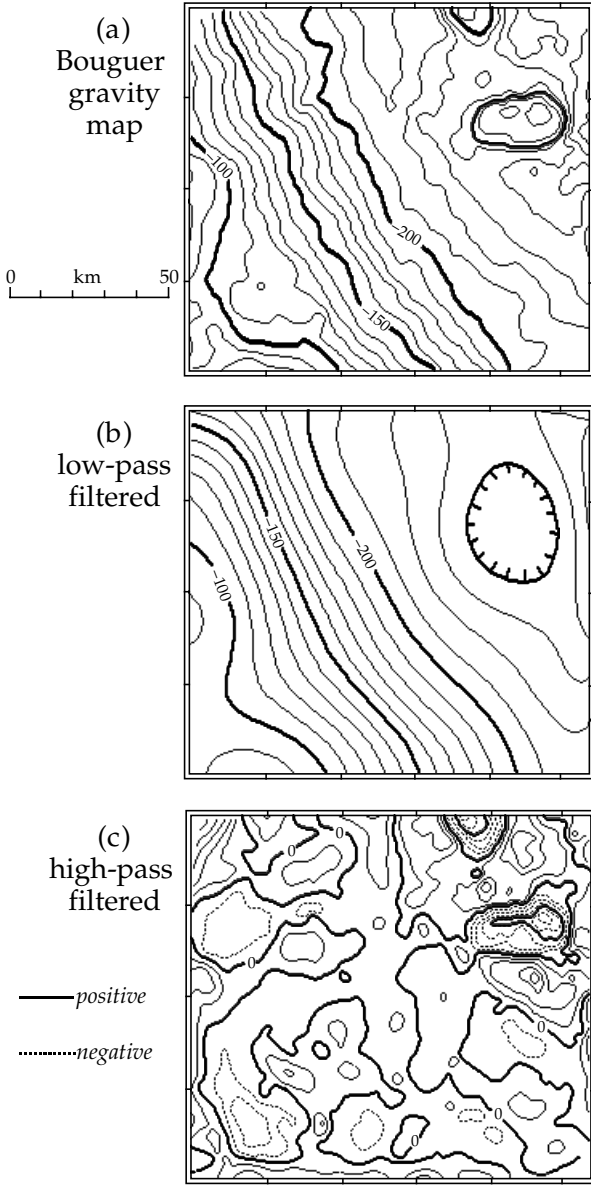


Fig. 2.49 The use of wavelength filtering to emphasize selected anomalies in the Sierra Nevada, California: (a) unfiltered Bouguer gravity map, (b) low-pass filtered gravity map with long-wavelength regional anomalies, and (c) high-pass filtered gravity map enhancing short-wavelength local anomalies. Contour interval: (a) and (b) 10 mgal, (c) 5 mgal (after Dobrin and Savit, 1988).

2.6.3.1 Uniform sphere: model for a diapir

Diapiric structures introduce material of different density into the host rock. A low-density salt dome ($\rho = 2150 \text{ kg m}^{-3}$) intruding higher-density carbonate rocks ($\rho_0 = 2500 \text{ kg m}^{-3}$) has a density contrast $\Delta\rho = -350 \text{ kg m}^{-3}$ and causes a negative gravity anomaly. A volcanic plug ($\rho = 2800 \text{ kg m}^{-3}$) intruding a granite body ($\rho_0 = 2600 \text{ kg m}^{-3}$) has a density contrast $\Delta\rho = +200 \text{ kg m}^{-3}$, which causes a positive gravity anomaly. The contour lines on a map of the anomaly are centered on the diapir, so all profiles across the center of the structure are equivalent. The anomalous body can be

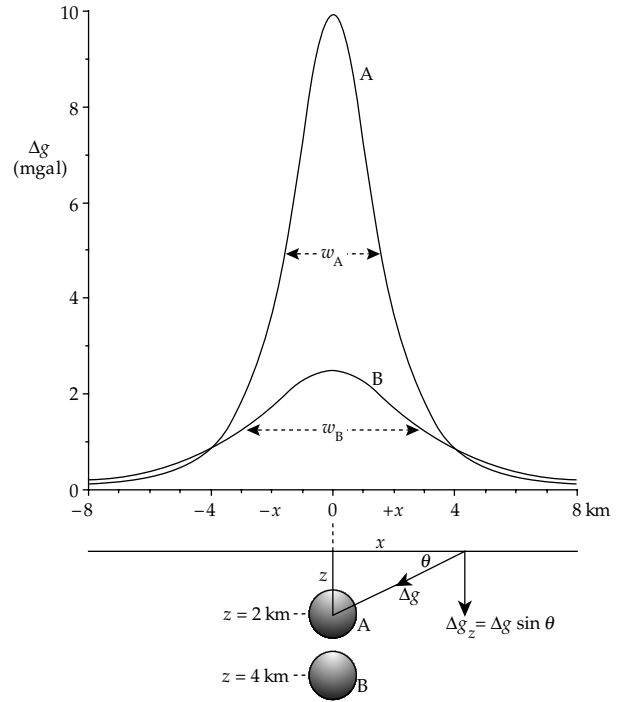


Fig. 2.50 Gravity anomalies for buried spheres with the same radius R and density contrast $\Delta\rho$ but with their centers at different depths z below the surface. The anomaly of the deeper sphere B is flatter and broader than the anomaly of the shallower sphere A.

modelled equally by a vertical cylinder or by a sphere, which we will evaluate here because of the simplicity of the model.

Assume a sphere of radius R and density contrast $\Delta\rho$ with its center at depth z below the surface (Fig. 2.50). The attraction Δg of the sphere is as though the anomalous mass M of the sphere were concentrated at its center. If we measure horizontal position from a point above its center, at distance x the vertical component Δg_z is given by

$$\Delta g_z = \Delta g \sin \theta = G \frac{M z}{r^2} \quad (2.82)$$

where

$$M = \frac{4}{3}\pi R^3 \Delta\rho \quad \text{and} \quad r^2 = z^2 + x^2$$

Substituting these expressions into Eq. (2.82) and rearranging terms gives

$$\begin{aligned} \Delta g_z &= \frac{4}{3}\pi G \Delta\rho R^3 \frac{z}{(z^2 + x^2)^{3/2}} \\ &= \frac{4}{3}\pi G \left(\frac{\Delta\rho R^3}{z^2} \right) \left[\frac{1}{(1 + (x/z)^2)} \right]^{3/2} \end{aligned} \quad (2.83)$$

The terms in the first pair of parentheses depend on the size, depth and density contrast of the anomalous sphere. They determine the maximum amplitude of the anomaly, Δg_0 , which is reached over the center of the sphere at $x = 0$. The peak value is given by

$$\Delta g_0 = \frac{4}{3}\pi G \left(\frac{\Delta\rho R^3}{z^2} \right) \quad (2.84)$$

This equation shows how the depth to the center of the sphere affects the peak amplitude of the anomaly; the greater the depth to the center, the smaller the amplitude (Fig. 2.50). For a given depth the same peak anomaly can be produced by numerous combinations of $\Delta\rho$ and R ; a large sphere with a low density contrast can give an identical anomaly to a small sphere with a high density contrast. The gravity data alone do not allow us to resolve this ambiguity.

The terms in the second pair of parentheses in Eq. (2.83) describe how the amplitude of the anomaly varies with distance along the profile. The anomaly is symmetrical with respect to x , reaches a maximum value Δg_0 over the center of the sphere ($x=0$) and decreases to zero at great distances ($x=\infty$). Note that the larger the depth z , the more slowly the amplitude decreases laterally with increasing x . A deep source produces a smaller but broader anomaly than the same source at shallower depth. The width w of the anomaly where the amplitude has one-half its maximum value is called the “half-height width.” The depth z to the center of the sphere is deduced from this anomaly width from the relationship $z = 0.652w$.

2.6.3.2 Horizontal line element

Many geologically interesting structures extend to great distances in one direction but have the same cross-sectional shape along the strike of the structure. If the length along strike were infinite, the two-dimensional variation of density in the area of cross-section would suffice to model the structure. However, this is not really valid as the lateral extent is never infinite. As a general rule, if the length of the structure normal to the profile is more than twenty times its width or depth, it can be treated as two-dimensional (2D). Otherwise, the end effects due to the limited lateral extent of the structure must be taken into account in computing its anomaly. An elongate body that requires end corrections is sometimes referred to as a 2.5D structure. For example, the mass distribution under elongated bodies like anticlines, synclines and faults should be modelled as 2.5D structures. Here, we will handle the simpler two-dimensional models of these structures.

Let an infinitely long linear mass distribution with mass m per unit length extend horizontally along the y -axis at depth z (Fig. 2.51). The contribution $d(\Delta g_z)$ to the vertical gravity anomaly Δg_z at a point on the x -axis due to a small element of length dy is

$$d(\Delta g_z) = G \frac{m dy}{r^2} \sin \theta = G \frac{m dy}{r^2} \frac{z}{r} \quad (2.85)$$

The line element extends to infinity along the positive and negative y -axis, so its vertical gravity anomaly is found by integration:

$$\Delta g_z = Gmz \int_{-\infty}^{\infty} \frac{dy}{r^3} = Gmz \int_{-\infty}^{\infty} \frac{dy}{(u^2 + y^2)^{3/2}} \quad (2.86)$$

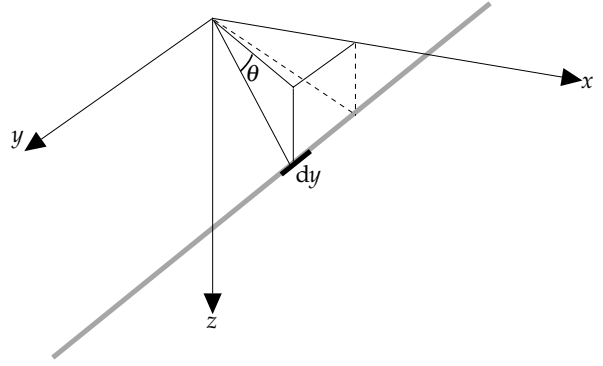


Fig. 2.51 Geometry for calculation of the gravity anomaly of an infinitely long linear mass distribution with mass m per unit length extending horizontally along the y -axis at depth z .

where $u^2 = x^2 + z^2$. The integration is simplified by changing variables, so that $y = u \tan \varphi$; then $dy = u \sec^2 \varphi d\varphi$ and $(u^2 + y^2)^{3/2} = u^3 \sec^3 \varphi$. This gives

$$\Delta g_z = \frac{Gmz}{u^2} \int_{-\pi/2}^{\pi/2} \cos \varphi d\varphi \quad (2.87)$$

which, after evaluation of the integral, gives

$$\Delta g_z = \frac{2Gmz}{z^2 + x^2} \quad (2.88)$$

This expression can be written as the derivative of a potential function Ψ :

$$\Delta g_z = Gm \frac{2z}{u^2} = -\frac{\partial \Psi}{\partial z} \quad (2.89)$$

$$\Psi = Gm \log_e \left(\frac{1}{u} \right) = Gm \log_e \left(\frac{1}{\sqrt{x^2 + z^2}} \right) \quad (2.90)$$

Ψ is called the *logarithmic potential*. Equations (2.88) and (2.90) are useful results for deriving formulas for the gravity anomaly of linear structures like an anticline (or syncline) or a fault.

2.6.3.3 Horizontal cylinder: model for anticline or syncline

The gravity anomaly of an anticline can be modelled by assuming that the upward folding of strata brings rocks with higher density nearer to the surface (Fig. 2.52a), thereby causing a positive density contrast. A syncline is modelled by assuming that its core is filled with strata of lower density that cause a negative density contrast. In each case the geometric model of the structure is an infinite horizontal cylinder (Fig. 2.52b).

A horizontal cylinder may be regarded as composed of numerous line elements parallel to its axis. The cross-sectional area of an element (Fig. 2.53) gives a mass anomaly per unit length, $m = \Delta\rho r d\theta dr$. The contribution $d\Psi$ of a line element to the potential at the surface is

$$d\Psi = 2G\Delta\rho \log_e \left(\frac{1}{u} \right) r dr d\theta \quad (2.91)$$

Fig. 2.52 Calculation of the gravity anomaly of an anticline: (a) structural cross-section, and (b) geometric model by an infinite horizontal cylinder.

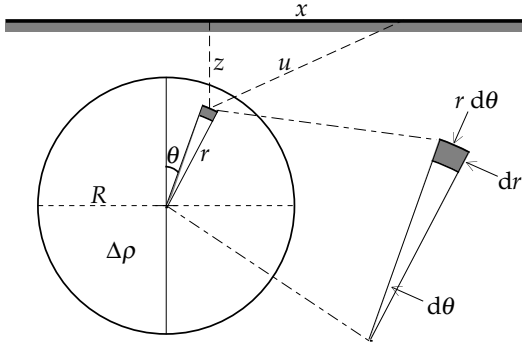
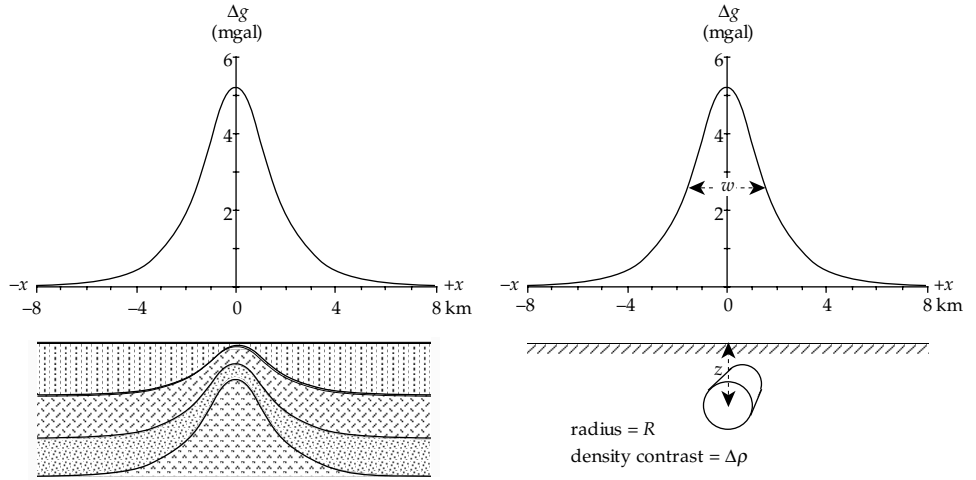


Fig. 2.53 Cross-sectional geometry for calculating the gravity anomaly of a buried horizontal cylinder made up of line elements parallel to its axis.

Integrating over the cross-section of the cylinder gives its potential Ψ ; the vertical gravity anomaly of the cylinder is then found by differentiating Ψ with respect to z . Noting that $du/dz = z/u$ we get

$$\begin{aligned} \Delta g_z &= -\frac{\partial \Psi}{\partial z} = -\frac{z}{u} \frac{\partial \Psi}{\partial u} \\ &= -\frac{2G\Delta\rho}{u} \int_0^{2\pi} \int_0^R \frac{\partial}{\partial u} \log_e \left(\frac{1}{u} \right) r dr d\theta \end{aligned} \quad (2.92)$$

After first carrying out the differentiation within the integral, this simplifies to

$$\Delta g_z = \frac{2G\Delta\rho z}{u^2} \int_0^{2\pi} \int_0^R r dr d\theta = \frac{2\pi GR^2\Delta\rho z}{x^2 + z^2} \quad (2.93)$$

Comparing Eq. (2.93) and Eq. (2.88) it is evident that the gravity anomaly of the cylinder is the same as that of a linear mass element concentrated along its axis, with mass $m = \pi R^2 \Delta\rho$ per unit length along the strike of the structure. The anomaly can be written

$$\Delta g_z = 2\pi G \left(\frac{\Delta\rho R^2}{z} \right) \left[\frac{1}{1 + (x/z)^2} \right] \quad (2.94)$$

The shape of the anomaly on a profile normal to the structure (Fig. 2.52b) resembles that of a sphere (Fig. 2.50). The central peak value Δg_0 is given by

$$\Delta g_0 = 2\pi G \left(\frac{\Delta\rho R^2}{z} \right) \quad (2.95)$$

The anomaly of a horizontal cylinder decreases laterally less rapidly than that of a sphere, due to the long extent of the cylinder normal to the profile. The “half-height width” w of the anomaly is again dependent on the depth z to the axis of the cylinder; in this case the depth is given by $z = 0.5w$.

2.6.3.4 Horizontal thin sheet

We next compute the anomaly of a thin horizontal ribbon of infinite length normal to the plane of the profile. This is done by fictively replacing the ribbon with numerous infinitely long line elements laid side by side. Let the depth of the sheet be z , its thickness t and its density contrast $\Delta\rho$ (Fig. 2.54a); the mass per unit length in the y -direction of a line element of width dx is $(\Delta\rho t dx)$. Substituting in Eq. (2.88) gives the gravity anomaly of the line element; the anomaly of the thin ribbon is then computed by integrating between the limits x_1 and x_2 (Fig. 2.54b)

$$\begin{aligned} \Delta g_z &= 2G\Delta\rho t z \int_{x_1}^{x_2} \frac{dx}{x^2 + z^2} \\ &= 2G\Delta\rho t \left[\tan^{-1} \left(\frac{x_2}{z} \right) - \tan^{-1} \left(\frac{x_1}{z} \right) \right] \end{aligned} \quad (2.96)$$

Writing $\tan^{-1}(x_1/z) = \phi_1$ and $\tan^{-1}(x_2/z) = \phi_2$ as in Fig. 2.54b the equation becomes

$$\Delta g_z = 2G\Delta\rho t [\phi_2 - \phi_1] \quad (2.97)$$

i.e., the gravity anomaly of the horizontal ribbon is proportional to the angle it subtends at the point of measurement.

The anomaly of a semi-infinite horizontal sheet is a limiting case of this result. For easier reference, the origin of x is moved to the edge of the sheet, so that distances to the left are negative and those to the right are positive (Fig. 2.54c). This makes $\phi_1 = -\tan^{-1}(x/z)$. The remote end of the sheet is at infinity, and $\phi_2 = \pi/2$. The gravity

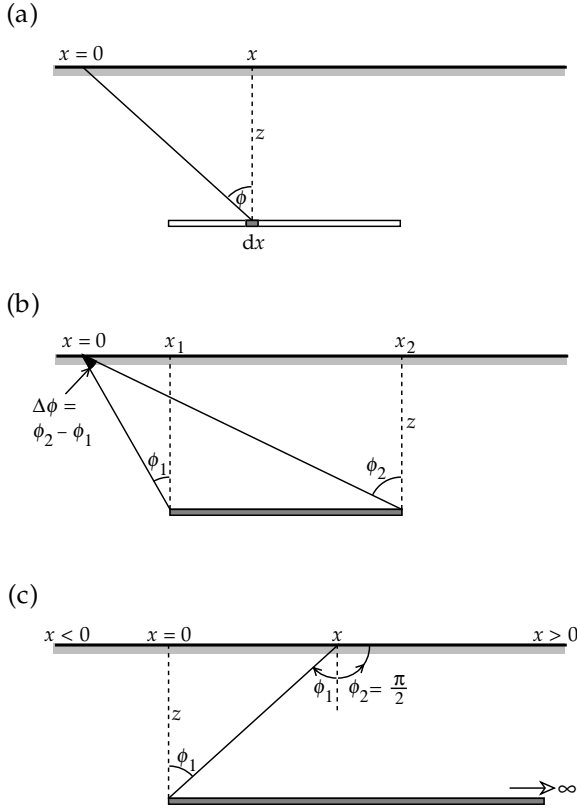


Fig. 2.54 Geometry for computation of the gravity anomaly across a horizontal thin sheet: (a) subdivision of the ribbon into line elements of width dx , (b) thin ribbon between the horizontal limits x_1 and x_2 , and (c) semi-infinite horizontal thin sheet.

anomaly is then

$$\Delta g_z = 2G\Delta\rho t \left[\frac{\pi}{2} + \tan^{-1} \left(\frac{x}{z} \right) \right] \quad (2.98)$$

A further example is the infinite horizontal sheet, which extends to infinity in the positive and negative x and y directions. With $\phi_2 = \pi/2$ and $\phi_1 = -\pi/2$ the anomaly is

$$\Delta g_z = 2\pi G\Delta\rho t \quad (2.99)$$

which is the same as the expression for the Bouguer plate correction (Eq. (2.73)).

2.6.3.5 Horizontal slab: model for a vertical fault

The gravity anomaly across a vertical fault increases progressively to a maximum value over the uplifted side (Fig. 2.55a). This is interpreted as due to the upward displacement of denser material, which causes a horizontal density contrast across a vertical step of height h (Fig. 2.55b). The faulted block can be modelled as a semi-infinite horizontal slab of height h and density contrast $\Delta\rho$ with its mid-point at depth z_0 (Fig. 2.55c).

Let the slab be divided into thin, semi-infinite horizontal sheets of thickness dz at depth z . The gravity anomaly of a given sheet is given by Eq. (2.98) with dz for the thickness t . The anomaly of the semi-infinite slab is found

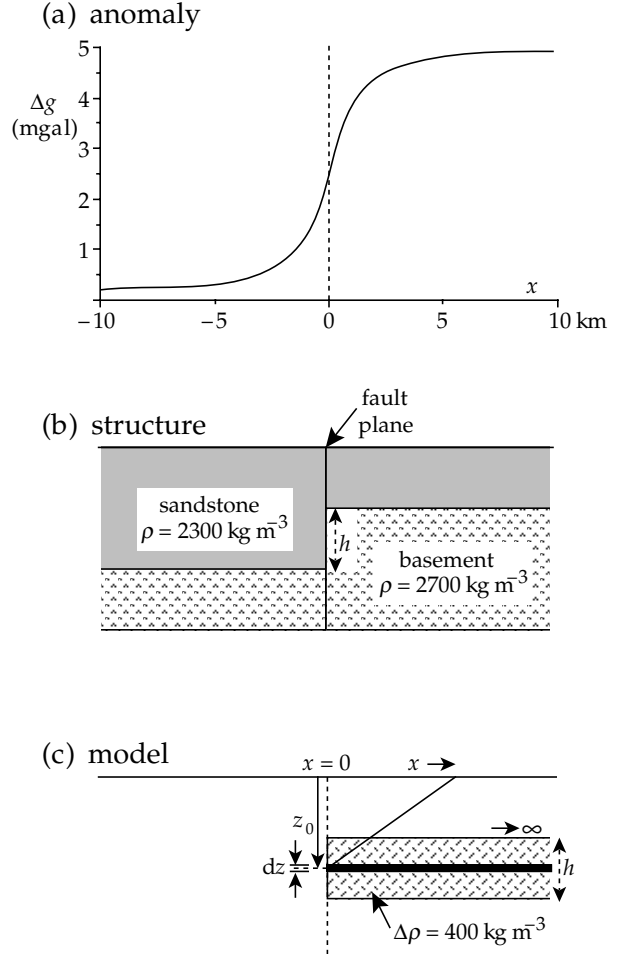


Fig. 2.55 (a) The gravity anomaly across a vertical fault; (b) structure of a fault with vertical displacement h , and (c) model of the anomalous body as a semi-infinite horizontal slab of height h .

by integrating with respect to z over the thickness of the slab; the limits of integration are $z - (h/2)$ and $z + (h/2)$. After slightly rearranging terms this gives

$$\Delta g_z = 2G\Delta\rho h \left[\frac{\pi}{2} + \frac{1}{h} \int_{z_0 - h/2}^{z_0 + h/2} \tan^{-1} \left(\frac{x}{z} \right) dz \right] \quad (2.100)$$

The second expression in the brackets is the mean value of the angle $\tan^{-1}(x/z)$ averaged over the height of the fault step. This can be replaced to a good approximation by the value at the mid-point of the step, at depth z_0 . This gives

$$\Delta g_z = 2G\Delta\rho h \left[\frac{\pi}{2} + \tan^{-1} \left(\frac{x}{z_0} \right) \right] \quad (2.101)$$

Comparison of this expression with Eq. (2.98) shows that the anomaly of the vertical fault (or a semi-infinite thick horizontal slab) is the same as if the anomalous slab were replaced by a thin sheet of thickness h at the mid-point of the vertical step. Equation (2.101) is called the “thin-sheet approximation.” It is accurate to about 2% provided that $z_0 > 2h$.

2.6.3.6 Iterative modelling

The simple geometric models used to compute the gravity anomalies in the previous sections are crude representations of the real anomalous bodies. Modern computer algorithms have radically changed modelling methods by facilitating the use of an *iterative* procedure. A starting model with an assumed geometry and density contrast is postulated for the anomalous body. The gravity anomaly of the body is then computed and compared with the residual anomaly. The parameters of the model are changed slightly and the computation is repeated until the discrepancies between the model anomaly and the residual anomaly are smaller than a determined value. However, as in the case of the simple models, this does not give a unique solution for the density distribution.

Two- and three-dimensional iterative techniques are in widespread use. The two-dimensional (2D) method assumes that the anomalous body is infinitely long parallel to the strike of the structure, but end corrections for the possibly limited horizontal extent of the body can be made. We can imagine that the cross-sectional shape of the body is replaced by countless thin rods or line elements aligned parallel to the strike. Each rod makes a contribution to the vertical component of gravity at the origin (Fig. 2.56a). The gravity anomaly of the structure is calculated by adding up the contributions of all the line elements; mathematically, this is an integration over the end surface of the body. Although the theory is beyond the scope of this chapter, the gravity anomaly has the following simple form:

$$\Delta g_z = 2G\Delta\rho \oint z \, d\theta \quad (2.102)$$

The angle θ is defined to lie between the positive x -axis and the radius from the origin to a line element (Fig. 2.56a), and the integration over the end-surface has been changed to an integration around its boundary. The computer algorithm for the calculation of this integral is greatly speeded up by replacing the true cross-sectional shape with an N -sided polygon (Fig. 2.56a). Apart from the assumed density contrast, the only important parameters for the computation are the (x, z) coordinates of the corners of the polygon. The origin is now moved to the next point on a profile across the structure. This move changes only the x -coordinates of the corners of the polygon. The calculations are repeated for each successive point on the profile. Finally, the calculated gravity anomaly profile across the structure is compared to the observed anomaly and the residual differences are evaluated. The coordinates of the corners of the polygon are adjusted and the calculation is reiterated until the residuals are less than a selected tolerance level.

The gravity anomaly of a three-dimensional (3D) body is modelled in a similar way. Suppose that we have a contour map of the body; the contour lines show the smooth outline of the body at different depths. We could construct a close replica of the body by replacing the

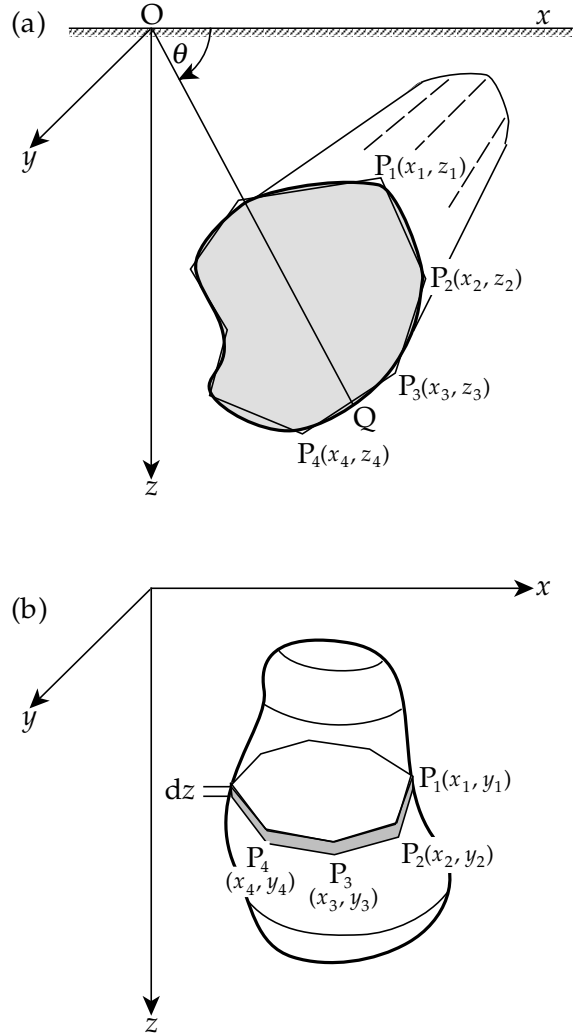


Fig. 2.56 Methods of computing gravity anomalies of irregular bodies: (a) the cross-section of a two-dimensional structure can be replaced with a multi-sided polygon (Talwani *et al.*, 1959); (b) a three-dimensional body can be replaced with thin horizontal laminae (Talwani and Ewing, 1960).

material between successive contour lines with thin laminae. Each lamina has the same outline as the contour line and has a thickness equal to the contour separation. As a further approximation the smooth outline of each lamina is replaced by a multi-sided polygon (Fig. 2.56b). The gravity anomaly of the polygon at the origin is computed as in the 2D case, using the (x, y) coordinates of the corners, the thickness of the lamina and an assumed density contrast. The gravity anomaly of the 3D body at the origin is found by adding up the contributions of all the laminae. As in the simpler 2D example, the origin is now displaced to a new point and the computation is repeated. The calculated and observed anomalies are now compared and the coordinates of the corners of the laminae are adjusted accordingly; the assumed density distribution can also be adjusted. The iterative procedure is repeated until the desired match between computed and observed anomalies is obtained.

2.6.4 Some important regional gravity anomalies

Without auxiliary information the interpretation of gravity anomalies is ambiguous, because the same anomaly can be produced by different bodies. An independent data source is needed to restrict the choices of density contrast, size, shape and depth in the many possible gravity models. The additional information may be in the form of surface geological observations, from which the continuation of structures at depth is interpreted. Seismic refraction or reflection data provide better constraints.

The combination of seismic refraction experiments with precise gravity measurements has a long, successful history in the development of models of crustal structure. Refraction seismic profiles parallel to the trend of elongate geological structures give reliable information about the vertical velocity distribution. However, refraction profiles normal to the structural trend give uncertain information about the tilt of layers or lateral velocity changes. Lateral changes of crustal structure can be interpreted from several refraction profiles more or less parallel to the structural trend or from seismic reflection data. The refraction results give layer velocities and the depths to refracting interfaces. To compute the gravity effect of a structure the velocity distribution must first be converted to a density model using a P-wave velocity–density relationship like the curve shown in Fig. 2.40. The theoretical gravity anomaly over the structure is computed using a 2D or 3D method. Comparison with the observed gravity anomaly (e.g., by calculating the residual differences point by point) indicates the plausibility of the model. It is always important to keep in mind that, because of the non-uniqueness of gravity modelling, a plausible structure is not necessarily the true structure. Despite the ambiguities, some characteristic features of gravity anomalies have been established for important regions of the Earth.

2.6.4.1 Continental and oceanic gravity anomalies

In examining the shape of the Earth we saw that the ideal reference figure is a spheroid, or ellipsoid of rotation. It is assumed that the reference Earth is in hydrostatic equilibrium. This is supported by observations of free-air and isostatic anomalies which suggest that, except in some unusual locations such as deep-sea trenches and island arcs in subduction zones, the continents and oceans are in approximate isostatic equilibrium with each other. By applying the concepts of isostasy (Section 2.7) we can understand the large-scale differences between Bouguer gravity anomalies over the continents and those over the oceans. In general, Bouguer anomalies over the continents are negative, especially over mountain ranges where the crust is unusually thick; in contrast, strongly positive Bouguer anomalies are found over oceanic regions where the crust is very thin.

The inverse relationship between Bouguer anomaly amplitude and crustal thickness can be explained with the

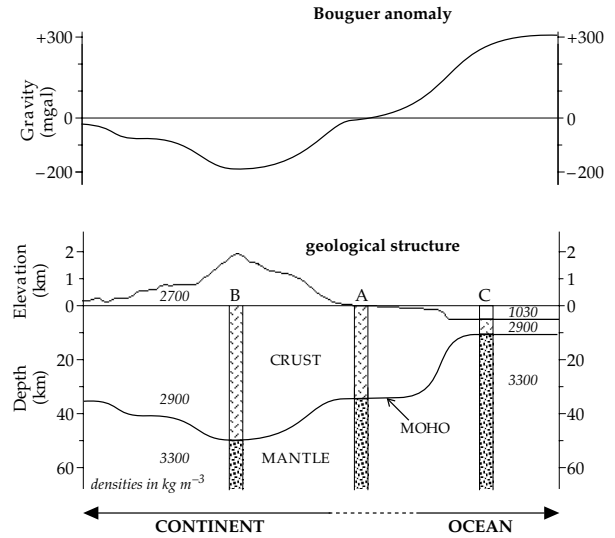
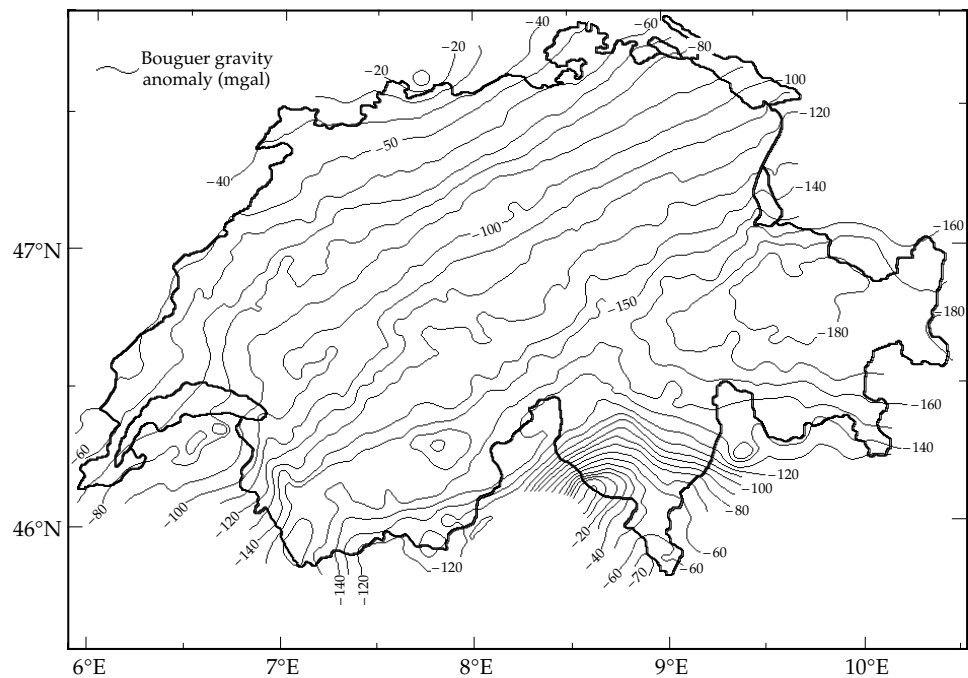


Fig. 2.57 Hypothetical Bouguer anomalies over continental and oceanic areas. The regional Bouguer anomaly varies roughly inversely with crustal thickness and topographic elevation (after Robinson and Çoruh, 1988).

aid of a hypothetical example (Fig. 2.57). Continental crust that has not been thickened or thinned by tectonic processes is considered to be “normal” crust. It is typically 30–35 km thick. Under location A on an undeformed continental coastal region a thickness of 34 km is assumed. The theoretical gravity used in computing a gravity anomaly is defined on the reference ellipsoid, the surface of which corresponds to mean sea-level. Thus, at coastal location A on normally thick continental crust the Bouguer anomaly is close to zero. Isostatic compensation of the mountain range gives it a root-zone that increases the crustal thickness at location B. Seismic evidence shows that continental crustal density increases with depth from about 2700 kg m^{-3} in the upper granitic crust to about 2900 kg m^{-3} in the lower gabbroic crust. Thus, the density in the root-zone is much lower than the typical mantle density of $3300\text{--}3400 \text{ kg m}^{-3}$ at the same depth under A. The low-density root beneath B causes a negative Bouguer anomaly, which typically reaches -150 to -200 mgal .

At oceanic location C the vertical crustal structure is very different. Two effects contribute to the Bouguer anomaly. A 5 km thick layer of sea-water (density 1030 kg m^{-3}) overlies the thin basic oceanic crust (density 2900 kg m^{-3}) which has an average thickness of only about 6 km. To compute the Bouguer anomaly the sea-water must be replaced by oceanic crustal rock. The attraction of the water layer is inherent in the measured gravity so the density used in correcting for the Bouguer plate and the topography of the ocean bottom is the reduced density of the oceanic crust (i.e., $2900 - 1030 = 1870 \text{ kg m}^{-3}$). However, a more important effect is that the top of the mantle is at a depth of only 11 km. In a vertical section below this depth the mantle has a density of $3300\text{--}3400 \text{ kg m}^{-3}$, much higher than the density of the continental crust at equivalent depths below site A.

Fig. 2.58 Bouguer gravity map of Switzerland (after Klingelé and Olivier, 1980).



The lower 23 km of the section beneath C represents a large excess of mass. This gives rise to a strong positive Bouguer anomaly, which can amount to 300–400 mgal.

2.6.4.2 Gravity anomalies across mountain chains

The typical gravity anomaly across a mountain chain is strongly negative due to the large low-density root-zone. The Swiss Alps provide a good example of the interpretation of such a gravity anomaly with the aid of seismic refraction and reflection results. A precise gravity survey of Switzerland carried out in the 1970s yielded an accurate Bouguer gravity map (Fig. 2.58). The map contains effects specific to the Alps. Most obviously, the contour lines are parallel to the trend of the mountain range. In the south a strong positive anomaly overrides the negative anomaly. This is the northern extension of the positive anomaly of the so-called Ivrea body, which is a high-density wedge of mantle material that was forced into an uplifted position within the western Alpine crust during an earlier continental collision. In addition, the Swiss gravity map contains the effects of low-density sediments that fill the Molasse basin north of the Alps, the Po plain to the south and the major Alpine valleys.

In the late 1980s a coordinated geological and geophysical study – the European Geotraverse (EGT) – was made along and adjacent to a narrow path stretching from northern Scandinavia to northern Africa. Detailed reflection seismic profiles along its transect through the Central Swiss Alps complemented a large amount of new and extant refraction data. The seismic results gave the depths to important interfaces. Assuming a velocity–density relationship, a model of the density distribution in the lithosphere under the traverse was obtained. Making appropriate

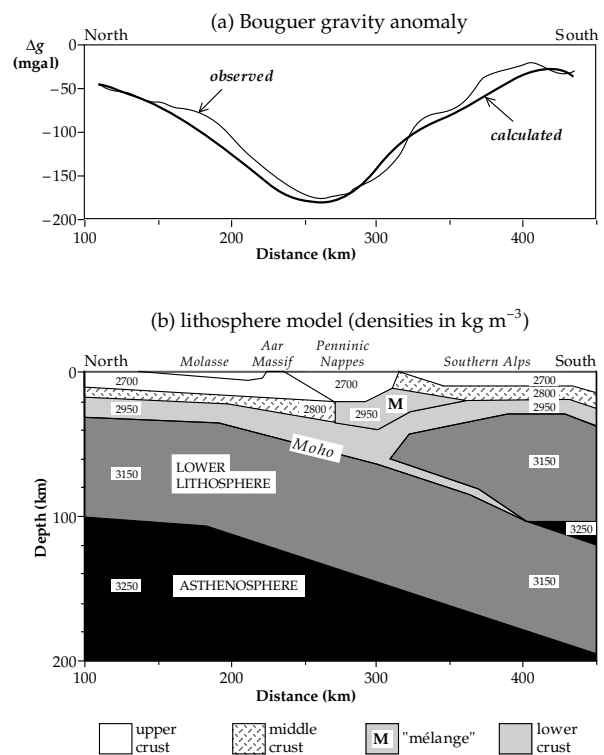
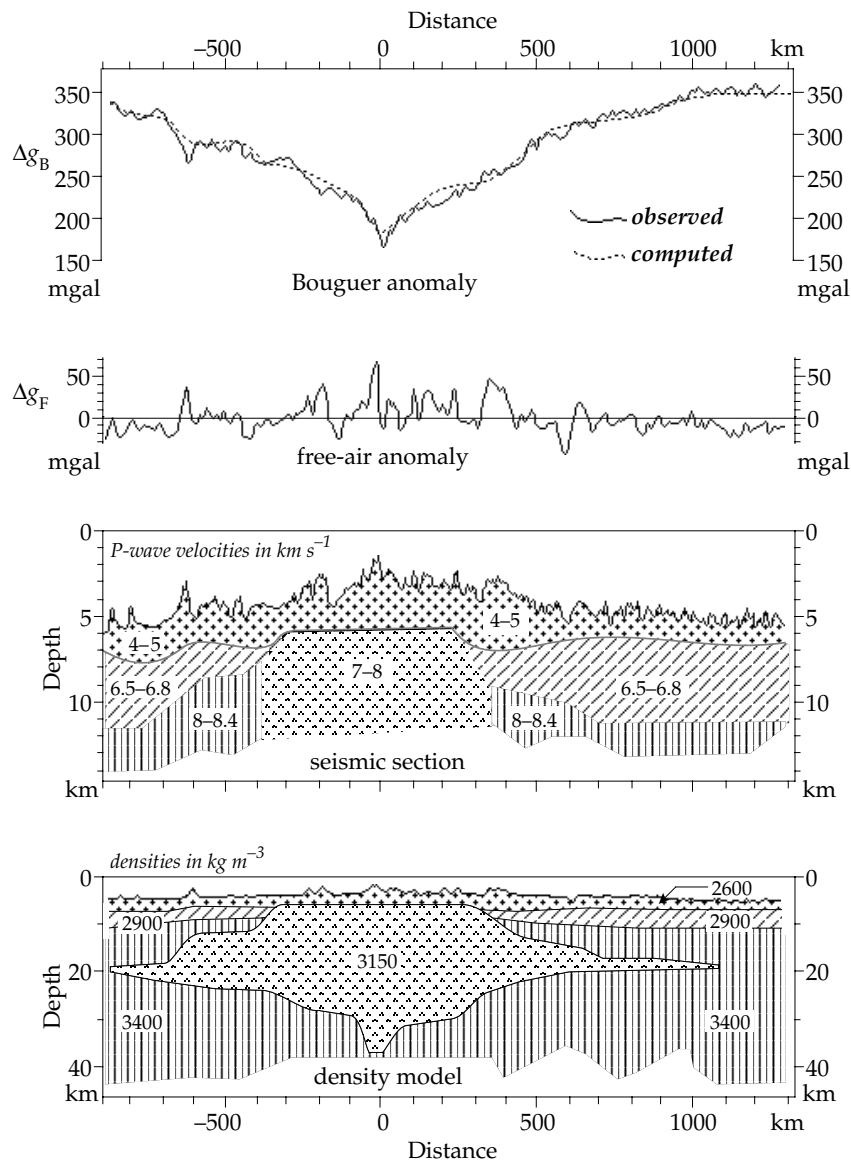


Fig. 2.59 Lithosphere density model for the Central Swiss Alps along the European Geotraverse transect, compiled from seismic refraction and reflection profiles. The 2.5D gravity anomaly calculated for this lithospheric structure is compared to the observed Bouguer anomaly after removal of the effects of the high-density Ivrea body and the low-density sediments in the Molasse basin, Po plain and larger Alpine valleys (after Holliger and Kissling, 1992).

Fig. 2.60 Bouguer and free-air gravity anomalies over the Mid-Atlantic Ridge near 32°N. The seismic section is projected onto the gravity profile. The gravity anomaly computed from the density model fits the observed anomaly well, but is non-unique (after Talwani *et al.*, 1965).

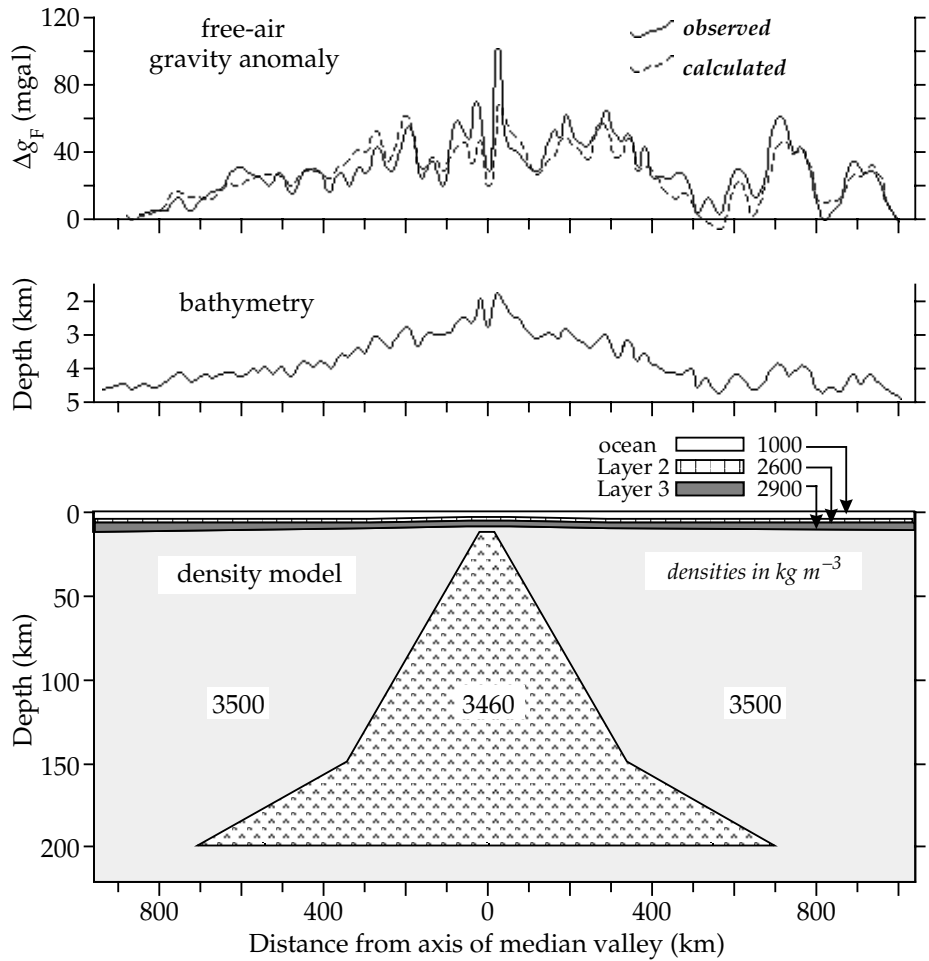


corrections for end effects due to limited extent along strike, a 2.5D gravity anomaly was calculated for this lithospheric structure (Fig. 2.59). After removal of the effects of the high-density Ivrea body and the low-density sediments, the corrected Bouguer gravity profile is reproduced well by the anomaly of the lithospheric model. Using the geometric constraints provided by the seismic data, the lithospheric gravity model favors a subduction zone, dipping gently to the south, with a high-density wedge of deformed lower crustal rock (“mélange”) in the middle crust. As already noted, the fact that a density model delivers a gravity anomaly that agrees well with observation does not establish the reality of the interpreted structure, which can only be confirmed by further seismic imaging. However, the gravity model provides an important check on the reasonableness of suggested models. A model of crustal or lithospheric structure that does not give an appropriate gravity anomaly can reasonably be excluded.

2.6.4.3 Gravity anomalies across an oceanic ridge

An oceanic ridge system is a gigantic submarine mountain range. The difference in depth between the ridge crest and adjacent ocean basin is about 3 km. The ridge system extends laterally for several hundred kilometers on either side of the axis. Gravity and seismic surveys have been carried out across several oceanic ridge systems. Some common characteristics of continuous gravity profiles across a ridge are evident on a WNW–ESE transect crossing the Mid-Atlantic Ridge at 32°N (Fig. 2.60). The free-air gravity anomalies are small, around 50 mgal or less, and correlate closely with the variations in ocean-bottom topography. This indicates that the ridge and its flanks are nearly compensated isostatically. As expected for an oceanic profile, the Bouguer anomaly is strongly positive. It is greater than 350 mgal at distances beyond 1000 km from the ridge, but decreases to less than 200 mgal over the axis of the ridge.

Fig. 2.61 Free-air gravity anomaly across the Mid-Atlantic Ridge near 46°N, and the lithospheric density model for the computed anomaly (after Keen and Tramontini, 1970).

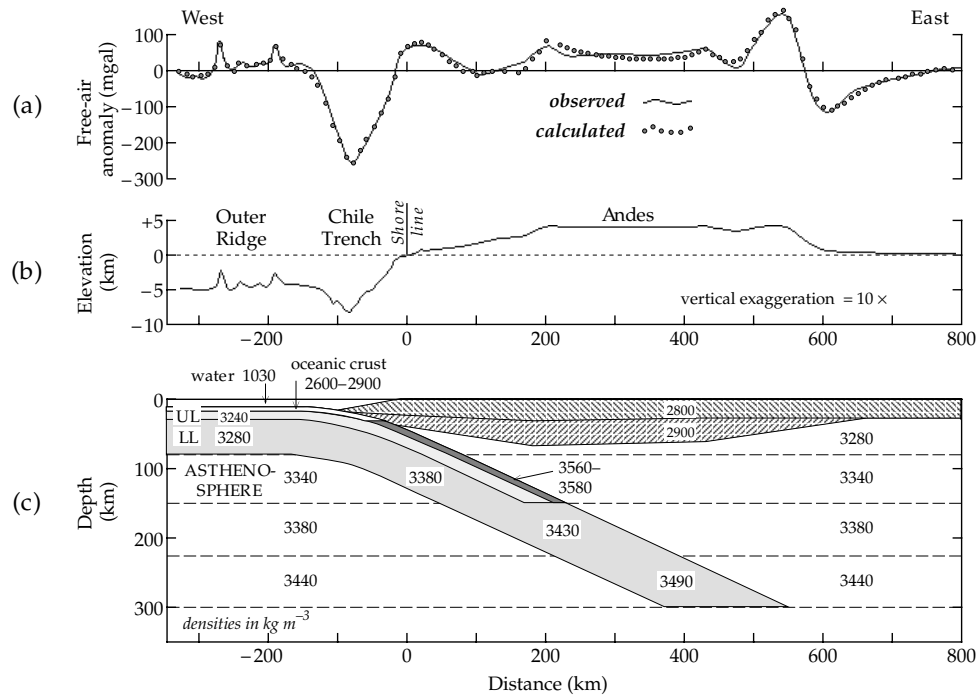


The depths to important refracting interfaces and the P-wave layer velocities are known from seismic refraction profiles parallel to the ridge. The seismic structure away from the ridge is layered, with P-wave velocities of 4–5 km s⁻¹ for the basalts and gabbros in oceanic Layer 2 and 6.5–6.8 km s⁻¹ for the meta-basalts and meta-gabbros in Layer 3; the Moho is at about 11 km depth, below which typical upper-mantle velocities of 8–8.4 km s⁻¹ are found. However, the layered structure breaks down under the ridge at distances less than 400 km from its axial zone. Unusual velocities around 7.3 km s⁻¹ occurred at several places, suggesting the presence of anomalous low-density mantle material at comparatively shallow depth beneath the ridge. The seismic structure was converted to a density model using a velocity–density relationship. Assuming a 2D structure, several density models were found that closely reproduced the Bouguer anomaly. However, to satisfy the Bouguer anomaly on the ridge flanks each model requires a flat body in the upper mantle beneath the ridge; it extends down to about 30 km depth and for nearly 1000 km on each side of the axis (Fig. 2.60). The density of the anomalous structure is only 3150 kg m⁻³ instead of the usual 3400 kg m⁻³. The model was proposed before the theory of plate tectonics was

accepted. The anomalous upper mantle structure satisfies the gravity anomaly but has no relation to the known physical structure of a constructive plate margin. A broad zone of low upper-mantle seismic velocities was not found in later experiments, but narrow low-velocity zones are sometimes present close to the ridge axis.

A further combined seismic and gravity study of the Mid-Atlantic Ridge near 46°N gave a contradictory density model. Seismic refraction results yielded P-wave velocities of 4.6 km s⁻¹ and 6.6 km s⁻¹ for Layer 2 and Layer 3, respectively, but did not show anomalous mantle velocities beneath the ridge except under the median valley. A simpler density model was postulated to account for the gravity anomaly (Fig. 2.61). The small-scale free-air anomalies are accounted for by the variations in ridge topography seen in the bathymetric plot. The large-scale free-air gravity anomaly is reproduced well by a wedge-shaped structure that extends to 200 km depth. Its base extends to hundreds of kilometers on each side of the axis. A very small density contrast of only –40 kg m⁻³ suffices to explain the broad free-air anomaly. The model is compatible with the thermal structure of an accreting plate margin. The low-density zone may be associated with hot material from the asthenosphere, which rises

Fig. 2.62 Observed and computed free-air gravity anomalies across a subduction zone. The density model for the computed anomaly is based on seismic, thermal and petrological data. The profile crosses the Chile trench and Andes mountains at 23°S (after Grow and Bowin, 1975).



beneath the ridge, melts and accumulates in a shallow magma chamber within the oceanic crust.

2.6.4.4 Gravity anomalies at subduction zones

Subduction zones are found primarily at continental margins and island arcs. Elongate, narrow and intense isostatic and free-air gravity anomalies have long been associated with island arcs. The relationship of gravity to the structure of a subduction zone is illustrated by the free-air anomaly across the Chile trench at 23°S (Fig. 2.62). Seismic refraction data define the thicknesses of the oceanic and continental crust. Thermal and petrological data are integrated to give a density model for the structure of the mantle and the subducting lithosphere.

The continental crust is about 65 km thick beneath the Andes mountains, and gives large negative Bouguer anomalies. The free-air gravity anomaly over the Andes is positive, averaging about +50 mgal over the 4 km high plateau. Even stronger anomalies up to +100 mgal are seen over the east and west boundaries of the Andes. This is largely due to the edge effect of the low-density Andean crustal block (see Fig. 2.44b and Section 2.5.6).

A strong positive free-air anomaly of about +70 mgal lies between the Andes and the shore-line of the Pacific ocean. This anomaly is due to the subduction of the Nazca plate beneath South America. The descending slab is old and cool. Subduction exposes it to higher temperatures and pressures, but the slab descends faster than it can be heated up. The increase in density accompanying greater depth and pressure outweighs the decrease in density due to hotter temperatures. There is a positive density contrast between the subducting lithosphere and the surrounding mantle. Also, petrological changes accompanying the sub-

duction result in mass excesses. Peridotite in the upper lithosphere changes phase from plagioclase-type to the higher-density garnet-type. When oceanic crust is subducted to depths of 30–80 km, basalt changes phase to eclogite, which has a higher density (3560–3580 kg m⁻³) than upper mantle rocks. These effects combine to produce the positive free-air anomaly.

The Chile trench is more than 2.5 km deeper than the ocean basin to the west. The sediments flooring the trench have low density. The mass deficiency of the water and sediments in the trench cause a strong negative free-air anomaly, which parallels the trench and has an amplitude greater than –250 mgal. A small positive anomaly of about +20 mgal is present about 100 km seaward of the trench axis. This anomaly is evident also in the mean level of the ocean surface as mapped by SEASAT (Fig. 2.28), which shows that the mean sea surface is raised in front of deep ocean trenches. This is due to upward flexure of the lithosphere before its downward plunge into the subduction zone. The flexure elevates higher-density mantle rocks and thereby causes the small positive free-air anomaly.

2.7 ISOSTASY

2.7.1 The discovery of isostasy

Newton formulated the law of universal gravitation in 1687 and confirmed it with Kepler’s laws of planetary motion. However, in the seventeenth and eighteenth centuries the law could not be used to calculate the mass or mean density of the Earth, because the value of the gravitational constant was not yet known (it was first determined by Cavendish in 1798). Meanwhile, eighteenth century scientists attempted to estimate the mean density of the Earth

by various means. They involved comparing the attraction of the Earth with that of a suitable mountain, which could be calculated. Inconsistent results were obtained.

During the French expedition to Peru in 1737–1740, Pierre Bouguer measured gravity with a pendulum at different altitudes, applying the elevation correction term which now bears his name. If the density of crustal rocks is ρ and the mean density of the Earth is ρ_0 , the ratio of the Bouguer-plate correction (see Section 2.5.4.3) for elevation h to mean gravity for a spherical Earth of radius R is

$$\frac{\Delta g_{BP}}{g} = \frac{2\pi G\rho h}{\frac{4}{3}\pi G\rho_0 R} = \frac{3}{2} \left(\frac{\rho}{\rho_0} \right) \left(\frac{h}{R} \right) \quad (2.103)$$

From the results he obtained near Quito, Bouguer estimated that the mean density of the Earth was about 4.5 times the density of crustal rocks.

The main method employed by Bouguer to determine the Earth's mean density consisted of measuring the deflection of the plumb-line (vertical direction) by the mass of a nearby mountain (Fig. 2.63). Suppose the elevation of a known star is measured relative to the local vertical direction at points N and S on the same meridian. The elevations should be α_N and α_S , respectively. Their sum is α , the angle subtended at the center of the Earth by the radii to N and S, which corresponds to the difference in latitude. If N and S lie on opposite sides of a large mountain, the plumb-line at each station is deflected by the attraction of the mountain. The measured elevations of the star are β_N and β_S , respectively, and their sum is β . The local vertical directions now intersect at the point D instead of at the center of the (assumed spherical) Earth. The difference $\delta = \beta - \alpha$ is the sum of the deviations of the vertical direction caused by the mass of the mountain.

The horizontal attraction f of the mountain can be calculated from its shape and density, with a method that resembles the computation of the topographic correction in the reduction of gravity measurements. Dividing the mountain into vertical cylindrical elements, the horizontal attraction of each element is calculated and its component ($G\rho h_i$) towards the center of mass of the mountain is found. Summing up the effects of all the cylindrical elements in the mountain gives the horizontal attraction f towards its center of mass. Comparing f with mean gravity g , we then write

$$\tan \delta = \frac{f}{g} = \frac{G\rho \sum h_i}{\frac{4}{3}\pi G\rho_0 R} = \frac{\sum h_i}{\frac{4}{3}\pi R} \left(\frac{\rho}{\rho_0} \right) \quad (2.104)$$

For very small angles, $\tan \delta$ is equal to δ , and so the deflection of the vertical is proportional to the ratio ρ/ρ_0 of the mean densities of the mountain and the Earth. Bouguer measured the deflection of the vertical caused by Mt. Chimborazo (6272 m), Ecuador's highest mountain. His results gave a ratio ρ/ρ_0 of around 12, which is unrealistically large and quite different from the values he had obtained near Quito. The erroneous result indicated

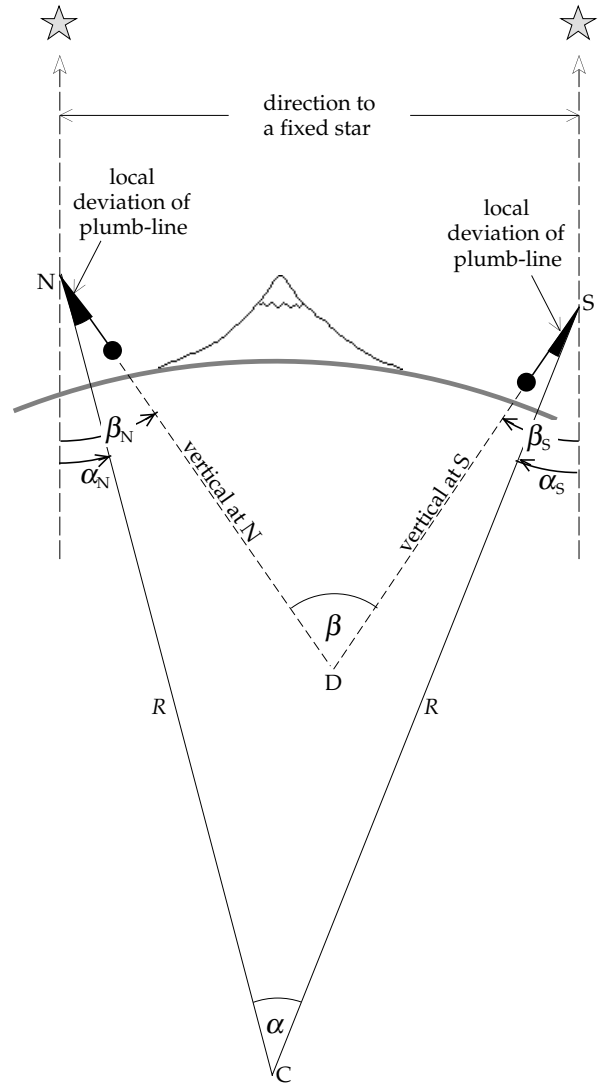


Fig. 2.63 Deviations of the local plumb-line at N and S on opposite sides of a large mountain cause the local vertical directions to intersect at the point D instead of at the center of the Earth.

that the deflection of the vertical caused by the mountain was much too small for its estimated mass.

In 1774 Bouguer's Chimborazo experiment was repeated in Scotland by Neville Maskelyne on behalf of the Royal Society of London. Measurements of the elevations of stars were made on the north and south flanks of Mt. Schiehallion at sites that differed in latitude by 42.9" of arc. The observed angle between the plumb-lines was 54.6". The analysis gave a ratio ρ/ρ_0 equal to 1.79, suggesting a mean density for the Earth of 4500 kg m^{-3} . This was more realistic than Bouguer's result, which still needed explanation.

Further results accumulated in the first half of the nineteenth century. From 1806 to 1843 the English geodesist George Everest carried out triangulation surveys in India. He measured by triangulation the separation of a site at Kalianpur on the Indo-Ganges plain from a site at Kaliana in the foothills of the Himalayas. The distance differed substantially from the separation of the sites

computed from the elevations of stars, as in Fig. 2.63. The discrepancy of 5.23" of arc (162 m) was attributed to deflection of the plumb-line by the mass of the Himalayas. This would affect the astronomic determination but not the triangulation measurement. In 1855 J. H. Pratt computed the minimum deflection of the plumb-line that might be caused by the mass of the Himalayas and found that it should be 15.89" of arc, about three times larger than the observed deflection. Evidently the attraction of the mountain range on the plumb-line was not as large as it should be.

The anomalous deflections of the vertical were first understood in the middle of the nineteenth century, when it was realized that there are regions beneath mountains – “root-zones” – in which rocks have a lower density than expected. The deflection of a plumb-line is not caused only by the horizontal attraction of the visible part of a mountain. The deficiency of mass at depth beneath the mountain means that the “hidden part” exerts a reduced lateral attraction, which partly offsets the effect of the mountain and diminishes the deflection of the vertical. In 1889 C. E. Dutton referred to the compensation of a topographic load by a less-dense subsurface structure as *isostasy*.

2.7.2 Models of isostasy

Separate explanations of the anomalous plumb-line deflections were put forward by G. B. Airy in 1855 and J. H. Pratt in 1859. Airy was the Astronomer Royal and director of the Greenwich Observatory. Pratt was an archdeacon of the Anglican church at Calcutta, India, and a devoted scientist. Their hypotheses have in common the compensation of the extra mass of a mountain above sea-level by a less-dense region (or root) below sea-level, but they differ in the way the compensation is achieved. In the Airy model, when isostatic compensation is complete, the mass deficiency of the root equals the excess load on the surface. At and below a certain *compensation depth* the pressure exerted by all overlying vertical columns of crustal material is then equal. The pressure is then *hydrostatic*, as if the interior acted like a fluid. Hence, isostatic compensation is equivalent to applying Archimedes' principle to the uppermost layers of the Earth.

The Pratt and Airy models achieve compensation locally by equalization of the pressure below vertical columns under a topographic load. The models were very successful and became widely used by geodesists, who developed them further. In 1909–1910, J. F. Hayford in the United States derived a mathematical model to describe the Pratt hypothesis. As a result, this theory of isostasy is often called the Pratt–Hayford scheme of compensation. Between 1924 and 1938 W. A. Heiskanen derived sets of tables for calculating isostatic corrections based on the Airy model. This concept of isostatic compensation has since been referred to as the Airy–Heiskanen scheme.

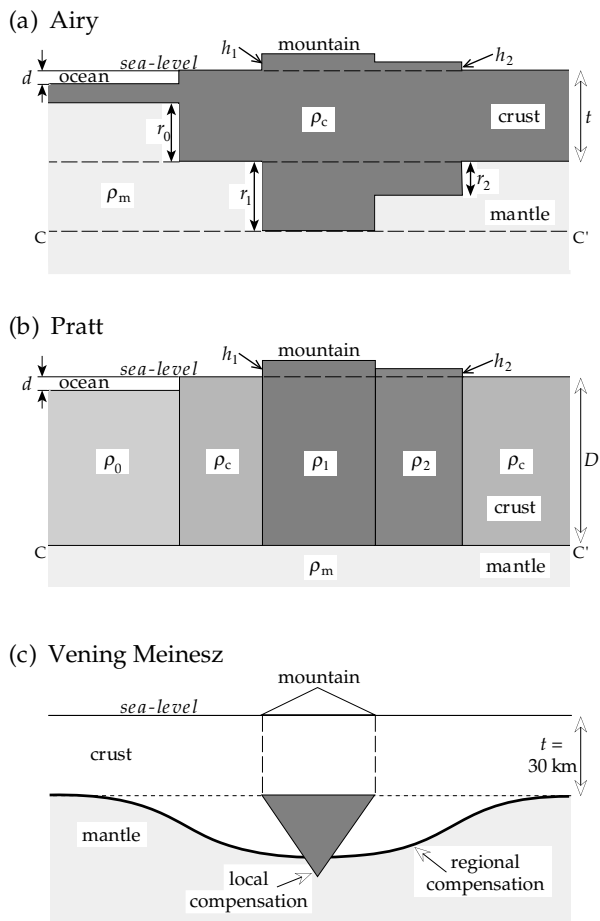


Fig. 2.64 Local isostatic compensation according to (a) the Airy–Heiskanen model and (b) the Pratt–Hayford model; (c) regional compensation according to the elastic plate model of Vening Meinesz.

It became apparent that both models had serious deficiencies in situations that required compensation over a larger region. In 1931 F. A. Vening Meinesz, a Dutch geophysicist, proposed a third model, in which the crust acts as an elastic plate. As in the other models, the crust floats buoyantly on a substratum, but its inherent rigidity spreads topographic loads over a broader region.

2.7.2.1 The Airy–Heiskanen model

According to the Airy–Heiskanen model of isostatic compensation (Fig. 2.64a) an upper layer of the Earth “floats” on a denser magma-like substratum, just as icebergs float in water. The upper layer is equated with the crust and the substratum with the mantle. The height of a mountain above sea-level is much less than the thickness of the crust underneath it, just as the visible tip of an iceberg is much smaller than the subsurface part. The densities of the crust and mantle are assumed to be constant; the thickness of the root-zone varies in proportion to the elevation of the topography.

The analogy to an iceberg is not exact, because under land at sea-level the “normal” crust is already about 30–35 km thick; the compensating root-zone of a

mountain lies below this depth. Oceanic crust is only about 10 km thick, thinner than the “normal” crust. The mantle between the base of the oceanic crust and the normal crustal depth is sometimes called the *anti-root* of the ocean basin.

The Airy–Heiskanen model assumes local isostatic compensation, i.e., the root-zone of a mountain lies directly under it. Isostasy is assumed to be complete, so that hydrostatic equilibrium exists at the compensation depth, which is equivalent to the base of the deepest mountain root. The pressure at this level is due to the weight of the rock material in the overlying vertical column (of basal area one square meter) extending to the Earth’s surface. The vertical column for the mountain of height h_1 in Fig. 2.64a contains only crustal rocks of density ρ_c . The pressure at CC’ due to the mountain, “normal” crust of thickness t , and a root-zone of thickness r_1 amounts to $(h_1 + t + r_1)\rho_c$. The vertical column below the “normal” crust contains a thickness t of crustal rocks and thickness r_1 of mantle rocks; it exerts a pressure of $(t\rho_c + r_1\rho_m)$. For hydrostatic equilibrium the pressures are equal. Equating, and noting that each expression contains the term $t\rho_c$, we get

$$r_1 = \frac{\rho_c}{\rho_m - \rho_c} h_1 \quad (2.105)$$

with a similar expression for the root of depth r_2 under the hill of height h_2 . The thickness r_0 of the anti-root of the oceanic crust under an ocean basin of water depth d and density ρ_w is given by

$$r_0 = \frac{\rho_c - \rho_w}{\rho_m - \rho_c} d \quad (2.106)$$

The Airy–Heiskanen model assumes an upper layer of constant density floating on a more dense substratum. It has root-zones of variable thickness proportional to the overlying topography. This scenario agrees broadly with seismic evidence for the thickness of the Earth’s crust (see Section 3.7). The continental crust is much thicker than the oceanic crust. Its thickness is very variable, being largest below mountain chains, although the greatest thickness is not always under the highest topography. Airy-type compensation suggests hydrostatic balance between the crust and the mantle.

2.7.2.2 The Pratt–Hayford model

The Pratt–Hayford isostatic model incorporates an outer layer of the Earth that rests on a weak magmatic substratum. Differential expansion of the material in vertical columns of the outer layer accounts for the surface topography, so that the higher the column above a common base the lower the mean density of rocks in it. The vertical columns have constant density from the surface to their base at depth D below sea-level (Fig. 2.64b). If the rock beneath a mountain of height h_i ($i = 1, 2, \dots$) has density ρ_i , the pressure at CC’ is $\rho_i(h_i + D)$.

Beneath a continental region at sea-level the pressure of the rock column of density ρ_c is $\rho_c D$. Under an ocean basin the pressure at CC’ is due to water of depth d and density ρ_w on top of a rock column of thickness $(D - d)$ and density ρ_0 ; it is equal to $\rho_w d + \rho_0(D - d)$. Equating these pressures, we get

$$\rho_i = \frac{D}{h_i + D} \rho_c \quad (2.107)$$

for the density below a topographic elevation h_i , and

$$\rho_0 = \frac{\rho_c D - \rho_w d}{D - d} \quad (2.108)$$

for the density under an oceanic basin of depth d . The compensation depth D is about 100 km.

The Pratt–Hayford and Airy–Heiskanen models represent *local isostatic compensation*, in which each column exerts an equal pressure at the compensation level. At the time these models were proposed very little was yet known about the internal structure of the Earth. This was only deciphered after the development of seismology in the late nineteenth and early twentieth century. Each model is idealized, both with regard to the density distributions and the behavior of Earth materials. For example, the upper layer is assumed to offer no resistance to shear stresses arising from vertical adjustments between adjacent columns. Yet the layer has sufficient strength to resist stresses due to horizontal differences in density. It is implausible that small topographic features require compensation at large depths; more likely, they are entirely supported by the strength of the Earth’s crust.

2.7.2.3 Vening Meinesz elastic plate model

In the 1920s F. A. Vening Meinesz made extensive gravity surveys at sea. His measurements were made in a submarine to avoid the disturbances of wave motions. He studied the relationship between topography and gravity anomalies over prominent topographic features, such as the deep sea trenches and island arcs in southeastern Asia, and concluded that isostatic compensation is often not entirely local. In 1931 he proposed a model of *regional isostatic compensation* which, like the Pratt–Hayford and Airy–Heiskanen models, envisages a light upper layer that floats on a denser fluid substratum. However, in the Vening Meinesz model the upper layer behaves like an elastic plate overlying a weak fluid. The strength of the plate distributes the load of a surface feature (e.g., an island or seamount) over a horizontal distance wider than the feature (Fig. 2.64c). The topographic load bends the plate downward into the fluid substratum, which is pushed aside. The buoyancy of the displaced fluid forces it upward, giving support to the bent plate at distances well away from the central depression. The bending of the plate which accounts for the regional compensation in the Vening Meinesz model depends on the elastic properties of the lithosphere.

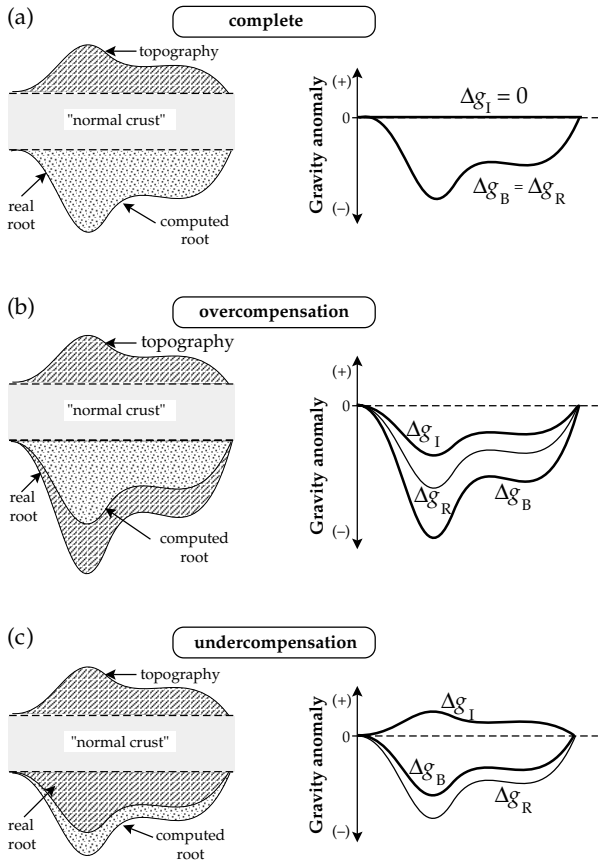


Fig. 2.65 Explanation of the isostatic gravity anomaly (Δg_I) as the difference between the Bouguer gravity anomaly (Δg_B) and the computed anomaly (Δg_R) of the root-zone estimated from the topography for (a) complete isostatic compensation, (b) isostatic overcompensation and (c) isostatic undercompensation.

2.7.3 Isostatic compensation and vertical crustal movements

In the Pratt–Hayford and Airy–Heiskanen models the lighter crust floats freely on the denser mantle. The system is in hydrostatic equilibrium, and local isostatic compensation is a simple application of Archimedes’ principle. A “normal” crustal thickness for sea-level coastal regions is assumed (usually 30–35 km) and the additional depths of the root-zones below this level are exactly proportional to the elevations of the topography above sea-level. The topography is then *completely compensated* (Fig. 2.65a). However, isostatic compensation is often incomplete. The geodynamic imbalance leads to vertical crustal movements.

Mountains are subject to erosion, which can disturb isostatic compensation. If the eroded mountains are no longer high enough to justify their deep root-zones, the topography is isostatically *overcompensated* (Fig. 2.65b). Buoyancy forces are created, just as when a wooden block floating in water is pressed downward by a finger; the underwater part becomes too large in proportion to the amount above the surface. If the finger pressure is removed, the block rebounds in order to restore hydrostatic equilibrium. Similarly, the buoyancy forces that result from overcom-

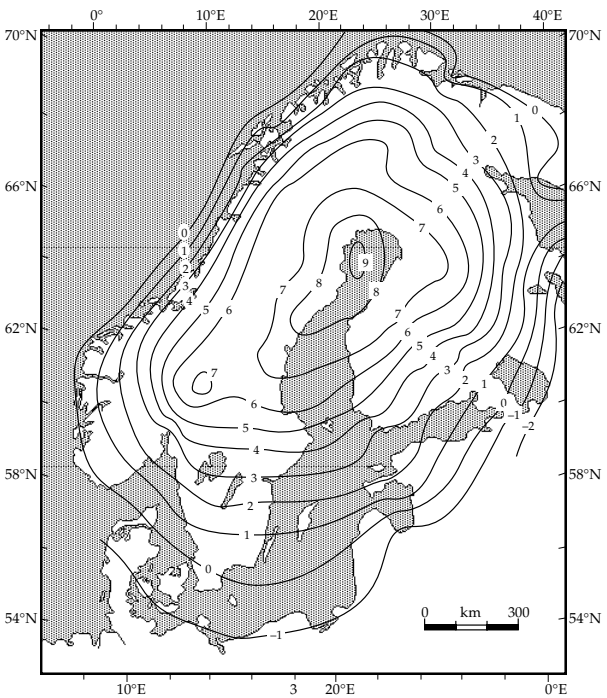


Fig. 2.66 Fennoscandian rates of vertical crustal movement (in mm yr⁻¹) relative to mean sea-level. Positive rates correspond to uplift, negative rates to subsidence (after Kakkuri, 1992).

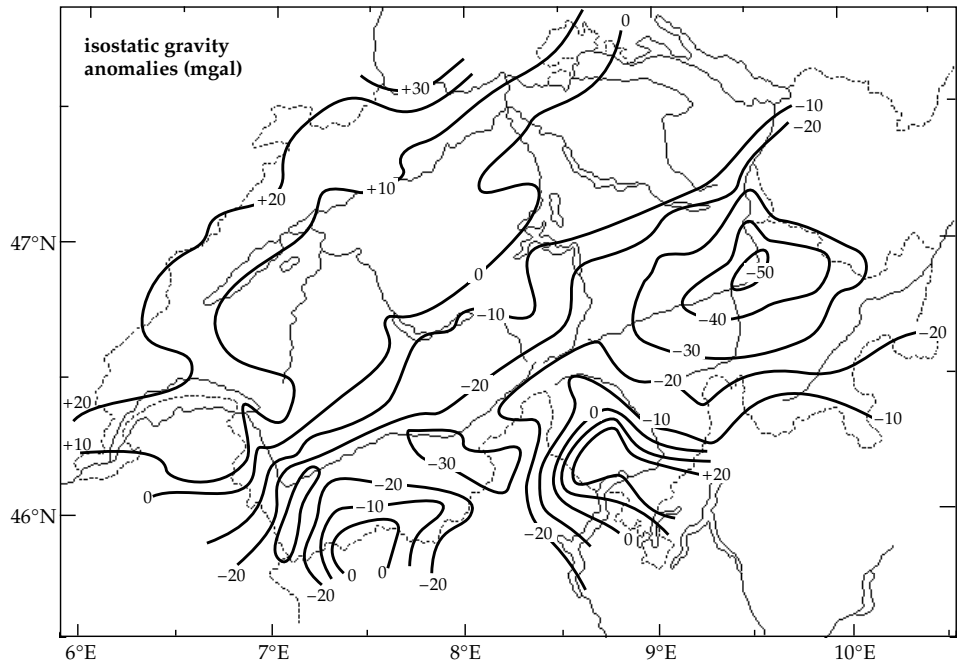
pensation of mountainous topography cause vertical *uplift*. The opposite scenario is also possible. When the visible topography has roots that are too small, the topography is isostatically *undercompensated* (Fig. 2.65c). This situation can result, for example, when tectonic forces thrust crustal blocks on top of each other. Hydrostatic equilibrium is now achieved by *subsidence* of the uplifted region.

The most striking and best-observed examples of vertical crustal movements due to isostatic imbalance are related to the phenomenon of glacial rebound observed in northern Canada and in Fennoscandia. During the latest ice-age these regions were covered by a thick ice-cap. The weight of ice depressed the underlying crust. Subsequently melting of the ice-cap removed the extra load on the crust, and it has since been rebounding. At stations on the Fennoscandian shield, modern tide-gauge observations and precision levelling surveys made years apart allow the present uplift rates to be calculated (Fig. 2.66). The contour lines of equal uplift rate are inexact over large areas due to the incompleteness of data from inaccessible regions. Nevertheless, the general pattern of glacial rebound is clearly recognizable, with uplift rates of up to 8 mm yr⁻¹.

2.7.4 Isostatic gravity anomalies

The different degrees of isostatic compensation find expression in gravity anomalies. As explained in Section 2.5.6 the free-air gravity anomaly Δg_F is small near the center of a large region that is isostatically compensated; the Bouguer anomaly Δg_B is strongly negative. Assuming complete isostatic compensation, the size and shape of the root-zone

Fig. 2.67 Isostatic gravity anomalies in Switzerland (after Klingelé and Kissling, 1982), based on the national gravity map (Klingelé and Olivier, 1980), corrected for the effects of the Molasse basin and the Ivrea body.



can be determined from the elevations of the topography. With a suitable density contrast the gravity anomaly Δg_R of the modelled root-zone can be calculated; because the root-zone has lower density than adjacent mantle rocks Δg_R is also negative. The *isostatic gravity anomaly* Δg_I is defined as the difference between the Bouguer gravity anomaly and the computed anomaly of the root-zone, i.e.,

$$\Delta g_I = \Delta g_B - \Delta g_R \quad (2.109)$$

Examples of the isostatic gravity anomaly for the three types of isostatic compensation are shown schematically in Fig. 2.65. When isostatic compensation is complete, the topography is in hydrostatic equilibrium with its root-zone. Both Δg_B and Δg_R are negative but equal; consequently, the isostatic anomaly is everywhere zero ($\Delta g_I = 0$). In the case of overcompensation the eroded topography suggests a root-zone that is smaller than the real root-zone. The Bouguer anomaly is caused by the larger real root, so Δg_B is numerically larger than Δg_R . Subtracting the smaller negative anomaly of the computed root-zone leaves a negative isostatic anomaly ($\Delta g_I < 0$). On the other hand, with undercompensation the topography suggests a root-zone that is larger than the real root-zone. The Bouguer anomaly is caused by the smaller real root, so Δg_B is numerically smaller than Δg_R . Subtracting the larger negative anomaly of the root-zone leaves a positive isostatic anomaly ($\Delta g_I > 0$).

A national gravity survey of Switzerland carried out in the 1970s gave a high-quality map of Bouguer gravity anomalies (see Fig. 2.58). Seismic data gave representative parameters for the Central European crust and mantle: a crustal thickness of 32 km without topography, and mean densities of 2670 kg m^{-3} for the topography, 2810 kg m^{-3} for the crust and 3310 kg m^{-3} for the mantle. Using the Airy–Heiskanen model of compensation, a

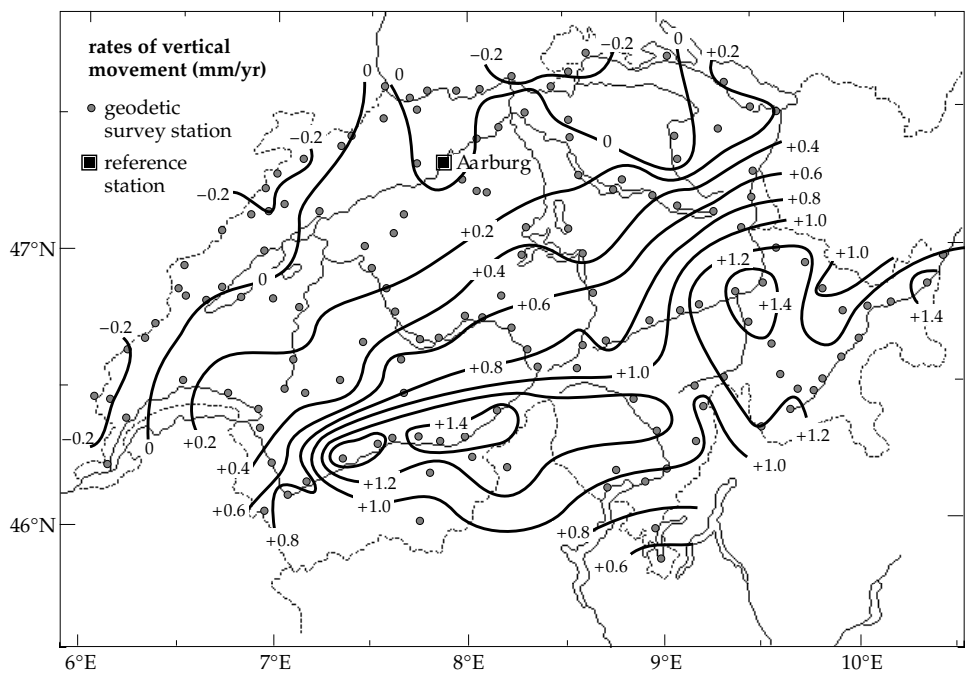
map of isostatic gravity anomalies in Switzerland was derived (Fig. 2.67) after correcting the gravity map for the effects of low-density sediments in the Molasse basin north of the Alps and high-density material in the anomalous Ivrea body in the south.

The pattern of isostatic anomalies reflects the different structures beneath the Jura mountains, which do not have a prominent root-zone, and the Alps, which have a low-density root that extends to more than 55 km depth in places. The dominant ENE–WSW trend of the isostatic gravity anomaly contour lines is roughly parallel to the trends of the mountain chains. In the northwest, near the Jura mountains, positive isostatic anomalies exceed 20 mgal. In the Alps isostatic anomalies are mainly negative, reaching more than -50 mgal in the east.

A computation based on the Vening Meinesz model gave an almost identical isostatic anomaly map. The agreement of maps based on the different concepts of isostasy is somewhat surprising. It may imply that vertical crustal columns are not free to adjust relative to one another without friction as assumed in the Airy–Heiskanen model. The friction is thought to result from horizontal compressive stresses in the Alps, which are active in the on-going mountain-building process.

Comparison of the isostatic anomaly map with one of recent vertical crustal movements (Fig. 2.68) illustrates the relevance of isostatic gravity anomalies for tectonic interpretation. Precise levelling surveys have been carried out since the early 1900s along major valleys transecting and parallel to the mountainous topography of Switzerland. Relative rates of uplift or subsidence are computed from the differences between repeated surveys. The results have not been tied to absolute tide-gauge observations and so are relative to a base station at Aargau in the canton of Aargau, in the northeast.

Fig. 2.68 Rates of vertical crustal motion in Switzerland deduced from repeated precise levelling. Broken contour lines indicate areas in which geodetic data are absent or insufficient. Positive rates correspond to uplift, negative rates to subsidence (data source: Gubler, 1991).



The rates of relative vertical movement in northeastern Switzerland are smaller than the confidence limits on the data and may not be significant, but the general tendency suggests subsidence. This region is characterized by mainly positive isostatic anomalies. The rates of vertical movement in the southern part of Switzerland exceed the noise level of the measurements and are significant. The most notable characteristic of the recent crustal motions is vertical uplift of the Alpine part of Switzerland relative to the central plateau and Jura mountains. The Alpine uplift rates are up to 1.5 mm yr^{-1} , considerably smaller than the rates observed in Fennoscandia. The most rapid uplift rates are observed in the region where isostatic anomalies are negative. The constant erosion of the mountain topography relieves the crustal load and the isostatic response is uplift. However, the interpretation is complicated by the fact that compressive stresses throughout the Alpine region acting on deep-reaching faults can produce non-isostatic uplift of the surface. The separation of isostatic and non-isostatic vertical crustal movements in the Alps will require detailed and exact information about the structure of the lithosphere and asthenosphere in this region.

2.8 RHEOLOGY

2.8.1 Brittle and ductile deformation

Rheology is the science of the deformation and flow of solid materials. This definition appears at first sight to contradict itself. A solid is made up of particles that cohere to each other; it is rigid and resists a change of shape. A fluid has no rigidity; its particles can move about comparatively freely. So how can a solid flow? In fact, the way in which a solid reacts to stress depends on how large the stress is and the length of time for which it is applied.

Provided the applied stress does not exceed the yield stress (or elastic limit) the short-term behavior is elastic. This means that any deformation caused by the stress is completely recoverable when the stress is removed, leaving no permanent change in shape. However, if the applied stress exceeds the yield stress, the solid may experience either brittle or ductile deformation.

Brittle deformation consists of rupture without other distortion. This is an abrupt process that causes faulting in rocks and earthquakes, accompanied by the release of elastic energy in the form of seismic waves. Brittle fracture occurs at much lower stresses than the intrinsic strength of a crystal lattice. This is attributed to the presence of cracks, which modify the local internal stress field in the crystal. Fracture occurs under either extension or shear. Extensional fracture occurs on a plane at right angles to the direction of maximum tension. Shear fracture occurs under compression on one of two complementary planes which, reflecting the influence of internal friction, are inclined at an angle of less than 45° (typically about 30°) to the maximum principal compression. Brittle deformation is the main mechanism in tectonic processes that involve the uppermost 5–10 km of the lithosphere.

Ductile deformation is a slow process in which a solid acquires strain (i.e., it changes shape) over a long period of time. A material may react differently to a stress that is applied briefly than to a stress of long duration. If it experiences a large stress for a long period of time a solid can slowly and permanently change shape. The time-dependent deformation is called plastic flow and the capacity of the solid to flow is called its *ductility*. The ductility of a solid above its yield stress depends on temperature and confining pressure, and materials that are brittle under ordinary conditions may be ductile at high temperature and pressure. The behavior of rocks and minerals in

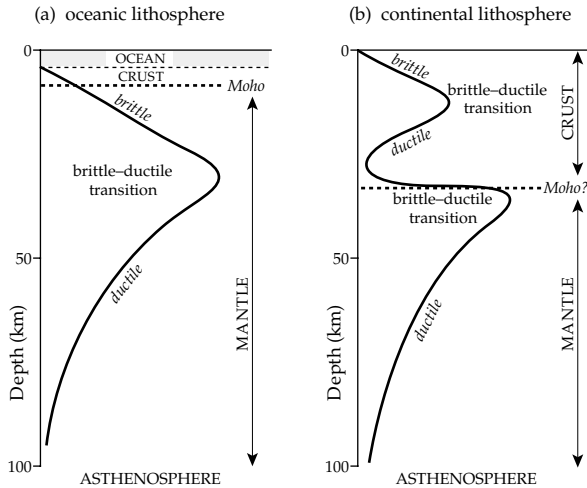


Fig. 2.69 Hypothetical vertical profiles of rigidity in (a) oceanic lithosphere and (b) continental lithosphere with the estimated depths of brittle–ductile transitions (after Molnar, 1988).

the deep interior of the Earth is characterized by ductile deformation.

The transition from brittle to ductile types of deformation is thought to occur differently in oceanic and continental lithosphere (Fig. 2.69). The depth of the transition depends on several parameters, including the composition of the rocks, the local geothermal gradient, initial crustal thickness and the strain rate. Consequently it is sensitive to the vertically layered structure of the lithosphere. The oceanic lithosphere has a thin crust and shows a gradual increase in strength with depth, reaching a maximum in the upper mantle at about 30–40 km depth. At greater depths the lithosphere gradually becomes more ductile, eventually grading into the low-rigidity asthenosphere below about 100 km depth. The continental crust is much thicker than the oceanic crust and has a more complex layering. The upper crust is brittle, but the minerals of the lower crust are weakened by high temperature. As a result the lower crust becomes ductile near to the Moho at about 30–35 km depth. In the upper mantle the strength increases again, leading to a second brittle–ductile transition at about 40–50 km depth. The difference in rheological layering of continental and oceanic lithosphere is important in collisions between plates. The crustal part of the continental lithosphere may detach from the mantle. The folding, underthrusting and stacking of crustal layers produce folded mountain ranges in the suture zone and thickening of the continental crust. For example, great crustal thicknesses under the Himalayas are attributed to underthrusting of crust from the Indian plate beneath the crust of the Eurasian plate.

2.8.2 Viscous flow in liquids

Consider the case when a liquid or gas flows in thin layers parallel to a flat surface (Fig. 2.70). The *laminar flow* exists as long as the speed stays below a critical value,

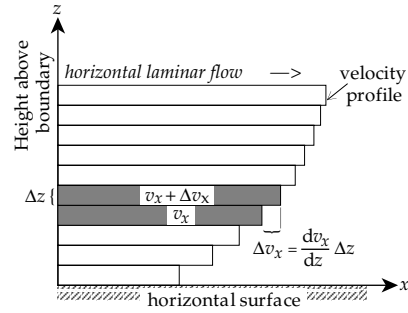


Fig. 2.70 Schematic representation of laminar flow of a fluid in infinitesimally thin layers parallel to a horizontal surface.

above which the flow becomes turbulent. Turbulent flow does not interest us here, because the rates of flow in solid earth materials are very slow.

Suppose that the velocity of laminar flow along the horizontal x -direction increases with vertical height z above the reference surface. The molecules of the fluid may be regarded as having two components of velocity. One component is the velocity of flow in the x -direction, but in addition there is a random component with a variable velocity whose root-mean-square value is determined by the temperature (Section 4.2.2). Because of the random component, one-sixth of the molecules in a unit volume are moving upward and one-sixth downward on average at any time. This causes a transfer of molecules between adjacent layers in the laminar flow. The number of transfers per second depends on the size of the random velocity component, i.e., on temperature. The influx of molecules from the slower-moving layer reduces the momentum of the faster-moving layer. In turn, molecules transferred downward from the faster-velocity layer increase the momentum of the slower-velocity layer. This means that the two layers do not move freely past each other. They exert a shear force – or drag – on each other and the fluid is said to be *viscous*.

The magnitude of the shear force F_{xz} (Section 3.2) depends on how much momentum is transferred from one layer to the next. If all the molecules in a fluid have the same mass, the momentum transfer is determined by the change in the velocity v_x between the layers; this depends on the vertical gradient of the flow velocity (dv_x/dz). The momentum exchange depends also on the number of molecules that cross the boundary between adjacent layers and so is proportional to the surface area, A . We can bring these observations together, as did Newton in the seventeenth century, and derive the following proportionality relationship for F_{xz} :

$$F_{xz} \propto A \frac{dv_x}{dz} \quad (2.110)$$

If we divide both sides by the area A , the left side becomes the shear stress σ_{xz} . Introducing a proportionality constant η we get the equation

$$\sigma_{xz} = \eta \frac{dv_x}{dz} \quad (2.111)$$

This equation is Newton's law of viscous flow and η is the *coefficient of viscosity*. If η is constant, the fluid is called a *Newtonian fluid*. The value of η depends on the transfer rate of molecules between layers and so on temperature. Substituting the units of stress (pascal) and velocity gradient $((\text{ms}^{-1})/\text{m} = \text{s}^{-1})$ we find that the unit of η is a pascal-second (Pa s). A shear stress applied to a fluid with low viscosity causes a large velocity gradient; the fluid flows easily. This is the case in a gas (η in air is of the order 2×10^{-5} Pa s) or in a liquid (η in water is 1.005×10^{-3} Pa s at 20°C). The same shear stress applied to a very viscous fluid (with a large value of η) produces only a small velocity gradient transverse to the flow direction. The layers in the laminar flow are reluctant to move past each other. The viscous fluid is “sticky” and it resists flow. For example, η in a viscous liquid like engine oil is around $0.1\text{--}10$ Pa s, three or four orders of magnitude higher than in water.

2.8.3 Flow in solids

The response of a solid to an applied load depends upon whether the stress exceeds the elastic limit (Section 3.2.1) and for how long it is applied. When the yield stress (elastic limit) is reached, a solid may deform continuously without further increase in stress. This is called *plastic deformation*. In *perfectly plastic* behavior the stress–strain curve has zero slope, but the stress–strain curves of plastically deformed materials usually have a small positive slope (see Fig. 3.2a). This means that the stress must be increased above the yield stress for plastic deformation to advance. This effect is called *strain-hardening*. When the stress is removed after a material has been strain-hardened, a permanent residual strain is left.

Consider the effects that ensue if a stress is suddenly applied to a material at time t_0 , held constant until time t_1 and then abruptly removed (Fig. 2.71a). As long as the applied stress is lower than the yield stress, the solid deforms elastically. The elastic strain is acquired immediately and remains constant as long as the stress is applied. Upon removal of the stress, the object at once recovers its original shape and there is no permanent strain (Fig. 2.71b).

If a constant load greater than the yield stress is applied, the resulting strain consists of a constant elastic strain and a changing *plastic* strain, which increases with time. After removal of the load at t_1 the plastic deformation does not disappear but leaves a permanent strain (Fig. 2.71c). In some plastic materials the deformation increases slowly at a decreasing rate, eventually reaching a limiting value for any specific value of the stress. This is called *viscoelastic* deformation (Fig. 2.71d). When the load is removed at t_1 , the elastic part of the deformation is at once restored, followed by a slow decrease of the residual strain. This phase is called *recovery* or *delayed elasticity*. Viscoelastic behavior is an important rheological process deep in the Earth, for example in the asthenosphere and deeper mantle.

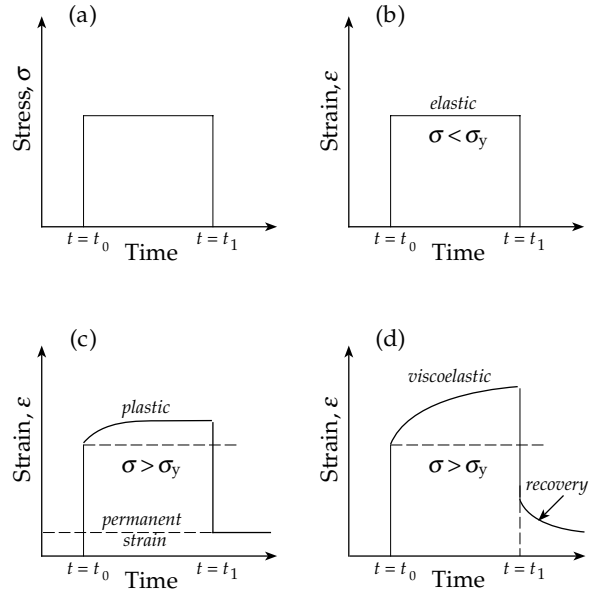


Fig. 2.71 (a) Application of a constant stress σ to a solid between times t_0 and t_1 ; (b) variation of elastic strain below the yield point; (c) plastic strain and (d) viscoelastic deformation at stresses above the yield point σ_y .

The analogy to the viscosity of liquids is apparent by inspection of Eq. (2.111). Putting $v_x = dx/dt$ and changing the order of differentiation, the equation becomes

$$\sigma_{xz} = \eta \frac{d}{dz} \frac{dx}{dt} = \eta \frac{d}{dt} \frac{dx}{dz} = \eta \frac{d}{dt} \epsilon_{xz} \quad (2.112)$$

This equation resembles the elastic equation for shear deformation, which relates stress and strain through the shear modulus (see Eq. (3.16)). However, in the case of the “viscous flow” of a solid the shear stress depends on the strain *rate*. The parameter η for a solid is called the *viscosity modulus*, or *dynamic viscosity*. It is analogous to the viscosity coefficient of a liquid but its value in a solid is many orders of magnitude larger. For example, the viscosity of the asthenosphere is estimated to be of the order of $10^{20}\text{--}10^{21}$ Pa s. Plastic flow in solids differs from true flow in that it only occurs when the stress exceeds the yield stress of the solid. Below this stress the solid does not flow. However, internal defects in a metal or crystal can be mobilized and reorganized by stresses well below the yield stress. As a result the solid may change shape over a long period of time.

2.8.3.1 Viscoelastic model

Scientists have tried to understand the behavior of rocks under stress by devising models based on mechanical analogs. In 1890 Lord Kelvin modelled viscoelastic deformation by combining the characteristics of a perfectly elastic solid and a viscous liquid. An applied stress causes both elastic and viscous effects. If the elastic strain is ϵ , the corresponding elastic part of the stress is $E\epsilon$, where E is Young's modulus (Section 3.2.4). Similarly, if the rate of change of strain with time is $d\epsilon/dt$, the viscous part of

the applied stress is $\eta \, d\varepsilon/dt$, where η is the viscosity modulus. The applied stress σ is the sum of the two parts and can be written

$$\sigma = E\varepsilon + \eta \frac{d\varepsilon}{dt} \quad (2.113)$$

To solve this equation we first divide throughout by E , then define the *retardation time* $\tau = \eta/E$, which is a measure of how long it takes for viscous strains to exceed elastic strains. Substituting and rearranging the equation we get

$$\varepsilon + \tau \frac{d\varepsilon}{dt} = \frac{\sigma}{E} \quad (2.114)$$

$$\frac{\varepsilon}{\tau} + \frac{d\varepsilon}{dt} = \frac{\varepsilon_m}{\tau} \quad (2.115)$$

where $\varepsilon_m = \sigma/E$. Multiplying throughout by the integrating factor $e^{t/\tau}$ gives

$$\frac{\varepsilon}{\tau} e^{t/\tau} + \frac{d\varepsilon}{dt} e^{t/\tau} = \frac{\varepsilon_m}{\tau} e^{t/\tau} \quad (2.116)$$

$$\frac{d}{dt}(\varepsilon e^{t/\tau}) = \frac{\varepsilon_m}{\tau} e^{t/\tau} \quad (2.117)$$

Integrating both sides of this equation with respect to t gives

$$\varepsilon e^{t/\tau} = \varepsilon_m e^{t/\tau} + C \quad (2.118)$$

where C is a constant of integration determined by the boundary conditions. Initially, the strain is zero, i.e., at $t = 0$, $\varepsilon = 0$; substituting in Eq. (2.118) gives $C = -\varepsilon_m$. The solution for the strain at time t is therefore

$$\varepsilon = \varepsilon_m (1 - e^{-t/\tau}) \quad (2.119)$$

The strain rises exponentially to a limiting value given by $\varepsilon_m = \sigma/E$. This is characteristic of viscoelastic deformation.

2.8.4 Creep

Many solid materials deform slowly at room temperature when subjected to small stresses well below their brittle strength for long periods of time. The slow time-dependent deformation is known as *creep*. This is an important mechanism in the deformation of rocks because of the great intervals of time involved in geological processes. It is hard enough to approximate the conditions of pressure and temperature in the real Earth, but the time factor is an added difficulty in investigating the phenomenon of creep in laboratory experiments.

The results of observations on rock samples loaded by a constant stress typically show three main regimes of creep (Fig. 2.72a). At first, the rock at once strains elastically. This is followed by a stage in which the strain initially increases rapidly (i.e., the strain rate is high) and then levels off. This stage is known as *primary creep* or *delayed elastic creep*. If the stress is removed within this regime, the deformation drops rapidly as the elastic strain recovers, leaving a

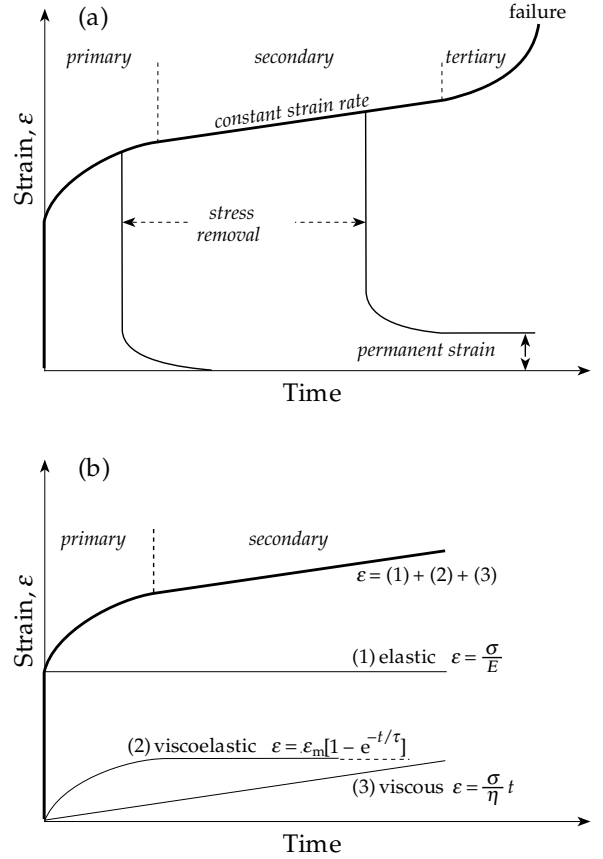


Fig. 2.72 Hypothetical strain-time curve for a material exhibiting creep under constant stress, and (b) model creep curve that combines elastic, viscoelastic and viscous elements (after Ramsay, 1967).

deformation that sinks progressively to zero. Beyond the primary stage creep progresses at a slower and nearly constant rate. This is called *secondary creep* or *steady-state creep*. The rock deforms plastically, so that if the stress is removed a permanent strain is left after the elastic and delayed elastic recoveries. After the secondary stage the strain rate increases ever more rapidly in a stage called *tertiary creep*, which eventually leads to failure.

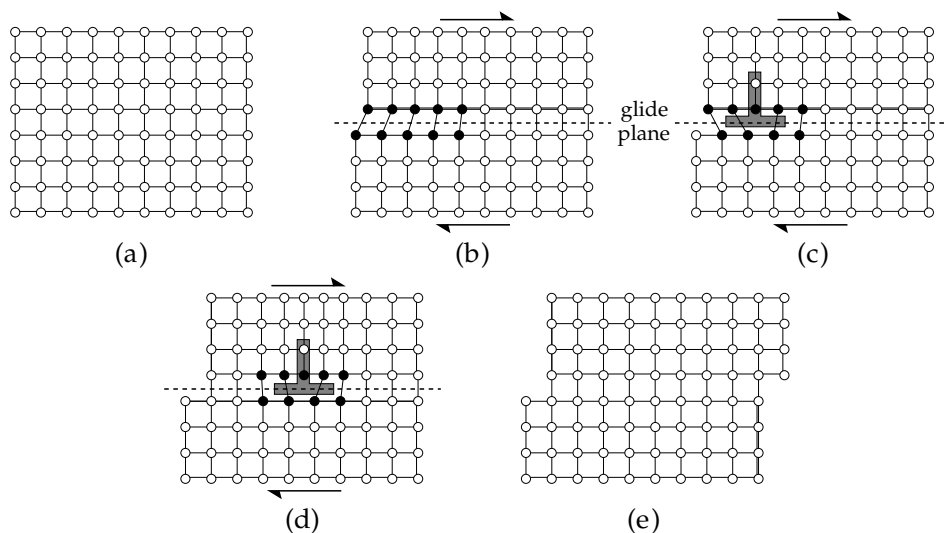
The primary and secondary stages of the creep curve can be modelled by combining elastic, viscoelastic and viscous elements (Fig. 2.72b). Below the yield stress only elastic and delayed elastic (viscoelastic) deformation occur and the solid does not flow. The strain flattens off at a limiting value $\varepsilon_m = \sigma/E$. The viscous component of strain rises linearly with time, corresponding to a constant strain rate. In practice the stress must exceed the yield stress σ_y for flow to occur. In this case the viscous component of strain is proportional to the excess stress $(\sigma - \sigma_y)$ and to the time t . Combining terms gives the expression

$$\varepsilon = \frac{\sigma}{E} + \varepsilon_m (1 - e^{-t/\tau}) + \frac{(\sigma - \sigma_y)}{\eta} t \quad (2.120)$$

This simple model explains the main features of experimentally observed creep curves. It is in fact very difficult to ensure that laboratory observations are representative of creep in the Earth. Conditions of pressure and temperature

Fig. 2.73 Permanent (plastic) shear deformation produced by motion of a dislocation through a crystal in response to shear stress:

(a) undeformed crystal lattice, (b) entry of dislocation at left edge, (c) accommodation of dislocation into lattice, (d) passage of dislocation across the crystal, and (e) sheared lattice after dislocation leaves the crystal.



in the crust and upper mantle can be achieved or approximated. The major problems arise from the differences in timescales and creep rates. Even creep experiments conducted over months or years are far shorter than the lengths of time in a geological process. The creep rates in nature (e.g., around 10^{-14} s^{-1}) are many orders of magnitude slower than the slowest strain rate used in laboratory experiments (around 10^{-8} s^{-1}). Nevertheless, the experiments have provided a better understanding of the rheology of the Earth's interior and the physical mechanisms active in different depths.

2.8.4.1 Crystal defects

Deformation in solids does not take place homogeneously. Laboratory observations on metals and minerals have shown that crystal defects play an important role. The atoms in a metal or crystal are arranged regularly to form a lattice with a simple symmetry. In some common arrangements the atoms are located at the corners of a cube or a hexagonal prism, defining a *unit cell* of the crystal. The lattice is formed by stacking unit cells together. Occasionally an imperfect cell may lack an atom. The space of the missing atom is called a vacancy. Vacancies may be distributed throughout the crystal lattice, but they can also form long chains called dislocations.

There are several types of dislocation, the simplest being an edge dislocation. It is formed when an extra plane of atoms is introduced in the lattice (Fig. 2.73). The edge dislocation terminates at a plane perpendicular to it called the *glide plane*. It clearly makes a difference if the extra plane of atoms is above or below the glide plane, so edge dislocations have a sign. This can be represented by a T-shape, where the cross-bar of the T is parallel to the glide plane and the stalk is the extra plane of atoms. If oppositely signed edge dislocations meet, they form a complete plane of atoms and the dislocations annihilate each other. The displacement of atoms in the vicinity of a dislocation increases the local internal stress in the crystal. As a result,

the application of a small external stress may be enough to mobilize the dislocation, causing it to glide through the undisturbed part of the crystal (Fig. 2.73). If the dislocation glide is not blocked by an obstacle, the dislocation migrates out of the crystal, leaving a shear deformation.

Another common type of dislocation is the screw dislocation. It also is made up of atoms that are displaced from their regular positions, in this case forming a spiral about an axis.

The deformation of a crystal lattice by dislocation glide requires a shear stress; hydrostatic pressure does not cause plastic deformation. The shear stress needed to actuate dislocations is two or three orders of magnitude less than the shear stress needed to break the bonds between layers of atoms in a crystal. Hence, the mobilization of dislocations is an important mechanism in plastic deformation at low stress. As deformation progresses it is accompanied by an increase in the dislocation density (the number of dislocations per unit area normal to their lengths). The dislocations move along a glide plane until it intersects the glide plane of another set of dislocations. When several sets of dislocations are mobilized they may interfere and block each other, so that the stress must be increased to mobilize them further. This is manifest as strain-hardening, which is a thermodynamically unstable situation. At any stage of strain-hardening, given enough time, the dislocations redistribute themselves to a configuration with lower energy, thereby reducing the strain. The time-dependent strain relaxation is called *recovery*.

Recovery can take place by several processes, each of which requires thermal energy. These include the annihilation of oppositely signed edge dislocations moving on parallel glide planes (Fig. 2.74a) and the climb of edge dislocations past obstacles against which they have piled up (Fig. 2.74b). Edge dislocations with the same sign may align to form walls between domains of a crystal that have low dislocation density (Fig. 2.74c), a process called polygonization. These are some of the ways in which thermal energy promotes the migration of lattice defects,

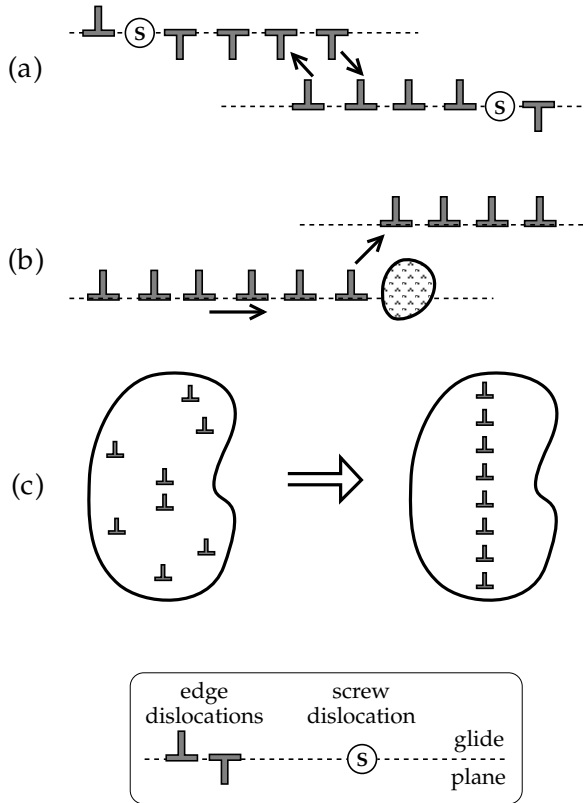


Fig. 2.74 Thermally activated processes that assist recovery: (a) annihilation of oppositely signed edge dislocations moving on parallel glide planes, (b) climb of edge dislocations past obstacles, and (c) polygonization by the alignment of edge dislocations with the same sign to form walls separating regions with low dislocation density (after Ranalli, 1987).

eventually driving them out of the crystal and leaving an annealed lattice.

2.8.4.2 Creep mechanisms in the Earth

Ductile flow in the Earth's crust and mantle takes place by one of three mechanisms: low-temperature plastic flow; power-law creep; or diffusion creep. Each mechanism is a thermally activated process. This means that the strain rate depends on the temperature T according to an exponential function with the form $e^{-E_a/kT}$. Here k is Boltzmann's constant, while E_a is the energy needed to activate the type of flow; it is called the *activation energy*. At low temperatures, where $T \ll E_a/k$, the strain rate is very slow and creep is insignificant. Because of the exponential function, the strain rate increases rapidly with increasing temperature above $T = E_a/k$. The type of flow at a particular depth depends on the local temperature and its relationship to the melting temperature T_{mp} . Above T_{mp} the interatomic bonds in the solid break down and it flows as a true liquid.

Plastic flow at low temperature takes place by the motion of dislocations on glide planes. When the dislocations encounter an internal obstacle or a crystal boundary they pile up and some rearrangement is necessary.

The stress must be increased to overcome the obstacle and reactuate dislocation glide. Plastic flow can produce large strains and may be an important mechanism in the bending of the oceanic lithosphere near some subduction zones. It is likely to be most effective at depths below the brittle–ductile transition.

Power-law creep, or hot creep, also takes place by the motion of dislocations on glide planes. It occurs at higher temperatures than low-temperature plastic flow, so internal obstacles to dislocation migration are thermally activated and diffuse out of the crystal as soon as they arise. The strain rate in power-law creep is proportional to the n th power of the stress σ and has the form

$$\frac{d\varepsilon}{dt} = A \left(\frac{\sigma}{\mu} \right)^n e^{-E_d/kT} \quad (2.121)$$

where μ is the rigidity modulus and A is a constant with the dimensions of strain rate; typically $n \geq 3$. This relationship means that the strain rate increases much more rapidly than the stress. From experiments on metals, power-law creep is understood to be the most important mechanism of flow at temperatures between $0.55T_{mp}$ and $0.85T_{mp}$. The temperature throughout most of the mantle probably exceeds half the melting point, so power-law creep is probably the flow mechanism that permits mantle convection. It is also likely to be the main form of deformation in the lower lithosphere, where the relationship of temperature to melting point is also suitable.

Diffusion creep consists of the thermally activated migration of crystal defects in the presence of a stress field. There are two main forms. *Nabarro–Herring creep* consists of diffusion of defects through the body of a grain; *Coble creep* takes place by migration of the defects along grain boundaries. In each case the strain rate is proportional to the stress, as in a Newtonian fluid. It is therefore possible to regard ductile deformation by diffusion creep as the slow flow of a very viscous fluid. Diffusion creep has been observed in metals at temperatures $T > 0.85T_{mp}$. In the Earth's mantle the temperature approaches the melting point in the asthenosphere. As indicated schematically in Fig. 2.69 the transition from the rigid lithosphere to the soft, viscous underlying asthenosphere is gradational. There is no abrupt boundary, but the concept of rigid lithospheric plates moving on the soft, viscous asthenosphere serves well as a geodynamic model.

2.8.5 Rigidity of the lithosphere

Lithospheric plates are thin compared to their horizontal extents. However, they evidently react rigidly to the forces that propel them. The lithosphere does not easily buckle under horizontal stress. A simple analogy may be made to a thin sheet of paper resting on a flat pillow. If pushed on one edge, the page simply slides across the pillow without crumpling. Only if the leading edge encounters an obstacle does the page bend, buckling upward some distance in

front of the hindrance, while the leading edge tries to burrow under it. This is what happens when an oceanic lithospheric plate collides with another plate. A small forebulge develops on the oceanic plate and the leading edge bends downward into the mantle, forming a subduction zone.

The ability to bend is a measure of the rigidity of the plate. This is also manifest in its reaction to a local vertical load. If, in our analogy, a small weight is placed in the middle of the page, it is pressed down into the soft pillow. A large area around the weight takes part in this process, which may be compared with the Vening Meinesz type of regional isostatic compensation (Section 2.7.2.3). The weight of a seamount or chain of islands has a similar effect on the oceanic lithosphere. By studying the flexure due to local vertical loads, information is obtained about a *static property* of the lithosphere, namely its resistance to bending.

In our analogy the locally loaded paper sheet would not bend if it lay on a hard flat table. It is only able to flex if it rests on a soft, yielding surface. After the weight is removed, the page is restored to its original flat shape. The restoration after unloading is a measure of the properties of the pillow as well as the page. A natural example is the rebound of regions (such as the Canadian shield or Fennoscandia) that have been depressed by now-vanished ice-sheets. The analysis of the rates of glacial rebound provides information about a *dynamic property* of the mantle beneath the lithosphere. The depression of the surface forces mantle material to flow away laterally to make way for it; when the load is removed, the return mantle flow presses the concavity back upward. The ease with which the mantle material flows is described by its dynamic viscosity.

The resistance to bending of a thin elastic plate overlying a weak fluid is expressed by an elastic parameter called the *flexural rigidity* and denoted D . For a plate of thickness h ,

$$D = \frac{E}{12(1 - \nu^2)} h^3 \quad (2.122)$$

where E is Young's modulus and ν is Poisson's ratio (see Sections 3.2.3 and 3.2.4 for the definition of these elastic constants). The dimensions of E are Nm^{-2} and ν is dimensionless; hence, the dimensions of D are those of a bending moment (Nm). D is a fundamental parameter of the elastic plate, which describes how easily it can be bent; a large value of D corresponds to a stiff plate.

Here we consider two situations of particular interest for the rigidity of the oceanic lithosphere. The first is the bending of the lithosphere by a topographic feature such as an oceanic island or seamount; only the vertical load on the elastic plate is important. The second is the bending of the lithosphere at a subduction zone. In this case vertical and horizontal forces are located along the edge of the plate and the plate experiences a bending moment which deflects it downward.

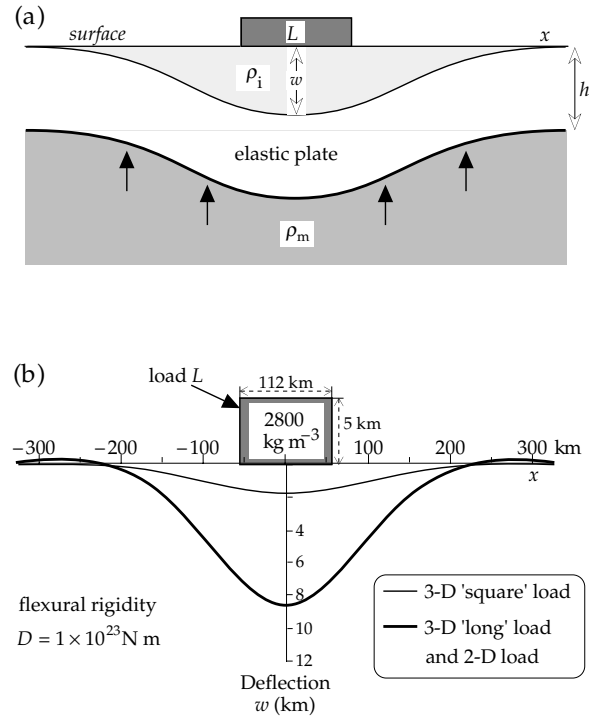


Fig. 2.75 (a) Geometry for the elastic bending of a thin plate of thickness h supported by a denser substratum: the surface load L causes a downward bending w . (b) Comparison of 2D and 3D elastic plate models. The load is taken to be a topographic feature of density 2800 kg m^{-3} , height 5 km and cross-sectional width 112 km (after Watts *et al.*, 1975).

2.8.5.1 Lithospheric flexure caused by oceanic islands

The theory for elastic bending of the lithosphere is derived from the bending of thin elastic plates and beams. This involves a fourth-order differential equation, whose derivation and solution are beyond the scope of this book. However, it is instructive to consider the forces involved in setting up the equation, and to examine its solution in a simplified context.

Consider the bending of a thin isotropic elastic plate of thickness h carrying a surface load $L(x, y)$ and supported by a substratum of density ρ_m (Fig. 2.75a). Let the deflection of the plate at a position (x, y) relative to the center of the load be $w(x, y)$. Two forces act to counteract the downward force of the load. The first, arising from Archimedes' principle, is a buoyant force equal to $(\rho_m - \rho_i)gw$, where ρ_i is the density of the material that fills in the depression caused by the deflection of the plate. The second force arises from the elasticity of the beam. Elasticity theory shows that this produces a restoring force proportional to a fourth-order differential of the deflection w . Balancing the elastic and buoyancy forces against the deforming load leads to the equation

$$D \left[\frac{\partial^4 w}{\partial x^4} + 2 \frac{\partial^4 w}{\partial x^2 \partial y^2} + \frac{\partial^4 w}{\partial y^4} \right] + (\rho_m - \rho_i)gw = L(x, y) \quad (2.123)$$

For a linear topographic feature, such as a mountain range, oceanic island or chain of seamounts, the bending geometry is the same in any cross-section normal to its length and the problem reduces to the two-dimensional elastic bending of a thin beam. If the load is a linear feature in the y -direction, the variation of w with y disappears and the differential equation becomes:

$$D \frac{\partial^4 w}{\partial x^4} + (\rho_m - \rho_i)gw = L \quad (2.124)$$

An important example is a linear load L concentrated along the y -axis at $x = 0$. The solution is a damped sinusoidal function:

$$w = w_0 e^{-x/\alpha} \left(\cos \frac{x}{\alpha} + \sin \frac{x}{\alpha} \right) \quad (2.125)$$

where w_0 is the amplitude of the maximum deflection underneath the load (at $x = 0$). The parameter α is called the *flexural parameter*; it is related to the flexural rigidity D of the plate by

$$\alpha^4 = \frac{4D}{(\rho_m - \rho_i)g} \quad (2.126)$$

The elasticity of the plate (or beam) distributes the load of the surface feature over a large lateral distance. The fluid beneath the load is pushed aside by the penetration of the plate. The buoyancy of the displaced fluid forces it upward, causing uplift of the surface adjacent to the central depression. Equation (2.125) shows that this effect is repeated with increasing distance from the load, the wavelength of the fluctuation is $\lambda = 2\pi\alpha$. The amplitude of the disturbance diminishes rapidly because of the exponential attenuation factor. Usually it is only necessary to consider the central depression and the first uplifted region. The wavelength λ is equal to the distance across the central depression. Substituting in Eq. (2.126) gives D , which is then used with the parameters E and ν in Eq. (2.122) to obtain h , the thickness of the elastic plate. The computed values of h are greater than the thickness of the crust, i.e., the elastic plate includes part of the upper mantle. The value of h is equated with the thickness of the elastic lithosphere.

The difference between the deflection caused by a two-dimensional load (i.e., a linear feature) and that due to a three-dimensional load (i.e., a feature that has limited extent in the x - and y -directions) is illustrated in Fig. 2.75b. If the length of the three-dimensional load normal to the cross-section is more than about ten times its width, the deflection is the same as for a two-dimensional load. A load with a square base (i.e., extending the same distance along both x - and y -axes) causes a central depression that is less than a quarter the effect of the linear load.

The validity of the lithospheric flexural model of isostasy can be tested by comparing the computed gravity effect of the model with the observed free-air gravity anomaly Δg_F . The isostatic compensation of the Great Meteor seamount in the North Atlantic provides a suitable test for a three-dimensional model (Fig. 2.76). The

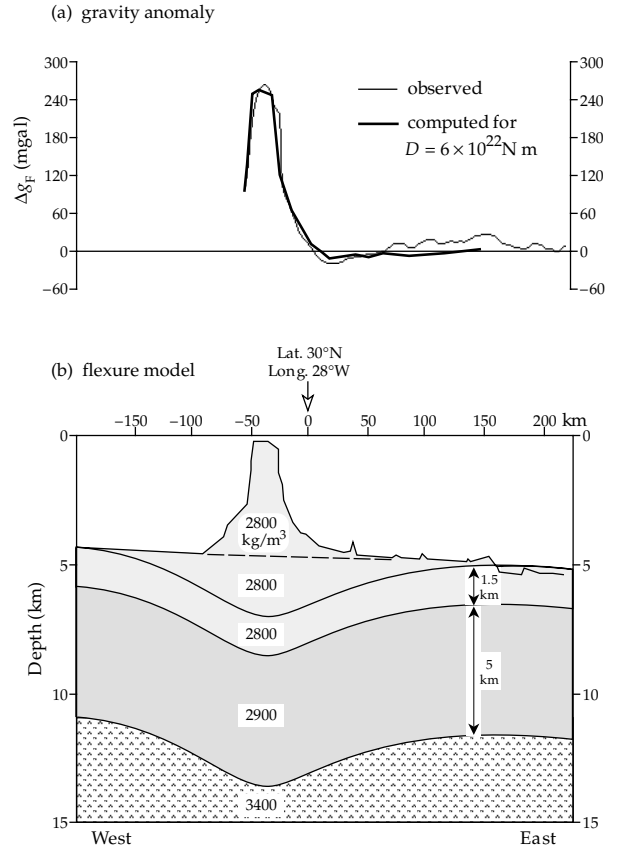


Fig. 2.76 (a) Comparison of observed free-air gravity anomaly profile across the Great Meteor seamount with the anomaly computed for (b) a lithospheric flexure model of isostatic compensation (after Watts *et al.*, 1975).

shape of the free-air gravity anomaly obtained from detailed marine gravity surveys was found to be fitted best by a model in which the effective flexural rigidity of the deformed plate was assumed to be $6 \times 10^{22} \text{ N m}$.

2.8.5.2 Lithospheric flexure at a subduction zone

The bathymetry of an oceanic plate at a subduction zone is typified by an oceanic trench, which can be many kilometers deep (Fig. 2.77a). Seaward of the trench axis the plate develops a small upward bulge (the outer rise) which can extend for 100–150 km away from the trench and reach heights of several hundred meters. The lithospheric plate bends sharply downward in the subduction zone. This bending can also be modelled with a thin elastic plate.

In the model, a horizontal force P pushes a plate of thickness h toward the subduction zone, the leading edge carries a vertical load L , and the plate is bent by a bending moment M (Fig. 2.77b). The horizontal force P is negligible in comparison to the effects of M and L . The vertical deflection of the plate must satisfy Eq. (2.124), with the same parameters as in the previous example. Choosing the origin to be the point nearest the trench where the deflection is zero simplifies the form of the solution, which is

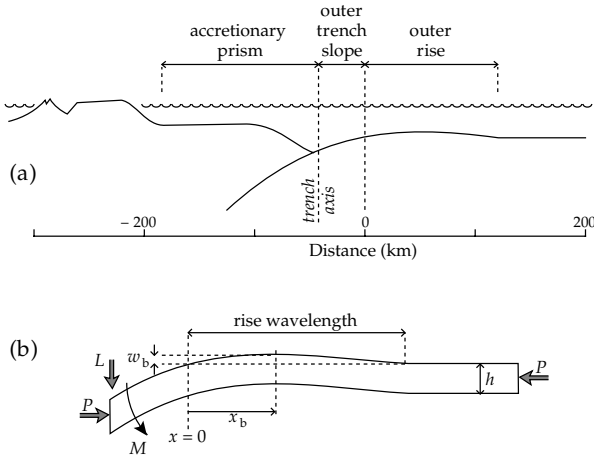


Fig. 2.77 (a) Schematic structural cross-section at a subduction zone (after Caldwell and Turcotte, 1979), and (b) the corresponding thin-plate model (after Turcotte *et al.*, 1978).

$$w = Ae^{-x/\alpha} \sin \frac{x}{\alpha} \quad (2.127)$$

where A is a constant and α is the flexural parameter as in Eq. (2.126). The value of the constant A is found from the position x_b of the forebulge, where $dw/dx = 0$:

$$\frac{dw}{dx} = A \left\{ -\frac{1}{\alpha} e^{-x/\alpha} \sin \frac{x}{\alpha} + \frac{1}{\alpha} e^{-x/\alpha} \cos \frac{x}{\alpha} \right\} = 0 \quad (2.128)$$

from which

$$x_b = \frac{\pi}{4} \alpha \quad \text{and} \quad A = w_b \sqrt{2} e^{\pi/4} \quad (2.129)$$

It is convenient to normalize the horizontal distance and vertical displacement: writing $x' = x/x_b$ and $w' = w/w_b$, the generalized equation for the elastic bending at an oceanic trench is obtained:

$$w' = \sqrt{2} \sin \left(\frac{\pi}{4} x' \right) \exp \left[\frac{\pi}{4} (1 - x') \right] \quad (2.130)$$

The theoretical deflection of oceanic lithosphere at a subduction zone obtained from the elastic bending model agrees well with the observed bathymetry on profiles across oceanic trenches (Fig. 2.78a, b). The calculated thicknesses of the elastic lithosphere are of the order 20–30 km and the flexural rigidity is around 10^{23} N m. However, at some trenches the assumption of a completely elastic upper lithosphere is evidently inappropriate. At the Tonga trench the model curve deviates from the observed bathymetry inside the trench (Fig. 2.78c). It is likely that the elastic limit is exceeded at parts of the plate where the curvature is high. These regions may yield, leading to a reduction in the effective rigidity of the plate. This effect can be taken into account by assuming that the inelastic deformation is perfectly plastic. The deflection calculated within the trench for an elastic-perfectly plastic model agrees well with the observed bathymetry.

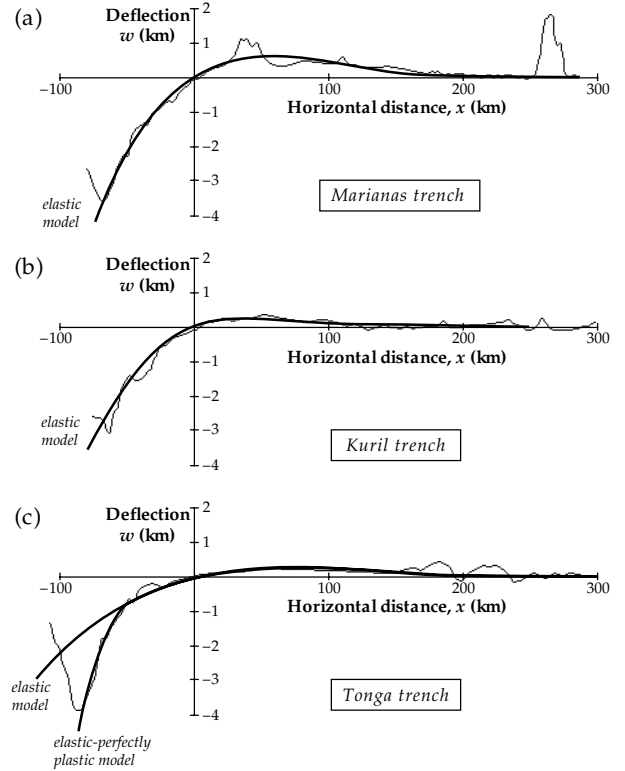


Fig. 2.78 Observed (solid) and theoretical (dashed) bathymetric profiles for elastic flexure of the lithosphere at (a) the Marianas trench and (b) the Kuril trench. The flexure at (c) the Tonga trench is best explained by an elastic-perfectly plastic model (after Turcotte *et al.*, 1978).

2.8.5.3 Thickness of the lithosphere

The rheological response of a material to stress depends on the duration of the stress. The reaction to a short-lasting stress, as experienced during the passage of a seismic wave, may be quite different from the reaction of the same material to a steady load applied for a long period of time. This is evident in the different thicknesses obtained for the lithosphere in seismic experiments and in elastic plate modelling. Long-period surface waves penetrate well into the upper mantle. Long wavelengths are slowed down by the low rigidity of the asthenosphere, so the dispersion of surface waves allows estimates of the seismic thickness of the lithosphere. For oceanic lithosphere the seismic thickness increases with age of the lithosphere (i.e., with distance from the spreading center), increasing to more than 100 km at ages older than 100 Ma (Fig. 2.79). The lithospheric thicknesses obtained from elastic modelling of the bending caused by seamounts and island chains or at subduction zones also increase with distance from the ridge, but are only one-third to one-half of the corresponding seismic thickness. The discrepancy shows that only the upper part of the lithosphere is elastic. Indeed, if the entire lithosphere had the flexural rigidity found in elastic models ($D \approx 10^{21}$ – 10^{23} N m), it would bend by only small amounts under topographic loads or at subduction zones. The base of the elastic lithosphere agrees well with the modelled depths of the 300–600 °C oceanic isotherms. At greater

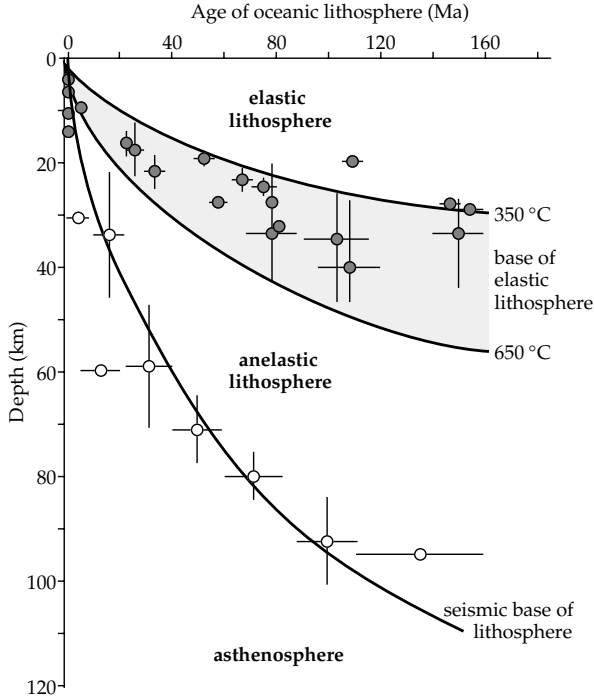


Fig. 2.79 Seismic and elastic thicknesses of oceanic lithosphere as a function of age (after Watts *et al.*, 1980).

depths the increase in temperature results in anelastic behavior of the lower lithosphere.

The elastic thickness of the continental lithosphere is much thicker than that of the oceanic lithosphere, except in rifts, passive continental margins and young orogenic belts. Precambrian shield areas generally have a flexural thickness greater than 100 km and a high flexural rigidity of around 10^{25} N m. Rifts, on the other hand, have a flexural thickness less than 25 km. Both continental and oceanic lithosphere grade gradually into the asthenosphere, which has a much lower rigidity and is able to flow in a manner determined by mantle viscosity.

2.8.6 Mantle viscosity

As illustrated by the transition from brittle to ductile behavior (Section 2.8.1), the Earth's rheology changes with depth. The upper part of the lithosphere behaves elastically. It has a constant and reversible response to both short-term and long-term loads. The behavior is characterized by the rigidity or shear modulus μ , which relates the strain to the applied shear stress and so has the dimensions of stress (N m^{-2} , or Pa). The resistance of the lithosphere to flexure is described by the flexural rigidity D , which has the dimensions of a bending moment (N m). In a subduction zone the tight bending may locally exceed the elastic limit, causing parts of the plate to yield.

The deeper part of the lithosphere does not behave elastically. Although it has an elastic response to abrupt stress changes, it reacts to long-lasting stress by ductile flow. This kind of rheological behavior also characterizes the asthenosphere and the deeper mantle. Flow takes

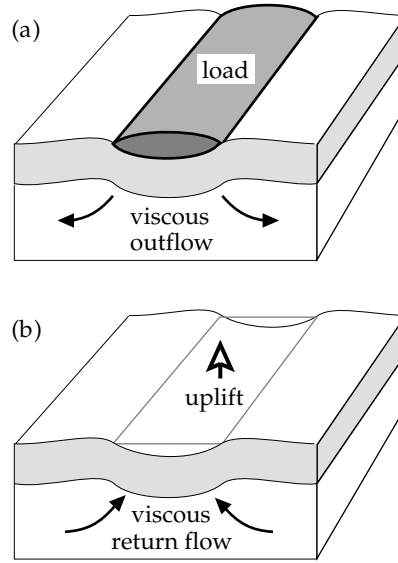


Fig. 2.80 (a) Depression of the lithosphere due to a surface load (ice-sheet) and accompanying viscous outflow in the underlying mantle; (b) return flow in the mantle and surface uplift after removal of the load.

place with a strain rate that is proportional to the stress or a power thereof. In the simplest case, the deformation occurs by Newtonian flow governed by a viscosity coefficient η , whose dimensions (Pa s) express the time-dependent nature of the process.

Under a surface load, such as an ice-sheet, the elastic lithosphere is pushed down into the viscous mantle (Fig. 2.80a). This causes an outflow of mantle material away from the depressed region. When the ice-sheet melts, removing the load, hydrostatic equilibrium is restored and there is a return flow of the viscous mantle material (Fig. 2.80b). Thus, in modelling the time-dependent reaction of the Earth to a surface load at least two and usually three layers must be taken into account. The top layer is an elastic lithosphere up to 100 km thick; it has infinite viscosity (i.e., it does not flow) and a flexural rigidity around 5×10^{24} N m. Beneath it lies a low-viscosity “channel” 75–250 km thick that includes the asthenosphere, with a low viscosity of typically 10^{19} – 10^{20} Pa s. The deeper mantle which makes up the third layer has a higher viscosity around 10^{21} Pa s.

Restoration of the surface after removal of a load is accompanied by uplift, which can be expressed well by a simple exponential relaxation equation. If the initial depression of the surface is w_0 , the deflection $w(t)$ after time t is given by

$$w(t) = w_0 e^{-t/\tau} \quad (2.131)$$

Here τ is the relaxation time, which is related to the mantle viscosity η by

$$\tau = \frac{4\pi}{\rho_m g \lambda} \eta \quad (2.132)$$

where ρ_m is the mantle density, g is gravity at the depth of the flow and λ is the wavelength of the depression, a

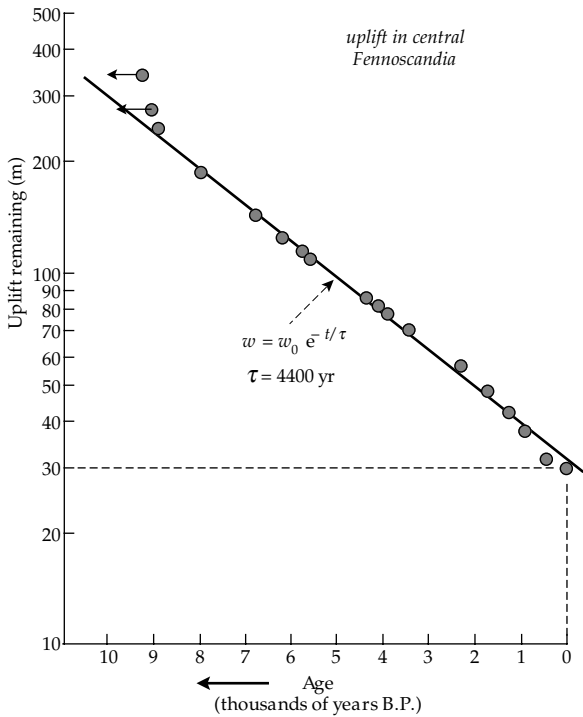


Fig. 2.81 Uplift in central Fennoscandia since the end of the last ice age illustrates exponential viscous relaxation with a time constant of 4400 yr (after Cathles, 1975).

dimension appropriate to the scale of the load (as will be seen in examples below). A test of these relationships requires data that give surface elevations in the past. These data come from analyses of sea-level changes, present elevations of previous shorelines and directly observed rates of uplift. The ancient horizons have been dated by radiometric methods as well as by sedimentary methods such as varve chronology, which consists of counting the annually deposited pairs of silt and clay layers in laminated sediments. A good example of the exponential restoration of a depressed region is the history of uplift in central Fennoscandia since the end of the last ice age some 10,000 yr ago (Fig. 2.81). If it is assumed that about 30 m of uplift still remain, the observed uplift agrees well with Eq. (2.131) and gives a relaxation time of 4400 yr.

An important factor in modelling uplift is whether the viscous response is caused by the mantle as a whole, or whether it is confined to a low-viscosity layer (or “channel”) beneath the lithosphere. Seismic shear-wave velocities are reduced in a low-velocity channel about 100–150 km thick, whose thickness and seismic velocity are however variable from one locality to another. Interpreted as the result of softening or partial melting due to temperatures near the melting point, the seismic low-velocity channel is called the asthenosphere. It is not sharply bounded, yet it must be represented by a distinct layer in viscosity models, which indicate that its viscosity must be at least 25 times less than in the deeper mantle.

The larger the ice-sheet (or other type of surface load), the deeper the effects reach into the mantle. By

studying the uplift following removal of different loads, information is obtained about the viscosity at different depths in the mantle.

2.8.6.1 Viscosity of the upper mantle

About 18,000–20,000 years ago Lake Bonneville, the predecessor of the present Great Salt Lake in Utah, USA, had a radius of about 95 km and an estimated depth of about 305 m. The water mass depressed the lithosphere, which was later uplifted by isostatic restoration after the lake drained and dried up. Observations of the present heights of ancient shorelines show that the central part of the lake has been uplifted by about 65 m. Two parameters are involved in the process: the flexural rigidity of the lithosphere, and the viscosity of the mantle beneath. The elastic response of the lithosphere is estimated from the geometry of the depression that would be produced in isostatic equilibrium. The maximum flexural rigidity that would allow a 65 m deflection under a 305 m water load is found to be about 5×10^{23} N m.

The surface load can be modelled as a heavy vertical right cylinder with radius r , which pushes itself into the soft mantle. The underlying viscous material is forced aside so that the central depression is surrounded by a circular uplifted “bulge.” After removal of the load, restorative uplift takes place in the central depression, the peripheral bulge subsides and the contour of zero-uplift migrates outward. The wavelength λ of the depression caused by a load with this geometry has been found to be about 2.6 times the diameter of the cylindrical load. In the case of Lake Bonneville $2r = 192$ km, so λ is about 500 km. The mantle viscosity is obtained by assuming that the response time of the lithosphere was short enough to track the loading history quite closely. This implies that the viscous relaxation time τ must have been 4000 yr or less. Substitution of these values for λ and τ in Eq. (2.132) suggests a maximum mantle viscosity η of about 2×10^{20} Pa s in the top 250 km of the mantle. A lower value of η would require a thinner low-viscosity channel beneath the lithosphere.

The Fennoscandian uplift can be treated in the same way (Fig. 2.82), but the weight and lateral expanse of the load were much larger. The ice-cap is estimated to have been about 1100 m thick and to have covered most of Norway, Sweden and Finland. Although the load was therefore somewhat elongate, it is possible to model it satisfactorily by a vertical cylinder of radius $r = 550$ km centered on the northern Gulf of Bothnia (Fig. 2.83). The load was applied for about 20,000 yr before being removed 10,000 yr ago. This caused an initial depression of about 300 m, which has subsequently been relaxing (Fig. 2.82). The uplift rates allow upper-mantle viscosities to be estimated. The data are compatible with different models of mantle structure, two of which are compared in Fig. 2.83. Each model has an elastic lithosphere with a flexural rigidity of 5×10^{24} N m underlain by a low-viscosity channel

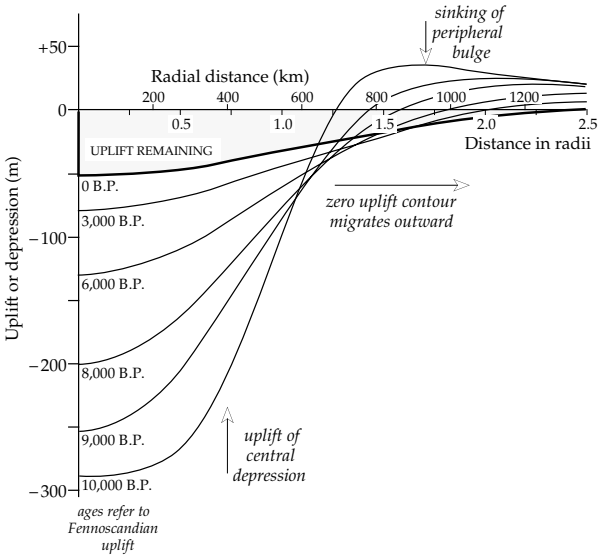


Fig. 2.82 Model calculations of the relaxation of the deformation caused by the Fennoscandian ice-sheet following its disappearance 10,000 yr ago (after Cathles, 1975).

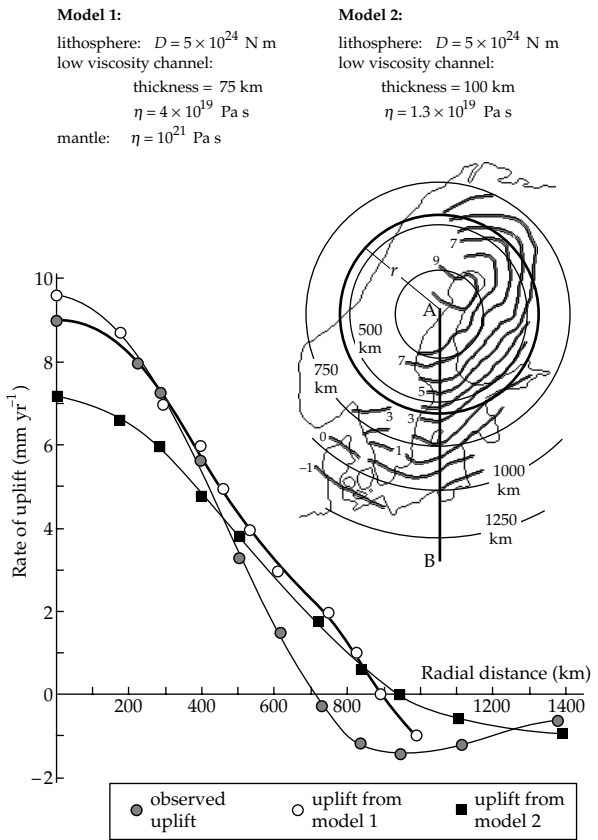


Fig. 2.83 Comparison of Fennoscandian uplift rates interpreted along profile AB (*inset*) with uplift rates calculated for two different models of mantle viscosity, assuming the ice-sheet can be represented by a vertical right cylindrical load centered on the northern Gulf of Bothnia (after Cathles, 1975).

and the rest of the mantle. The first model has a 75 km thick channel ($\eta = 4 \times 10^{19}$ Pa s) over a viscous mantle ($\eta = 10^{21}$ Pa s). The alternative model has a 100 km thick channel (viscosity coefficient $\eta = 1.3 \times 10^{19}$ Pa s) over a

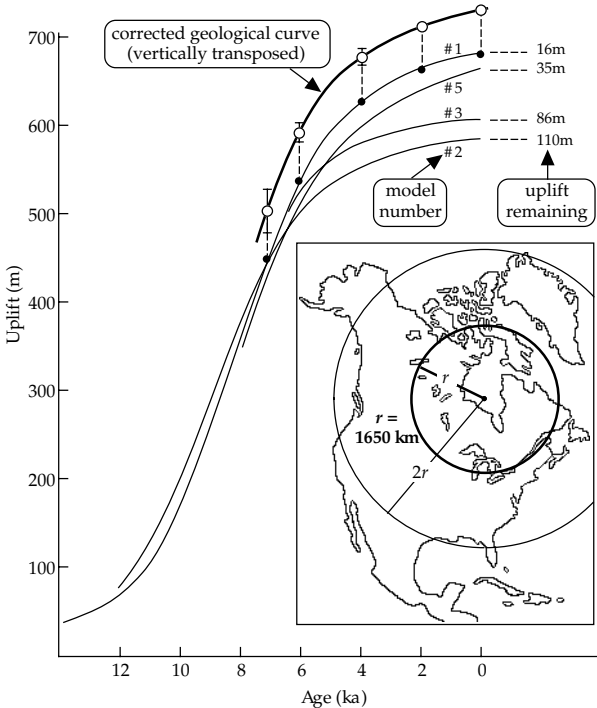


Fig. 2.84 Comparison of uplift history in the James Bay area with predicted uplifts for various Earth models after disappearance of the Wisconsin ice-sheet over North America, represented as a vertical right cylindrical load with radius 1650 km as in the inset (after Cathles, 1975). For details of model parameters see Table 2.2.

rigid mantle. Both models yield uplift rates that agree quite well with the observed uplift rates. However, results from North America indicate that the lower mantle is not rigid and so the first model fits the data better.

2.8.6.2 Viscosity of the lower mantle

Geologists have developed a coherent picture of the last glacial stage, the Wisconsin, during which much of North America was covered by an ice-sheet over 3500 m thick. The ice persisted for about 20,000 yr and melted about 10,000 yr ago. It caused a surface depression of about 600 m. The subsequent history of uplift in the James Bay area near the center of the feature has been reconstructed using geological indicators and dated by the radiocarbon method (see Section 4.1.4.1). For modelling purposes the ice-sheet can be represented as a right cylindrical load with radius $r = 1650$ km (Fig. 2.84, *inset*). A load as large as this affects the deep mantle. The central uplift following removal of the load has been calculated for several Earth models. Each model has elastic parameters and density distribution obtained from seismic velocities, including a central dense core. The models differ from each other in the number of viscous layers in the mantle and the amount by which the density gradient departs from the adiabatic gradient (Table 2.2). The curvature of the observed uplift curve only fits models in which the viscosity of the lower mantle is around 10^{21} Pa s (Fig. 2.84). A highly viscous lower mantle ($\eta = 10^{23}$ Pa s, model 4) is

Table 2.2 *Parameters of Earth models used in computing uplift rates. All models are for an elastic Earth with a dense core (after Cathles, 1975)*

Model	Density gradient	Viscosity [10 ²¹ Pa s]	Depth interval
1	adiabatic	1	entire mantle
2	adiabatic, except 335–635 km	1	entire mantle
3	adiabatic, except 335–635 km	0.1	0–335 km
		1	335 km to core
4	adiabatic	1	0–985 km
		100	985 km to core
5	adiabatic	1	0–985 km
		2	985–2185 km
		3	2185 km to core

incompatible with the observed uplift history. The best fit is obtained with model 1 or 5. Each has an adiabatic density gradient, but model 1 has a uniform lower mantle with viscosity 10²¹ Pa s, and model 5 has η increasing from 10²¹ Pa s below the lithosphere to 3 × 10²¹ Pa s just above the core.

The viscoelastic properties of the Earth’s interior influence the Earth’s rotation, causing changes in the position of the instantaneous rotation axis. The motion of the axis is traced by repeated photo-zenith tube measurements, in which the zenith is located by photographing the stars vertically above an observatory. Photo-zenith tube measurements reveal systematic movements of the rotation axis relative to the axis of the figure (Fig. 2.85). Decomposed into components along the Greenwich meridian (X-axis) and the 90°W meridian (Y-axis), the polar motion exhibits a fluctuation with cyclically varying amplitude superposed on a linear trend. The amplitude modulation has a period of approximately seven years and is due to the interference of the 12-month annual wobble and the 14-month Chandler wobble. The linear trend represents a slow drift of the pole toward northern Canada at a rate of 0.95 degrees per million years. It is due to the melting of the Fennoscandian and Laurentide ice-sheets. The subsequent uplift constitutes a redistribution of mass which causes modifications to the Earth’s moments and products of inertia, thus affecting the rotation.

The observed polar drift can be modelled with different viscoelastic Earth structures. The models assume a 120 km thick elastic lithosphere and take into account different viscosities in the layers bounded by the seismic discontinuities at 400 km and 670 km depths (Section 3.7, Table 3.4) and the core–mantle boundary. An Earth that is homogeneous below the lithosphere (without a core) is found to give imperceptible drift. Inclusion of the core, with a density jump across the core–mantle boundary and assuming the mantle viscosity to be around 1 × 10²¹ Pa s, gives a drift that is perceptible but much slower than that

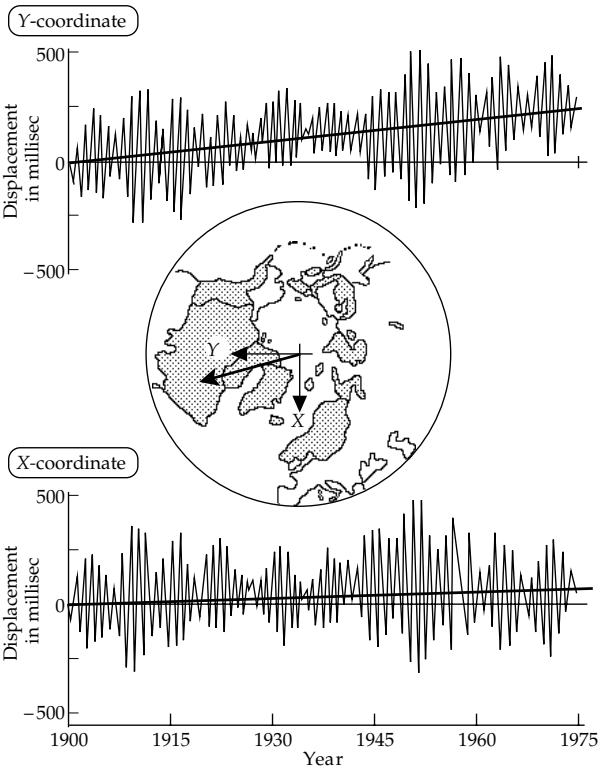


Fig. 2.85 Changes in the position of the instantaneous rotation axis from 1900 to 1975 relative to axes defined in the inset (after Peltier, 1989).

observed. Introduction of a density change at the 670 km discontinuity increases the drift markedly; the 400 km discontinuity does not noticeably change the drift further. The optimum model has an upper-mantle viscosity of about 1 × 10²¹ Pa s and a lower-mantle viscosity of about 3 × 10²¹ Pa s. The model satisfies both the rate of drift and its direction (Fig. 2.85). The viscosities are comparable to values found by modelling post-glacial uplift (Table 2.2, model 5).

2.9 SUGGESTIONS FOR FURTHER READING

Introductory level

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Massonnet, D. 1997. Satellite radar interferometry. *Sci. Am.*, **276**, 46–53.

Mussett, A. E. and Khan, M. A. 2000. *Looking into the Earth: An Introduction to Geological Geophysics*, Cambridge: Cambridge University Press.

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Intermediate level

- Dobrin, M. B. and Savit, C. H. 1988. *Introduction to Geophysical Prospecting*, 4th edn, New York: McGraw-Hill.
- Fowler, C. M. R. 2004. *The Solid Earth: An Introduction to Global Geophysics*, 2nd edn, Cambridge: Cambridge University Press.
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- Turcotte, D. L. and Schubert, G. 2002. *Geodynamics*, 2nd edn, Cambridge: Cambridge University Press.

Advanced level

- Blakely, R. J. 1995. *Potential Theory in Gravity and Magnetic Applications*, Cambridge: Cambridge University Press.
- Bullen, K. E. 1975. *The Earth's Density*, London: Chapman and Hall.
- Cathles, L. M. 1975. *The Viscosity of the Earth's Mantle*, Princeton, NJ: Princeton University Press.
- Officer, C. B. 1974. *Introduction to Theoretical Geophysics*, New York: Springer.
- Ranalli, G. 1995. *Rheology of the Earth*, 2nd edn, London: Chapman and Hall.
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- Watts, A. B. 2001. *Isostasy and Flexure of the Lithosphere*, Cambridge: Cambridge University Press.

9. What are the Bouguer plate and free-air gravity corrections?
10. What is a free-air gravity anomaly? How does it differ from a Bouguer anomaly?
11. Sketch how the Bouguer gravity anomaly might vary on a continuous profile that extends from a continental mountain range to an oceanic ridge.
12. What is the Coriolis acceleration? How does it originate? How does it affect wind patterns in the North and South hemispheres?
13. What is the Eötvös gravity correction? When is it needed? How does it originate?
14. Describe *two* borehole methods for determining the density of rocks around the borehole.
15. Describe and explain the Nettleton profile method to determine the optimum density for interpreting a gravity survey.
16. Explain how to calculate the position of the common center of mass of the Earth–Moon system.
17. Explain with the aid of diagrams that show the forces involved, why there are two lunar tides per day.
18. Why are the lunar tides almost equal on opposite sides of the Earth? Why are they not exactly equal?
19. What is isostasy? What is an isostatic gravity anomaly?
20. Why is the free-air gravity anomaly close to zero at the middle of a large crustal block that is in isostatic equilibrium?
21. Describe the three models of isostasy, and explain how they differ from each other.
22. What would be the geodynamic behavior of a region that is characterized by a negative isostatic gravity anomaly?

2.10 REVIEW QUESTIONS

1. Describe the principle of operation of a gravimeter.
2. Explain why a gravimeter only gives relative measurements of gravity.
3. What is the *geoid*? What is the *reference ellipsoid*? How and why do they differ?
4. What is a geoid anomaly? Explain how a positive (or negative) anomaly arises.
5. What is *normal gravity*? What does the word *normal* imply? Which surface is involved?
6. Gravitational acceleration is directed toward a center of mass. With the aid of a sketch that shows the directions of gravity's components, explain why *gravity* is not a centrally directed acceleration.
7. Write down the general expression for the normal gravity formula. Explain which geophysical parameters determine each of the constants in the formula?
8. What is the topographic correction in the reduction of gravity data? Why is it needed?

2.11 EXERCISES

1. Using the data in Table 1.1 calculate the gravitational acceleration on the surface of the Moon as a percentage of that on the surface of the Earth.
2. An Olympic high-jump champion jumps a record height of 2.45 m on the Earth. How high could this champion jump on the Moon?
3. (a) Calculate the escape velocity of an object on the Earth, assuming a mean gravitational acceleration of 9.81 m s^{-2} and mean Earth radius of 6371 km.
(b) What is the escape velocity of the same object on the Moon?
4. The equatorial radius of the Earth is 6378 km and gravity at the equator is 9.780 m s^{-2} . Compute the ratio m of the centrifugal acceleration at the equator to the *gravitational acceleration* at the equator. If the ratio m is written as $1/k$, what is the value of k ?

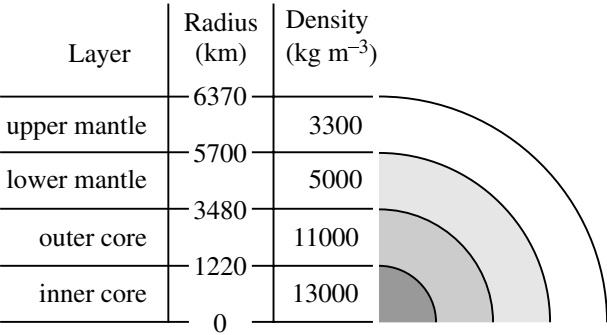
5. Given that the length of a month is 27.32 days, the mean gravity on Earth is 9.81 m s^{-2} and the Earth's radius is 6371 km, calculate the radius of the Moon's orbit.
6. A communications satellite is to be placed in a geo-stationary orbit.
- (a) What must the period and orientation of the orbit be?
 - (b) What is the radius of the orbit?
 - (c) If a radio signal is sent to the satellite from a transmitter at latitude 45°N , what is the shortest time taken for its reflection to reach the Earth?
7. Calculate the centrifugal acceleration due to the Earth's rotation of an object at rest on the Earth's surface in Paris, assuming a latitude of $48^\circ 52' \text{ N}$. Express the result as a percentage of the gravitational attraction on the object.
8. A solid angle (Ω) is defined as the quotient of the area (A) of the part of a spherical surface subtended by the angle, divided by the square of the spherical radius (r): i.e., $\Omega = A/r^2$ (see Box 5.4). Show with the aid of a diagram that the gravitational acceleration at any point inside a thin homogeneous spherical shell is zero.
9. Assuming that the gravitational acceleration inside a homogeneous spherical shell is zero, show that the gravitational acceleration inside a homogenous uniform *solid* sphere is proportional to the distance from its center.
10. Show that the gravitational potential U_G inside a homogenous uniform solid sphere of radius R at a distance r from its center is given by
- $$U_G = -\frac{2\pi}{3}G\rho(3R^2 - r^2)$$
11. Sketch the variations of gravitational acceleration and potential inside and outside a homogeneous solid sphere of radius R .
12. A thin borehole is drilled through the center of the Earth, and a ball is dropped into the borehole. Assume the Earth to be a homogenous solid sphere. Show that the ball will oscillate back and forth from one side of the Earth to the other. How long does it take to traverse the Earth and reach the other side?
13. The Roche limit is the closest distance an object can approach a planet before being torn apart by the tidal attraction of the planet. For a rigid spherical moon the Roche limit is given by Eq. (6) in Box 2.1.
- (a) Using the planetary dimensions in Table 1.1, calculate the Roche limit for the Moon with respect to the Earth. Express the answer as a multiple of the Earth's radius.
 - (b) Show that, for a planet whose mean density is less

than half that of its rigid moon, the moon would collide with the planet before being torn apart by its gravity.

- (c) Given that the Sun's mass is $1.989 \times 10^{30} \text{ kg}$ and that its radius is 695,500 km, calculate the Roche limit for the Earth with respect to the Sun.
 - (d) The mean density of a comet is about 500 kg m^{-3} . What is the Roche limit for comets that might collide with the Earth?
 - (e) The mean density of an asteroid is about 2000 kg m^{-3} . If an asteroid on collision course with the Earth has a velocity of 15 km s^{-1} , how much time will elapse between the break-up of the asteroid at the Roche limit and the impact of the remnant pieces on the Earth's surface, assuming they maintain the same velocity as the asteroid?
14. The mass M and moment of inertia C of a thick shell of uniform density ρ , with internal radius r and external radius R are given by

$$M = \frac{4}{3}\pi\rho(R^3 - r^3) \quad C = \frac{8}{15}\pi\rho(R^5 - r^5)$$

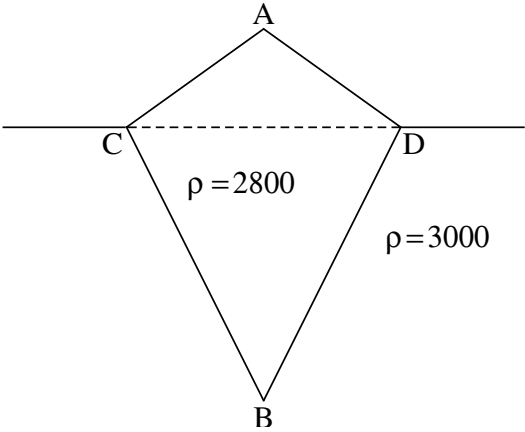
The Earth has an internal structure consisting of concentric spherical shells. A simple model with uniform density in each shell is given in the following figure.



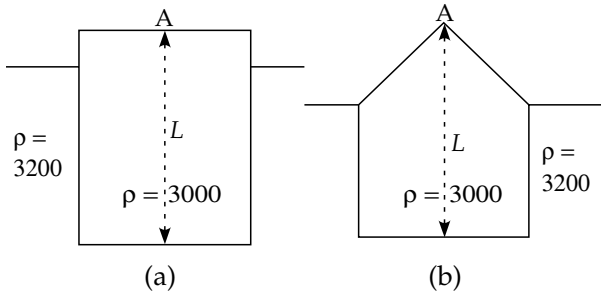
- (a) Compute the mass and moment of inertia of each spherical shell.
 - (b) Compute the total mass and total moment of inertia of the Earth.
 - (c) If the moment of inertia can be written $C = kMR^2$, where M is Earth's mass and R its radius, what is the value of k ?
 - (d) What would the value of k be if the density were uniform throughout the Earth?
15. By differentiating the normal gravity formula given by Eq. (2.56) develop an expression for the change in gravity with latitude. Calculate the gravity change in milligals per kilometer of northward displacement at latitude 45° .
16. The following gravity measurements were made on a traverse across a rock formation. Use the combined elevation correction to compute the apparent density of the rock.

Elevation [m]	Gravity [mgal]
100	-39.2
150	-49.5
235	-65.6
300	-78.1
385	-95.0
430	-104.2

17. Show that the “half-width” w of the gravity anomaly over a sphere and the depth z to the center of the sphere are related by $z = 0.652w$.
18. Assume the “thin-sheet approximation” (Eq. (2.101)) for the gravity anomaly over a vertical fault of density contrast $\Delta\rho$ and height h with mid-point at depth z_0 .
- (a) What is the maximum slope of the anomaly and where does it occur?
- (b) Determine the relationship between the depth z_0 and the horizontal distance w between the positions where the slope of the anomaly is one-half the maximum slope.
19. Calculate the maximum gravity anomaly at ground level over a buried anticlinal structure, modelled by a horizontal cylinder with radius 1000 m and density contrast 200 kg m^{-3} , when the depth of the cylinder axis is (a) 1500 m and (b) 5000 m.
20. The peak A of a mountain is 1000 meters above the level CD of the surrounding plain, as in the diagram. The density of the rocks forming the mountain is 2800 kg m^{-3} , that of the surrounding crust is 3000 kg m^{-3} . Assuming that the mountain and its “root” are symmetric about A and that the system is in local isostatic equilibrium, calculate the depth of B below the level CD.



21. A crustal block with mean density 3000 kg m^{-3} is initially in isostatic equilibrium with the surrounding rocks whose density is 3200 kg m^{-3} , as in the figure (a). After subsequent erosion the above-surface topography is as shown in (b). The distance L remains constant (i.e. there is no erosion at the highest point A) and Airy-type isostatic equilibrium is maintained. Calculate in terms of L the amount by which the height of A is changed. Explain why A moves in the sense given by your answer.



22. An idealized mountain-and-root system, as in the figure, is in isostatic equilibrium. The densities in kg m^{-3} are as shown. Express the height H of the point A above the horizontal surface RS in terms of the depth D of the root B below this surface.

